Class 6
Mathematics

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1. Knowing Our Numbers

Solutions

LEVEL- I

1. Correct option: (A) >

Number 102394 is greater than 99887.

102394 > 99887

2. Correct option: (B) <

Number 2507324 is less than 2517324.

2507324 < 2517324

3. Correct option: (B) 6539542, 8354208, 12345678, 63547201, 102345694

Given numbers are 63547201, 102345694, 6539542, 8354208, 12345678.

On arranging the numbers in ascending order,

6539542 < 8354208 < 12345678 < 63547201 < 102345694

Hence, Option (B) is the correct answer.

4. Correct option: (C) 056944000, 56943300, 56943201, 5695440, 5694437

Given numbers are 056944000, 56943201, 5695440, 56943300, 5694437.

On arranging the numbers in descending order,

056944000 > 56943300 > 56943201 > 5695440 > 5694437

Hence, Option (C) is the correct answer.

5. Correct option: (C) 85,210 and 10,258

Numbers are 5, 0, 2, 8 and 1.

To form the greatest 5-digit number, the biggest number is taken.

So, the greatest number formed is 85210.

And to form the smallest 5-digit number, smallest number 0 is taken.

The number formed is 01258. But the zero on the left side doesn't have any significance. So, it will be considered as a 4-digit number.

But the smaller 5-digit number will be 10,258.

Hence, the correct option is (C)

6. Correct option: (D) 71,23,400; 64,11,060

Number 71,23,400 is greater than 7,12,340.

Subtract the number to get value by how much number 7123400 is greater than 712340.

$$=7123400 - 712340$$

$$= 64,11,060$$

Hence, the correct option is (D)

7. Correct option: (B)

The smallest 5-digit number is 10000 and largest 5-digit number is 99999.

So, 5-digit number is given by, $99999 - 10000 + 1 = 90000$

8. Correct option: (D) 1001

The smallest 4-digit number from the given option is 1001. If the digits are written in the reverse order, it remains the same.

Hence, the correct answer is (D)

9. Correct option: (A)

The nearest tens for each number are:

21396 is 21400.

27808 is 27810.

42504 is 42500.

Sum of these numbers is, $21400 + 27810 + 42500 = 91710$

10. Correct option: (B)

i) Rounding off 1335 to greatest place, which is 1000
Rounding off 8657 to greatest place, which is 9000.
Estimated sum = 9000 + 1000 =10000

ii) Rounding off 2675 to greatest place, which is 3000.
Rounding off 9087 to greatest place, which is 9000.
Estimate sum = 3000 + 9000 = 12000.

11. Correct option: (D) 100, 1000

i) Rounding off 275 to greatest place, which is 300.
Rounding off 167 to greatest place, which is 200.
Estimated difference = $300 - 200 = 100$.

ii) Rounding off 9312 to greatest place, which is 9000.
Rounding off 7603 to greatest place, which is 8000.
Estimated difference = $9000 - 8000 = 1000$.

12. Correct option: (A)

The place value of 0 in 120945349 is 1000000.

13. Correct option: (A)

Sum of place value of 9 in 54936919 is, $900000 + 900 + 9$
$= 900909$

14. Correct option: (A)

Expanded form of 7303206 is,
$7 \times 1000000 + 3 \times 100000 + 0 \times 10000 + 3 \times 1000 + 2 \times 100 + 0 \times 10 + 6 \times 1$

15. Correct option: (C)

i) Four million five hundred twenty thousand six hundred seventy-eight in digits is 4,520,678

ii) Eight million nine thousand eight in digits is 8,009,008

16. Correct option: (B)

i) 4032897 with commas in international system of numeration is 4,032,897.

ii) 87594652 with commas in international system of numeration is 87,594,652

17. Correct option: (C) 100 lakhs

10 lakhs = 1 million
10 millions = 10 x 10 lakhs = 100 lakhs
Hence, the options (C) is the correct answer.

18. Correct option: (B)

10 crores = 100 million

19. Correct option: (C)

Mass of basket of fruits is 17000 g
Greatest mass of the basket is 17499 g

20. Correct option: (A)

Distance between school and house is 1km 875m == $1.875km$
Distance travelled by boy in a day is, $= 2 \times 1875 = 3.75$
Distance travelled in a week, $= 3.75 \times 7 = 26.25$

LEVEL-II

1. Correct option: (C)

20089405 is arranged to get 8-digit smallest number and largest number is 20004589 and 98542000 respectively.

2. Correct option: (A)

Three-digit numbers formed using number 0, 2 and 9 without repetition are 209, 290, 902 and 920.

3. Correct option: (B)

When digits in tens and one's place are interchanged with hundreds and thousands number becomes

a) 7865 is 5687
b) 8865 is 5688
c) 8765 is 5678
d) 7865 is 5687

So, 5688 is the largest.

4. Correct option: (B)

The value of $100 - 7 \times 1 = 100 - (7 \times 1) + a$

$100 - 7 = 100 - 7 + a$

$93 - 93 = a \; ; \; a = 0$

5. Correct option: (B)

The place value of 3 in 239766 is 30000.

The place value of 6 in 45655 is 600.

Then, $\frac{30000}{600} = 50$ So, the place value of 3 is 50 times place value of 6.

6. Correct option: (A)

The truck carries 70 boxes.

Boxes contain 45 small packages.

Small packages contain 25 toys.

Toys carried in truck,$= 70 \times 45 \times 25 = 78750$

7. Correct option: (C)

Numerator 2838 is rounded off as 2800.

Denominator 125 is rounded off as 100.

Then quotient is, $\frac{2800}{100} = 28$

8. Correct options: (A) 1173 > 73 and (B) 8110 < 9000

(A) 1173 > 73; four digits number is greater than the two-digit number

So, option (A) is correct.

Option (B): 8110 < 9000

Both numbers 8100 and 9000 have 4 digits.

Look at their respective digits from left to right.

Starting from the left, let's compare the first digit of both numbers from the left, we get, 8 < 9; So, option (B) is also correct.

Option (C): 117 > 977

Both numbers 117 and 977 have 3 digits.

Look at their respective digits from left to right.

Starting from the left, let's compare the first digit of both numbers from the left, we get, 1<9 117 < 977; So, option (C) is incorrect.

Hence the correct options are (A) and (B).

9. Correct options: (B), (C), and (D)

Option (A): The number 74 is a 2-digit number.

Option (B): The number 9 is a 1-digit number.

Option (C): The number 2 is a 1-digit number.

Option (D): The number 3 is a 1-digit number.

Hence the correct options are (B), (C) and (D).

10. Correct options: (A) and (B)

From the options.

Option (A): 100 lakhs = 100 × 1,00,000 = 1,00,00,000 = 1 crore

Option (B): 10,000 thousand = 10,000 × 1,000 = 1,00,00,000 = 1 crore

Option (C): 10,000 ≠ 1 crore

Hence the correct options are (A) and (B).

11. Correct options: (B)

The numbers, when arranged in ascending order: 8192, 8312, 8729, 8937, 8973.

The 2nd number, when the given set of numbers is arranged in ascending order, is 8312.

Hence the correct option is (B).

12. Correct options: (A)

The numbers, when arranged in descending order: 8973, 8937, 8729, 8312, 8192.

The middle number, when the given set of numbers is arranged in descending order, is 8729.

Hence the correct option is (A).

13. Correct options: (C)

The numbers, when arranged in ascending order:

8192, 8312, 8729, 8937, 8973

The 4th number, when the given set of numbers is arranged in ascending order, is 8937.

The 2nd number, when the given set of numbers is arranged in ascending order, is 8312.

Difference between the 4th and the 2nd number = 8937 – 8312 = 625

Hence the correct option is (C).

14. Correct Matching is: (a) → (Q); (b) →(R); (c) → (R); (d) → (T)

We know that, 1 km = 1,000 m = 1,00,000 cm = 10,00,000 mm

1 m = 100 cm = 1000 mm

1 cm = 10 mm

1 mm = $\frac{1}{10}$ cm = $\frac{1}{1000}$ m = $\frac{1}{1000000}$ km

1 cm = $\frac{1}{100}$ m = $\frac{1}{100000}$ km

1 m = $\frac{1}{1000}$ km

So, 10 mm = $\frac{10}{1000000}$ km = $\frac{1}{100000}$ km

100 cm = $\frac{100}{100000}$ km = $\frac{1}{1000}$ km

COLUMN-I	COLUMN-II
(a) 10 mm	(P) $\frac{1}{10000}$ km
(b) 100 cm	(Q) $\frac{1}{100000}$ km
(c) 1 m	(R) $\frac{1}{1000}$ km
(d) 1 km	(S) 100100 cm
	(T) 100000 cm

15. Correct Answer is: 1

The multiplication solution given in the question:

$$\begin{array}{cccccc} & & 2 & 6 & 7 \\ & & \times & 5 & 4 \\ \hline & 1 & 0 & 6 & 8 \\ 1 & 0 & 3 & 5 & \\ \hline 1 & 1 & 4 & 1 & 8 \end{array}$$

The correct calculation is given below:

$$\begin{array}{cccc} & 2 & 6 & 7 \\ \times & & 5 & 4 \\ \hline & 1 & 0 & 6 & 8 \\ 1 & 3 & 3 & 5 & \\ \hline 1 & 4 & 4 & 1 & 8 \end{array}$$

On comparing both solutions, the incorrect digit in the final solution is 1.

16. Correct Answer is: 6

$244 + (8 \times 4) - 318 \div a = 223$

By using BODMAS,

Step 1: Solve the expression in brackets, $244 + 32 - 318 \div a = 223$

Step 2: Simplify the expression in LHS. $276 - 318 \div a = 223$

Step 3: Rearrange the terms. $276 - 223 = 318 \div a$

Step 4: Simplify the expression in LHS. $53 = 318 \div a$

$53 = \frac{318}{a}$

Step 5: Rearrange the terms. $a = \frac{318}{53} = 6$

Hence the value of a is 6.

17. Correct Answer is: 0

Given the equation, $201 - 9 \times 1 = 201 - (9 \times 1) + a$

By using BODMAS: $201 - 9 = 201 - 9 + a$

The value of the expression on LHS and RHS will be equal if a = 0.

So, the value of a is 0.

18. Correct Answer is: 4

The number 23,54,125 is placed in a place value chart:

Number	T L	L	T Th	Th	H	T	O
23,54,125	2	3	5	4	1	2	5

From the place value chart, the digit in the thousands place is 4.

19. Correct Answer is: 6

Given equation, $-6 + __ = 0$

Since –6 is being added to the missing number, we can find the missing number by subtracting –6 from 0.

$0 - (-6) = 0 + 6 = 6$

Therefore, the missing number in the equation $-6 + __ = 0$ is 6.

20. Correct Answer is: 4

Since we have 3 digits to choose from, we can consider each digit as a separate position in the 3-digit number: hundreds, tens, and units.

For the hundreds place, we have 2 options (5 or 7).

For the tens place, we have 2 options (since we cannot repeat the digit used in the hundreds place).

For the unit's place, we have 1 option left.

Therefore, the total number of 3-digit numbers that can be formed without repeating any digit is:

2 options for the hundreds place × 2 options for the tens place × 1 option for the unit's place = 2 × 2 × 1 = 4

Hence, there are 4 different 3-digit numbers that can be formed using the digits 0, 5, and 7 without repeating any digit.

STUDENT PRACTICE QUESTIONS

1. Correct option: (A) >
Number 3572014 is greater than 1025310.
3572014 > 1025310
2. Correct option: (D) 181888, 190909, 1808088, 1808090, 16060666
Given numbers are 1020216, 1020308, 1021430, 893425, 893245.
On arranging the numbers in ascending order,
181888 < 190909 < 1808088 < 1808090 < 16060666
Hence, Option (D) is the correct answer.
3. Correct option: (B) 1021430, 1020308, 1020216, 893425, 893245
Given numbers are 1020216, 1020308, 1021430, 893425, 893245.
On arranging the numbers in descending order,
1021430 > 1020308 > 1020216 > 893425 > 893245.
Hence, the correction option is (B).
4. Correct option: (C)
The smallest three-digit number with unique number is 102.
5. Correct option: (D)
The greatest 7-digit number is 9999999 and smallest 5-digit number is 10000.
Difference of these number is, $9999999 - 10000 = 9989999$
6. Correct option: (A) 3,000, 5,00,000
i) Given product is 102 x 25 = 2550
Rounding off 2550 to greatest place, which is 3,000.
ii) Given product is 23502 x 23 = 5,40,546
Rounding off 5,40,546 to greatest place, which is 5,00,000
Hence, the correct answer is Option (A)
7. Correct option: (D)
Difference of place value and face value of 73810924 is,
$70000000 - 7 = 69999993$
8. Correct option: (A)
Ten thousand lakhs make-up a million
9. Correct option: (C)
The newspaper contains 24 pages.
10420 copies of the newspaper are published every day.
So, number of pages $= 10420 \times 24 = 250080$
10. Correct option: (B)
Mass of cylinder is 15kg 250 gm $= 15.250kg$
Mass of 28 cylinder is $= 28 \times 15.250 = 427kg$
11. Correct option: (A)
Number 8 has place value of thousand in 5,68,731.
12. Correct option: (D)
The digit in thousand place is 4 in 14,512.
13. Correct options: (B) and (C)
Best estimation of 5663×2234 to the tens value is 5660×2230.
Best estimation of 5663×2234 to the hundreds value is 5700×2200.
14. Correct options: (B) and (D)
Option (A): Sixty is in the word form.
Option (B): 74 is in the numeral.
Option (C): Ninety-five is in the word form.

Option (D): 8711 is in the numeral.
Hence the correct options are (B) and (D).

15. Correct options: (B) and (C)

From the options,
Option (A): 1001 ≠ 999 + 1
Option (B): 1000 = 999 + 1
Option (C): 1 + 999 = 1000 = 999 + 1
Hence the correct options are (B) and (C).

18. Correct options: (C)

The number of blue marbles with Johny = 36
The number of red marbles with Johny = 54
The total number of marbles with Johny in all
= Number of blue marbles + Number of red marbles
= 36 + 54
= 90
Hence the correct option is (C).

19. Correct option: (A)

The number of blue marbles with Johny = 36
The number of red marbles with Johny = 54
Difference = Number of red marbles + Number of blue marbles
= 54 – 36
= 18
The number of blue marbles is 18 less than the number of red marbles.
Hence the correct option is (A).

20. Correct Matching is: (a) → (T); (b) →(R); (c) → (S); (d) → (P)

To find the largest n-digit number, we need to choose the highest possible digit for each place value. Since each digit can range from 0 to 9, we choose the highest digit, which is 9, for each place value.

COLUMN-I	COLUMN-II
(a) Largest 4-digit number	(P) 999
(b) Largest 5-digit number	(Q) 99
(c) Largest 6-digit number	(R) 99999
(d) Largest 3-digit number	(S) 999999
	(T) 9999

21. Correct Answer is: 8

The number 1,38,463 is placed in a place value chart:

Number	L	T Th	Th	H	T	O
1,38,463	1	3	8	4	6	3

From the place value chart, the digit in the thousands place is 8.

22. Correct Answer is: 9

Subtracting a negative number is equivalent to adding the positive counterpart.
Therefore, −(−3) is the same as +3.
So, 6 – (−3) becomes: 6 + 3 = 9
Therefore, 6 – (−3) is equal to 9.

2. Integers, Fractions, and Decimals
Solutions

LEVEL – I

1. Correct option: (B)
The successor of a number is the next number that comes immediately after it in the number line.
In this case, the first successor of -42 is -41 because it is the next number that comes after -42 in the number line.
So, the correct answer is B) -41.

2. Correct option: (C)
Subtracting -7 from -10 can be solved as follows: $-10 - (-7) = -10 + 7 = -3$
So, the subtraction of -7 from -10 is equal to -3.
Therefore, the correct answer is C) -3.

3. Correct option: (B)
The additive inverse of a number is the number that, when added to it, results in zero. In this case, the additive inverse of 28 is -28.
So, the correct answer is B) -28.

4. Correct option: (B)
The predecessor of a number is the number that comes immediately before it in the number line. In the case of 0, the predecessor is -1.
So, the correct answer is B) -1.

5. Correct option: (A)
We have one integer as 135, and the sum of the two integers is -355. To find the other integer, we can use the equation: $135 + \text{Other Integer} = -355$
To solve for the Other Integer, we need to isolate it on one side of the equation:

$$\text{Other Integer} = -355 - 135$$
$$\text{Other Integer} = -490$$

So, the other integer is -490.
The correct answer is A) -490.

6. Correct option: (C)
To find the sum of the alternating sequence $1 - 2 + 3 - 4 + 5 - 6 + \ldots + 17 - 18$,
We can group the terms into pairs: $(1 - 2) + (3 - 4) + (5 - 6) + \ldots + (17 - 18)$
Now, simplify each pair: $-1 - 1 - 1 \ldots -1$
We have 9 pairs of -1, so the sum of the sequence is: $-1 \times 9 = -9$
So, the sum of the given sequence is -9.

7. Correct option: (B)
The symbol ">" represents "greater than."
In the given expression, "-14 > -24," which means that "-14" is greater than "-24."
So, the correct answer is B) >.

8. Correct option: (B)
When we calculate the absolute value of a number, we take its positive value.
So, |5| + |-12| = 5 + 12 = 17.
Therefore, the correct answer is B) 17.

9. Correct option: (B)
In this scenario, a lower temperature indicates colder weather.
So, since $-12℃$ is lower than $-8℃$, Kashmir is colder than Jammu on the same day.
The correct answer is B) Kashmir.

10. Correct option: (B)

The product of two negative numbers is B) Positive.

When we multiply two negative numbers together, the negative signs cancel out, resulting in a positive product.

11. Correct option: (C)

Geetha: 45 correct answers $\times$ 5 marks/answer $-$ 5 wrong answers $\times$ 3 marks/answer

$= 225 - 15 = 210$ marks

Sita: 42 correct answers $\times$ 5 marks/answer $-$ 8 wrong answers $\times$ 3 marks/answer $= 210 - 24$

$= 186$ marks

The difference in the total marks obtained by them is:

$$210 \text{ (Geetha)} - 186 \text{ (Sita)} = 24 \text{ marks}$$

So, the correct answer is C) 24.

12. Correct option: (B)

The other integer can be found by dividing the product (-1800) by the given integer (30):

$$-1800 \div 30 = -60$$

So, the other integer is -60.

The correct answer is B) -60.

13. Correct option: (A)

The height of the lighthouse from the seabed is the sum of the height of the lighthouse and the depth of the seabed: $50\text{ m} + 25\text{ m} = 75\text{ m}$

So, the height of the lighthouse from the seabed is 75 meters.

The correct answer is A) 75 m.

14. Correct option: (B)

To find Raja's net score, we need to calculate the total points he earned by hitting balloons and subtract the total points he lost by missing balloons.

Points earned by hitting balloons: 10 balloons $\times$ 100 points/balloon $= 1000$ points

Points lost by missing balloons: 25 balloons $\times$ 50 points/balloon $= -1250$ points

Now, calculate the net score:

$$\text{Net Score} = \text{Points earned} - \text{Points lost}$$

$$\text{Net Score} = 1000 \text{ points} - 1250 \text{ points}$$

$$\text{Net Score} = -250 \text{ points}$$

So, Raja's net score is -250 points, which corresponds to option B) -250.

15. Correct option: (C)

Let's break down the calculation step by step:

The highest two-digit number is 99.

The highest one-digit positive number is 9.

Now, we need to subtract the negative of 99 from 9:

$$9 - (-99) = 9 + 99 = 108$$

So, the result is 108, which corresponds to option C) 108.

16. Correct option: (C)

Given; $56 - \{52 \times 2 - (13 - 6 \div 2) \times (\overline{4 - 6})\} + 4$

To simplify the expression using the BODMAS (Brackets, Orders, Division, Multiplication, Addition, and Subtraction) rule, follow these steps:

1. Start with the innermost operations within parentheses:

$$56 - \{52 \times 2 - (13 - 6 \div 2) \times (\overline{4 - 6})\} + 4$$

$$= 56 - \{52 \times 2 - (13 - 3) \times (-2)\} + 4$$

2. Next, perform the operations within the innermost parentheses:

$$= 56 - \{104 - 10 \times (-2)\} + 4$$

3. Now, calculate the multiplication and subtraction within the curly braces:

$= 56 - \{104 + 20\} + 4$

4. Calculate the addition inside the curly braces: $= 56 - 124 + 4$

5. Finally, perform the subtraction and addition in the overall expression:

$$= (56 - 124) + 4$$
$$= -68 + 4$$
$$= -64$$

So, the simplified result is -64, which corresponds to option C) -64.

17. Correct option: (A)

The absolute value of a number is always positive, so when 'a' is greater than 3, the absolute value of (a-3) would be: $|a - 3| = a - 3$

This corresponds to option A) a - 3.

18. Correct option: (D)

To simplify the expression -87362 + 71943, we can add the two numbers together:

$$-87362 + 71943 = -15419$$

So, the simplified result is -15419, which corresponds to option D) -15419.

19. Correct option: (B)

To find the division with the greatest quotient, we need to calculate the quotient for each option:

A) $20 \div (-4) = -5$

B) $20 \div 4 = 5$

C) $(-20) \div (-5) = 4$

D) $20 \div (-5) = -4$

Among the given options, option B) 20 ÷ 4 = 5 has the greatest quotient.

So, the correct answer is B) 20 ÷ 4.

20. Correct option: (A)

If the product of three integers is odd, it means all the three integers must be odd. This is because the product of even numbers is always even and the product of an odd integer and an even integer is even.

Let's consider four different scenarios:

1. All three integers are odd:

Example: $3 \times 5 \times 7 = 105$, sum of these three integers is 3+5+7 = 15, which is odd.

These numbers combination can be anything like positive and negative integers.

LEVEL-II

1. Correct option: (D)

Let's follow the steps of the machine to calculate the output when the input is -2500:

1. Start with the input: -2500
2. Add 5000: $-2500 + 5000 = 2500$
3. Subtract 3000: $2500 - 3000 = -500$
4. Multiply by 6: $-500 \times 6 = -3000$
5. Divide by 20: $-3000 / 20 = -150$

So, the output when the input is -2500 is -150, which corresponds to option D) -150.

2. Correct option: (B)

To find the sum of the sequence $1 - 2 + 3 - 4 + \ldots + 99 - 100$, we can group the numbers into pairs: $(1 - 2) + (3 - 4) + \ldots + (99 - 100)$

Now, calculate the sum of each pair: $-1 - 1 - 1 - 1 - 1 \ldots -1$

There are a total of 50 pairs, and each pair sums to -1. Therefore, the sum of the entire sequence is: $\text{Sum} = -1 \times 50 = -50$

So, the sum of the sequence is -50, which corresponds to option B) -50.

3. Correct option: (C)

If the sum of two integers is -34, and one of them is 18, we can find the other integer by subtracting 18 from -34: $\text{Other integer} = -34 - 18 = -52$

So, the other integer is -52, which corresponds to option C) -52.

4. Correct option: (D)

The additive inverse of a number is the number that, when added to the original number, results in zero.

Let's find the additive inverse of the expression $(-3) \times 9 + 4 \times (-5) - 15$:

$$\text{Additive Inverse} = -((-3) \times 9 + 4 \times (-5) - 15)$$
$$= -(-27 - 20 - 15)$$
$$= -(-62) = 62$$

So, the additive inverse of $(-3) \times 9 + 4 \times (-5) - 15$ is 62, which corresponds to option D) 62.

5. Correct option: (B)

In this equation, if x is a positive integer, then $|x|$ will also be equal to x, and subtracting x from x will result in zero: $x - |x| = x - x = 0$

So, option B) $x - |x| = 0$ is the correct choice.

6. Correct option: (B)

If x is greater than 2, then $|2 - x|$ is equal to the positive difference between 2 and x.

In other words, it's the distance between 2 and x on the number line.

To represent this mathematically: $|2 - x| = x - 2$

So, the correct answer is B) x - 2.

7. Correct option: (C)

To find the result of the operation $7 \triangle (-4)$, we'll use the given operation definition:

$$a \triangle b = a - b - 2$$

Now, plug in the values: $7 \triangle (-4) = 7 - (-4) - 2$

$$7 + 4 - 2 = 11 - 2 = 9$$

So, $7 \triangle (-4)$ equals 9, which corresponds to option C) 9.

8. Correct option: (A, B)

Integers are whole numbers, including both positive and negative numbers, as well as zero. Among the options provided:

A) 5 – This is an integer (positive).

B) -2 – This is an integer (negative).

C) $3/2$ – This is not an integer; it's a fraction or a rational number.

D) $1/2$ – This is not an integer; it's a fraction or a rational number.

So, the integers among the options are A) 5 and B) -2.

9. Correct option: (B, C)

Among the given statements:

A) $-100 > -98$ - This statement is not true. -100 is less than -98.

B) $98 < 100$ - This statement is true. 98 is indeed less than 100.

C) $-2 > -200$ - This statement is true. -2 is greater than -200.

D) $-5 < -6$ - This statement is false. -5 is indeed greater than -6.

So, the true statements are B), C).

10. Correct option: (B)

To find the number of natural numbers between -20 and 50, we need to count all the positive integers from 1 to 49:

Number of natural numbers = 49

To find the number of integers between -20 and 50, we need to count all the integers between -20 to 50, including 0:

Number of integers = 49 +20 = 69

So, the number of natural numbers and integers between -20 and 50 are 49 and 69, respectively. This corresponds to option B) 49, 69.

11. Correct option: (B)

Given $105 \boxdot 250$

Using this; $a \boxdot b = -a + b + -5$

$$105 \boxdot 250 = -(105) + (250) + -(5)$$
$$= 250 - 110 = 140$$

Then the correct option is B)

12. Correct option: (D)

Given ; $230 \triangle 525$

Using this ; $a \triangle b = a - b + -15$

$$1230 \triangle 525 = 1230 - (525) + -(15)$$
$$= 1230 - 540 = 690$$

Then the correct option is D)

13. Correct option: (D)

Given ; $1510 \boxdot 780$

Using this; $a \boxdot b = -a + b + -5$

$$1510 \boxdot 780 = -(1510) + (780) + -(5)$$
$$= -1510 + 775 = -735$$

Then the correct option is D)

14. The correct matching is a→R; b→S; c→P; d→T

a) $|-7|$

The absolute value of -7 is 7.

So, | -7 | = 7. a)→(R)

b) $\mathbf{P \times 1 = 1 \times P = P}$

In mathematics, the number 1 is referred to as the multiplicative identity because it has the property that when we multiply any number by 1, the result is that number itself.

b)→(S)

c) $\mathbf{x \times (y + z) = (x \times y) + (x \times z)}$

This is an example of the distributive property in mathematics. The distributive property shows how multiplication distributes over addition (or subtraction) and is a fundamental property in algebra.

c)→(P)

d) $\mathbf{x \div 0}$

Division by zero is undefined in mathematics. It's a fundamental rule that we cannot divide any number (including x) by zero.

d)→(T)

15. Correct Answer is: 9

Given; $(-1) + (2) + (-3) + (4) + \cdots . + (-17) + 18$

We can first group the terms in pairs: $(-1 + 2) + (-3 + 4) + \ldots + (-17 + 18)$

This simplifies to: $1 + 1 + 1 + 1 + \ldots + 1 + 1$

We can see that there are 9 pairs of terms (from -1 to 18), and each pair sums to 1.

So, the sum of the series is: 9

16. Correct Answer is: 7

The additive inverse of a number is the value that, when added to the original number, results in 0.

For -7, the additive inverse is 7 because: $-7 + 7 = 0$

So, the additive inverse of -7 is 7.

17. Correct Answer is: 8
The difference between 27 and 19 is: $27 - 19 = 8$
18. Correct Answer is: 0
To find the sum of the values: $3 + (-3) + (-3) + 3 + (-3) + 3$
We can simplify it by adding the positive and negative numbers:
$(3 + 3 + 3) + (-3 - 3 - 3) = 9 + (-9)$
Now, add 9 and -9 together: $9 + (-9) = 0$
So, the sum of the given values is 0.
19. Correct Answer is: 3
Let's simplify the expression step by step: $(78 - 77) + (18 - 17) + (28 - 27)$
Now, calculate the differences within each set of parentheses: $(1) + (1) + (1)$
Now, add these values together: $1 + 1 + 1 = 3$
So, the sum of the expression $(78 - 77) + (18 - 17) + (28 - 27)$ is 3.
20. Correct Answer is: 1
Let's simplify the expression step by step: $81 - 91 - 101 + 100 + 5 + 7$
Now, add and subtract the numbers from left to right: $(81 - 91) - 101 + 100 + 5 + 7$
Now, perform the subtractions: $(-10) - 101 + 100 + 5 + 7$
Now, add the numbers: $(-10) - 101 + 100 + 5 + 7 = -10 - 101 + 100 + 5 + 7 = -10 - 101 + 112 = -111 + 112 = 1$
So, the sum of the expression $81 - 91 - 101 + 100 + 5 + 7$ is 1.

STUDENT PRACTICE QUESTIONS

1. Correct option: (A)
To subtract the sum of -23 and 33 from -38, we can perform the following calculation:

$$\text{sum of two numbers} = S = (-23 + 33) = 10$$

Now, subtract 10 from -38: $-38 - 10 = -48$
So, the result of subtracting the sum of -23 and 33 from -38 is equal to: A) -48
2. Correct option: (B)
When we subtract -15 from 0, we can think of it as adding 15 to 0. So:
0 - (-15) = 0 + 15 = 15
So, on subtracting -15 from 0, we get: B) 15
3. Correct option: (A)
The integer 8 more than -12 can be found by adding 8 to -12: $-12 + 8 = -4$
So, the integer 8 more than -12 is: A) -4
4. Correct option: (C)
To subtract a negative number, we can treat it as adding the positive of that number. So:
$450 - (-273) = 450 + 273 = 723$
So, $450 - (-273)$ equals 723.
5. Correct option: (A)
The product of -43×17 is: $-43 \times 17 = -731$
So, the correct answer is A) -731.
6. Correct option: (C)
The vertical distance from the mountain peak to the mine at the foot of the mountain is the sum of the height above sea level and the depth below sea level:

$$6300\text{ m} + 39060\text{ m} = 45360\text{ m}$$

So, the correct answer is C) 45360 m.

7. Correct option: (C)

The number of integers between 20 and 21 is 0 because there are no whole numbers between these two values. So, the correct answer is C) 0.

8. Correct option: (D)

Given;$\{200 \div 100 - 53 \times (-1) - (56 + 93 - 28) + 55\}$

Let's simplify the expression using the BODMAS (Brackets, Orders, Division and Multiplication, Addition and Subtraction) rule:

$$\{200 \div 100 - 53 \times (-1) - (56 + 93 - 28) + 55\}$$

First, perform operations inside the brackets: $\{200 \div 100 - 53 \times (-1) - (121) + 55\}$

Next, deal with division and multiplication: $\{2 - 53 \times (-1) - 121 + 55\}$

Now, perform the multiplication: $\{2 + 53 - 121 + 55\}$

Next, perform addition and subtraction from left to right:

$$2 + 53 = 55$$
$$55 - 121 = -66$$
$$-66 + 55 = -11$$

So, the simplified expression is -11, which corresponds to option D) -11.

9. Correct option: (D)

Given number line representation



For going forward we use +, for going backward we use −.

Firstly we are on place 0 going forward 5 steps, moving backward with 3 steps and then 2 steps then take forward 4 steps

We can write all above in equation format as $0 + 5 + -3 + -2 + 4$

Then the correct option is D)

10. Correct option: (C)

Let's simplify the expression step by step: $30 - (-14 + 22)$

$$30 - (-14 + 22) = 30 - 8$$
$$30 - 8 = 22$$

So, the expression is equivalent to 22.

Let's see which option matches this result:

A) $30 - 14 - 2 \times 3 = 30 - 14 - 6 = 16 - 6 = 10$ (Not the same)

B) $30 + 7 \times 2 - 6 + = 30 + 14 - 6 = 44 - 6 = 38$ (Not the same)

C) $30 - (14 - 6) = 30 - 8 = 22$ (This is the same)

D) $30 - 14 - 6 = 30 - 20 = 10$ (Not the same)

So, the expression 30 - (-14 + 22) is the same as option C) 30 - (14 - 6).

11. Correct option: (C)

To find the other integer when the product of two integers is 15612, and one of them is -12, we can set up an equation and solve for the unknown integer (let's call it x):

$$-12 \times x = 15612$$

Now, solve for x:

$$x = \frac{15612}{(-12)}$$
$$x = -1301$$

So, the other integer is -1301, which corresponds to option C) -1301.

12. Correct option: (B)

If the sum of two integers is 238, and one of the integers is -122, we can find the other integer by subtracting -122 from 238: Other integer $= 238 - (-122) = 238 + 122 = 360$

So, the other integer is 360, which corresponds to option B) 360.

13. Correct option: (A, B)

The expression $15 - (-5) - 2$ can be written in the following equivalent ways:

A) $18 = 15 - (-3)$ - This is the same as $15 - -3$.

B) $15 - (-3)$ - This is also the same as 15 - -3.

C) $10 - 2$ - This simplifies to 8.

D) $15 - 7$ - This is also 8.

So, the correct answers are A) 18 and B) 15 - -3.

14. Correct option: (B)

If the sum of two integers is -23, and one of them is 40, we can find the other integer by subtracting 40 from -23: $\text{Other integer} = -23 - 40 = -63$

So, the other integer is -63, which corresponds to option B) -63.

15. Correct option: (B, C, D)

Let's evaluate each statement:

A) $9 - -1 > 10$ - This simplifies to $9 + 1 > 10$, which is true (10 > 10 is not true).

B) $17 + -2 < 17$ - This simplifies to $17 - 2 < 17$, which is true (15 < 17 is true).

C) $19 < 20 - - 2$ - This simplifies to $19 < 20 + 2$, which is true (19 < 22 is true).

D) $15 + 5 < 13 - - 10$ - This simplifies to 20 < 13 + 10, which is true (20 < 23 is true).

So, the true statements are B), C), and D).

16. Correct option: (C)

Given ; $(-2)^3 + (-4)^2 + (-5)^3 =$

Using this; $-8 + 16 - 125 = -117$

Then the correct option is C)

17. Correct option: (A)

Given ; $P = (-1)^1 + (-1)^2 + \cdots . + (-1)^{2016}$

$Q = (-1)^1 - (-1)^2 + (-1)^3 - (-1)^4 + \cdots . - (-1)^{2016}$

We have to find the value of $P - Q$

$P = (-1)^1 + (-1)^2 + \cdots . + (-1)^{2016} = -1 + 1 - 1 + 1 \ldots - 1 + 1 = 0$

$Q = (-1)^1 - (-1)^2 + (-1)^3 - (-1)^4 + \cdots . - (-1)^{2016} = -1 + (-1) + (-1) + (-1) \ldots + (-1) = -2016$

Then , $P - Q = 0 - (-2016) = 2016$

Therefore, the correct option is A)

18. The Correct Matching is a→Q; b→T; c→R; d→P.

a) $(-\mathbf{a}) \times \mathbf{b} = \mathbf{b} \times (-\mathbf{a})$

The equation (−a) × b = b × (−a) demonstrates the commutative property of multiplication. The commutative property states that the order of multiplication does not affect the result. In this case, it shows that we can multiply -a by b in the same way we can multiply b by -a, and the result will be the same.

a)→(Q)

b) $(\mathbf{9} + \mathbf{5}) + (-\mathbf{2}) = \mathbf{9} + [\mathbf{5} + (-\mathbf{2})]$

The equation (9 + 5) + (−2) = 9 + [5 + (−2)] demonstrates the associative property of addition. Specifically, it shows that we can change the grouping of numbers within addition without changing the result.

b)→(T)

c) $\mathbf{0} + \mathbf{5} = \mathbf{5} + \mathbf{0} = \mathbf{5}$

In mathematics, 0 is referred to as the additive identity because it has the property that when we add it to any number, it doesn't change the value of that number.

c)→(R)

d) $(-\mathbf{709}) \div (\mathbf{709})$

The division of -709 by +709 can be calculated as follows:$(-709) \div (+709) = -1$

So, (−709) ÷ (+709) equals -1. d)→(P)

19. Correct Answer is: 0

Using the defined operation △: $a \triangle b = -a + b + (-2)$

Now, let's calculate $1 \triangle 3$: $1 \triangle 3 = -1 + 3 + (-2)$

$$1 \triangle 3 = 2 + (-2)$$

$$1 \triangle 3 = 0$$

So, $1 \triangle 3$ equals 0.

20. Correct Answer is: 1

Let's simplify the expression step by step: $1 - 10 + 9 - 7 + 6 + 2$

Now, add and subtract the numbers from left to right: $(1 - 10) + (9 - 7) + 6 + 2$

Now, perform the subtractions: $(-9) + (2) + 6 + 2$

Now, add the numbers: $-9 + 2 + 6 + 2 = -9 + 10 = 1$

So, the sum of the expression $1 - 10 + 9 - 7 + 6 + 2$ is 1.

Fractions and Decimals
Exercise – I

1. Correct option: (B)

To find the number of natural numbers between 3 and 18, we can count them:

Natural numbers between 3 and 18 are: $4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17$

There are 14 natural numbers between 3 and 18.

So, the correct answer is B) 14.

2. Correct option: (D)

To convert the fraction $\frac{226}{15}$ into a mixed fraction, we need to divide the numerator (226) by the denominator (15); $226 \div 15 = 15$ with a remainder of 1.

So, the mixed fraction is: $15\frac{1}{15}$

Therefore, $\frac{226}{15}$ as a mixed fraction is $15\frac{1}{15}$.

3. Correct option: (D)

To add the fractions $\frac{1}{3}$ and $\frac{1}{3}$ to the whole number 2, we can proceed as follows:

$$2 + \frac{1}{3} + \frac{1}{3} = 2 + \frac{2}{3} = \frac{8}{3}$$

So, $2 + \frac{1}{3} + \frac{1}{3}$ equals $\frac{8}{3}$ as a mixed number.

4. Correct option: (A)

Two fractions, $\frac{a}{b}$ and $\frac{c}{d}$, are equivalent if and only if their cross-products are equal. So, the correct answer is: A) $ad = bc$

5. Correct option: (A)

Rajesh's jump = Ramesh's jump - $\frac{2}{3}$ meters

Rajesh's jump = 8 meters - $\frac{2}{3}$ meters

Rajesh's jump = $\left(\frac{24}{3}\text{meters}\right) - \left(\frac{2}{3}\text{meters}\right)$

Rajesh's jump = $\frac{24-2}{3}$ meters

Rajesh's jump = $\frac{22}{3}$ meters

So, Rajesh's jump was $\frac{22}{3}$ meters.

6. Correct option: (D)

Let's call the number "x." According to the information given, the reciprocal of the number is $1\frac{4}{7}$, which can be written as the mixed fraction $\frac{11}{7}$.

So, we have the equation: $\frac{1}{x} = \frac{11}{7}$; $x = \frac{7}{11}$

So, the number is $\frac{7}{11}$.

7. Correct option: (D)

To find which of the given options is equal to $\frac{1}{3}$, we can evaluate each expression:

A) $\frac{5}{6} \div \frac{2}{3}$: $\left(\frac{5}{6}\right) \div \left(\frac{2}{3}\right) = \left(\frac{5}{6}\right) \times \left(\frac{3}{2}\right) = \frac{15}{12} = \frac{5}{4}$

B) $\frac{7}{12} - \frac{1}{6}$: $\left(\frac{7}{12}\right) - \left(\frac{1}{6}\right) = \left(\frac{7}{12}\right) - \left(\frac{2}{12}\right) = \frac{5}{12}$

C) $\frac{1}{2} \times \frac{1}{6}$: $\left(\frac{1}{2}\right) \times \left(\frac{1}{6}\right) = \frac{1}{12}$

D) $\frac{1}{6} + \frac{1}{6}$: $\left(\frac{1}{6}\right) + \left(\frac{1}{6}\right) = \frac{2}{6} = \frac{1}{3}$

So, option D) $\frac{1}{6} + \frac{1}{6}$ is equal to $\frac{1}{3}$.

8. Correct option: (C)

Total book content = 1 (as the whole book represents the entire content)

Teacher's teaching: $\frac{7}{12}$; Srinu's revision: $\frac{3}{12}$

Now, to find out how much more Srinu needs to revise, subtract what he has already covered from the total:

$$\text{Remaining revision} = \text{Total book content} - (\text{Teacher's teaching} + \text{Srinu's revision})$$

$$\text{Remaining revision} = 1 - \left(\frac{7}{12} + \frac{3}{12}\right)$$

$$\text{Remaining revision} = 1 - \left(\frac{10}{12}\right)$$

$$\text{Remaining revision} = \left(\frac{12}{12}\right) - \left(\frac{10}{12}\right)$$

$$\text{Remaining revision} = \frac{2}{12}$$

Then the correct option is C)

9. Correct option: (B)

7 hundredths can be written as a decimal fraction. To express it as a decimal, we can simply write it as 0.07. So, 7 hundredths is equal to 0.07.

Then the correct option is B) $\frac{7}{100}$

10. Correct option: (D)

To calculate the sum of the numbers: 7.73 + 0.07 + 7 + 7.86 + 7.153

Simply add them together: 7.73 + 0.07 + 7 + 7.86 + 7.153 = 29.813

So, the correct answer is D) 29.813.

11. Correct option: (B)

To find out how much cloth Raju has left after using 3.85 meters for stitching his dress,we can subtract 3.85 meters from the original length of the cloth (25.3 meters):

$$25.3 \text{ meters} - 3.85 \text{ meters} = 21.45 \text{ meters}$$

So, Raju has 21.45 meters of cloth left.

The correct answer is B) 21.45 m.

12. Correct option: (D)

To express 15 liters and 15 milliliters as a single volume in liters, we need to convert the milliliters to liters. There are 1000 milliliters in 1 liter.

So, 15 liters and 15 milliliters is equal to:

$$15 \text{ liters} + \left(\frac{15}{1000}\right) \text{ liters} = 15 \text{ liters} + 0.015 \text{ liters} = 15.015 \text{ liters}$$

Therefore, 15 liters and 15 milliliters is equal to D) 15.015 liters.

13. Correct option: (A)

$$2 + \left(\frac{3}{100}\right) + \left(\frac{5}{1000}\right)$$

To make the fractions have a common denominator of 1000: $\frac{2000}{1000} + \left(\frac{30}{1000}\right) + \left(\frac{5}{1000}\right)$

Now, add the fractions: 2 + 0.035 = 2.035

So, $2 + \frac{3}{100} + \frac{5}{1000}$ equals 2.035.

14. Correct option: (B)

To find the place value of 3 in the sum of 5.25 and 3.38, we can first add the two numbers and then determine the place value.

$$\begin{array}{r} 5.25 \\ +\ 3.38 \\ \hline 8.63 \end{array}$$

In the sum 8.63, the digit 3 is in the hundredths place. So, the place value of 3 in the sum is in the hundredths place as $\frac{3}{100}$.

Then the correct option is B)

15. Correct option: (A)

Given expression, $\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\ldots\ldots\left(1-\frac{1}{n}\right)$

Consider first three terms of expression, $\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right) = \left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(\frac{3}{4}\right) = \frac{1}{4}$

We can say that for every next term numerator will cancel out previous denominator which led to the expression, $= \frac{1}{n}$

Then the correct option is A)

16. Correct option: (A)

Given expression, $93.528 = 90 + 3 + \frac{5}{10} + \frac{1}{50} + \square$

First, simplify the right side of the equation:

$$90 + 3 + \frac{5}{10} + \frac{1}{50} = 90 + 3 + 0.5 + 0.02 = 93.52$$

Now, our equation becomes: $93.528 = 93.52 + \square$

$$0.008 = \square = \frac{8}{1000} = \frac{1}{125}$$

Then the correct option is A)

17. Correct option: (C)

In the number 843.543, let's compare the value of 4 in the tens place to the value of 4 in the hundredths place.

The value of 4 in the tens place is $4 \times 10 = 40$.

The value of 4 in the hundredths place is $4 \times \frac{1}{100} = \frac{4}{100} = 0.04$.

Now, let's calculate how many times the value of 4 in the tens place is greater than the value of 4 in the hundredths place: $\frac{40}{0.04} = 1000$

The correct answer is C) 1000.

18. Correct option: (D)

Each bag holds 12.4 kg of biscuits, which is equivalent to 12,400 g.

$$2\frac{1}{2} = \frac{5}{2} = 2.5$$

$$2.5 \text{ bags} \times 12{,}400 \text{ g/bag} = 31{,}000 \text{ g}$$

Now, divide the total weight by the weight of each packet:

$$31{,}000 \text{ g} / 62 \text{ g/packet} = 500 \text{ packets}$$

So, we can fill 500 packets from $2\frac{1}{2}$ bags. The correct answer is D) 500.

19. Correct option: (C)

Given figure.



We have to find the value of $P + Q - R$

From the figure we can conclude that the length of line segment between two points is 0.13.

Then the value of P Q and R are

$$P = 6.5 + 0.13, Q = P + 0.13, R = Q + 0.13$$

$$P + Q - R = 6.5 + 0.13 + Q - Q - 0.13 = 6.5$$

Therefore, the correct option is C)

20. Correct option: (D)

To find the simplest form of 55 minutes as a fraction of $2\frac{1}{4}$ hours, we need to convert both the minutes and hours into a common unit, such as minutes.

$$2\frac{1}{4}\text{hours} = 120 + \left(\frac{1}{4}\right) \times 60 = 120 + 15 = 135\text{ minutes.}$$

Now, we want to express 55 minutes as a fraction of 135 minutes: $\frac{55}{135}$

To simplify this fraction, we can find the greatest common divisor (GCD) of 55 and 135, which is 5.

$$\frac{55 \div 5}{135 \div 5} = \frac{11}{27}$$

Then the correct option is D)

Exercise-II

1. Correct option: (D)

To add the fractions $\frac{1}{6}, \frac{7}{12}$, and $\frac{2}{3}$, we first need to find a common denominator. In this case, the least common denominator is 12.

Now, rewrite each fraction with the common denominator:

$\frac{1}{6}$ can be written as $\left(\frac{2}{12}\right)$.

$\frac{7}{12}$ remains unchanged.

$\frac{2}{3}$ can be written as $\left(\frac{8}{12}\right)$.

Now, add the fractions: $\left(\frac{2}{12}\right) + \left(\frac{7}{12}\right) + \left(\frac{8}{12}\right) = \frac{2+7+8}{12} = \frac{17}{12}$

Therefore, sum of these given number is $\frac{17}{12}$ which is in mixed form as $1\frac{5}{12}$ matches with the option D.

2. Correct option: (B)

1. Convert the mixed numbers to improper fractions: $76\frac{1}{2} = \frac{76 \times 2 + 1}{2} = \frac{153}{2}$

$$11\frac{5}{6} = \frac{6 \times 11 + 5}{6} = \frac{71}{6}$$

2. Add the fractions: $\frac{153}{2} + \frac{71}{6} = \frac{153 \times 3}{6} + \frac{71}{6}$

3. Add the fractions with the same denominator: $\frac{459}{6} + \frac{71}{6} = \frac{459+71}{6} = \frac{530}{6} = \frac{265}{3}$

Therefore, sum of these given number is $\frac{265}{3} = 88\frac{1}{3}$

Then the correct option is B)

3. Correct option: (C)

1. Convert the mixed numbers to improper fractions: $4\frac{7}{10} = \frac{40+7}{10} = \frac{47}{10}$

$$3\frac{1}{10} = \frac{30+1}{10} = \frac{31}{10}$$

2. Subtract the fractions: $\frac{47}{10} - \frac{31}{10} = \frac{16}{10}$

Therefore, subtraction of these given number is $\frac{16}{10} = \frac{8}{5} = 1\frac{3}{5}$

Then the correct option is C)

4. Correct option: (B)

Subtract the given fraction as $\frac{5}{12} - \frac{3}{8}$

Make same denominator for both of the fraction as $\frac{5\times2}{12\times2} - \frac{3\times3}{8\times3} = \frac{10}{24} - \frac{9}{24} = \frac{1}{24}$

Therefore, subtraction of these given number is $\frac{1}{24}$

Then the correct option is B)

5. Correct option: (A)

Given multiplication of fraction; $\frac{1}{3} \times \frac{3}{4}$

To multiply the fractions, multiply the numerators together and the denominators together: $\left(\frac{1}{3}\right) \times \left(\frac{3}{4}\right) = \frac{1\times3}{3\times4} = \frac{3}{12}$

Now, simplify the fraction by dividing both the numerator and denominator by their greatest common divisor, which is 3: $\left(\frac{3\div3}{12\div3}\right) = 1/4$

Then the correct option is A)

6. Correct option: (B)

Step 1: Convert the mixed numbers to improper fractions. $3\frac{3}{4} = \frac{(4\times3)+3}{4} = \frac{12+3}{4} = \frac{15}{4}$

Step 2: Multiply the fractions. $\left(\frac{15}{4}\right) \times \left(\frac{4}{5}\right) = \frac{15\times4}{4\times5} = \frac{15}{5} = 3$

Then the correct option is B)

7. Correct option: (C)

Step 1: Convert the mixed numbers to improper fractions. $3\frac{5}{6} = \frac{23}{6}, 4\frac{2}{5} = \frac{22}{5}$

Step 2: Multiply the fractions using commutative property $\frac{23}{6} \times \frac{22}{5} = \frac{23\times11}{3\times5} = \frac{253}{15} = 16\frac{13}{15}$

Then the correct option is C)

8. Correct option: (A)

1. Create a fraction from the whole number:

Convert 3 into a fraction by placing it over 1, so it becomes $\frac{3}{1}$.

2. Invert the divisor: To divide fractions, we multiply the first fraction (dividend) by the reciprocal of the second fraction (divisor). The reciprocal of $\frac{3}{1}$ is $\frac{1}{3}$.

3. Multiply the fractions: Multiply the two fractions together.

$$\left(\frac{5}{8}\right) \div \left(\frac{3}{1}\right) = \left(\frac{5}{8}\right) \times \left(\frac{1}{3}\right) = \frac{5}{24}$$

Then the correct option is A)

9. Correct option: (A)

Given, $7 \div \frac{3}{8}$

Invert the second fraction: $\frac{3}{8} => \frac{8}{3}$

Now we put in the expression; $7 \times \frac{8}{3} = \frac{56}{3} = 18\frac{2}{3}$

Then the correct option is A)

10. Correct option: (A)

To compare the values of a and b we can simply compare the fractions with the same denominator: a $= \frac{2}{3} = \frac{6}{9}$; b $= \frac{4}{9}$

Now, let's compare them: $\frac{6}{9} > \frac{4}{9}$ (same denominator)

So, the correct answer is A) a > b.

11. Correct option: (C)

The fractions $\frac{1}{2}, \frac{2}{4}, \text{and} \frac{3}{6}$ are:

C) Equivalent fractions

These fractions represent the same portion of a whole, and they are all equal to each other when simplified. So, they are equivalent fractions.

12. Correct option: (C)

Given fraction: $=> \frac{24}{30}$

Simplify by dividing both numerator and denominator by their greatest common divisor, which is 6:

$$24 \div 6 = 4$$
$$30 \div 6 = 5$$

So, the simplest form of $\frac{24}{30}$ is $\frac{4}{5}$.

Then the correct option is C)

13. Correct option: (D)

Given fraction: $=> \frac{13}{25}$

Step 1: Multiply both the numerator and denominator by 100:

$$\frac{13 \times 100}{25 \times 100} = \frac{1300}{25 \times 100} = \frac{1300}{25} \times \frac{1}{100}$$

Step 2: Simplify the fraction: $\frac{1300}{25} = 52$

Step 3: Multiply the result by $\frac{1}{100}$:

$$52 \times \frac{1}{100} = 0.52$$
$$=> \frac{13}{25} = 0.52$$

Then the correct option is D)

14. Correct option: (A)

Given fractions: $=> \frac{3}{5}, \frac{3}{4}, \frac{3}{2}, \frac{5}{6}$

Let's convert all fractions to have a common denominator of 60:

$$\frac{3}{5} = \frac{36}{60}$$
$$\frac{3}{4} = \frac{45}{60}$$
$$\frac{3}{2} = \frac{90}{60}$$
$$\frac{5}{6} = \frac{50}{60}$$

Now it's clear that A) $\frac{3}{5}$ has the smallest value.

Then the correct option is A)

15. Correct option: (B)

Given: $=> 3\frac{7}{8}$

Convert to improper fraction: $3\frac{7}{8} = \frac{31}{8}$

Reciprocal: $=> \frac{8}{31}$

Then the correct option is B)

16. Correct option: (B)

An improper fraction is a fraction where the numerator is greater than or equal to the denominator. Let's check each of the given fractions:

$$A) \frac{22}{60} => \text{numerator} < \text{denominator}$$
$$B) \frac{66}{22} => \text{numerator} > \text{denominator}$$
$$C) \frac{90}{100} => \text{numerator} < \text{denominator}$$
$$D) \frac{1000}{2600} => \text{numerator} < \text{denominator}$$

Then the correct option is B)

17. Correct option: (A)

Given fractions: $=> \frac{3}{-5}, \frac{-7}{10}, \frac{8}{-10}, \frac{-17}{15}$

Step 1: Find the least common multiple (LCM) of the denominators, which is 30.

Step 2: Convert the fractions to have a denominator of 30:

$$\frac{3}{-5} = -\frac{18}{30}$$
$$-\frac{7}{10} = -\frac{21}{30}$$
$$\frac{8}{-10} = -\frac{24}{30}$$
$$-\frac{17}{15} = -\frac{34}{30}$$

Step 3: Arrange the fractions in ascending order:

$$=> -\frac{34}{30}, -\frac{24}{30}, -\frac{21}{30}, -\frac{18}{30} => \frac{-17}{15}, \frac{8}{-10}, \frac{-7}{10}, \frac{3}{-5}$$

Then the correct option is A)

18. Correct option: (B)

The formula to convert Celsius to Fahrenheit is: $F = \left(\frac{9}{5}\right)C + 32$

Given temperature: $35°C$

Substitute $C = 35$ into the formula: $F = \left(\frac{9}{5}\right) \times (35) + 32$

$$F = 63 + 32$$
$$F = 95°F$$

So, the equivalent of 35°C is 95°F.

Then the correct option is B)

19. Correct option: (C)

At the party, there are 3 large pizzas. Each pizza has been cut into 9 equal pieces. Here's how the portions have been eaten:

1. First pizza: Cut into 9 pieces, and 8 pieces have been eaten, leaving 1 piece.
2. Second pizza: Cut into 9 pieces, and $\frac{2}{3}$ of it (6 pieces) have been eaten, leaving 3 pieces.
3. Third pizza: Cut into 9 pieces, and 7 pieces have been eaten, leaving 2 pieces.

Total number of pieces: 9 (first pizza) + 9 (second pizza) + 9 (third pizza) = 27 pieces.

Portion of the pizzas eaten: 8 pieces + 6 pieces + 7 pieces = 21 pieces.

Portion of the pizzas left: 27 pieces - 21 pieces = 6 pieces.

The fraction of the pizzas left is $\frac{6}{27}$, which simplifies to $\frac{2}{9}$.
So, the fraction of the 3 pizzas that is left is indeed $\frac{2}{9}$.
Then the correct option is C)

20. Correct option: (C)

To find the fraction of the day that Bob spent on the phone, divide the time he spent on the phone (2 hours) by the total length of the workday (8 hours).

$$\text{Fraction} = \frac{\text{Time on phone}}{\text{Total workday time}}$$

$$\text{Fraction} = \frac{2 \text{ hours}}{8 \text{ hours}}$$

Simplify the fraction if possible: Fraction $= \frac{1}{4}$

So, Bob spent $\frac{1}{4}$ of the day on the phone.

Then the correct option is C)

Exercise - III

1. Correct option: (B)

Given: $=> \frac{35}{4}$

We can break down 35 into 32 and 3, so:

$$\frac{35}{4} = \frac{32 + 3}{4} = \frac{32}{4} + \frac{3}{4} = 8 + 0.75 = 8.75$$

So, the decimal equivalent of $\frac{35}{4}$ is indeed 8.75 .

Then the correct option is B)

2. Correct option: (D)

Given: $=> \frac{825}{125}$

We can break down 825 into 750 and 75, so:

$$\frac{825}{125} = \frac{750 + 75}{125} = \frac{750}{125} + \frac{75}{125} = 6 + 0.6 = 6.6$$

So, the decimal equivalent of $\frac{825}{125}$ is indeed 6.6.

Then the correct option is D)

3. Correct option: (D)

Given decimals: $0.088 , 0.091$

Add the decimals:

$$\begin{array}{r} 0.088 \\ +\ 0.091 \\ \hline 0.179 \end{array}$$

Then the correct option is D)

4. Correct option: (C)

To multiply decimals, we can follow these steps:

Given decimals: $0.63, 0.42$

Step 1: Multiply the decimals as if they were whole numbers:

$$0.63 \times 0.42 => 63 \times 42 = 2646$$

Step 2: Count the total number of decimal places in the original numbers (2 decimal places in each number).

Step 3: Place the decimal point in the product. The final answer is the number of decimal places in total, which are 4 in this case.

$$=> 0.63 \times 0.42 = 0.2646$$

Then the correct option is C)

5. Correct option: (A)

Given: $2.9 \div 0.8758$

Step 1: Count the total number of decimal places in the numerator and denominator:

Numerator (2.9) has 1 decimal place.

Denominator (0.8758) has 4 decimal places.

Step 2: Compare the number of decimal places. Since the numerator has fewer decimal places than the denominator ($1 < 4$), multiply the numerator by $10^{4-1} = 10^3 = 1000$.

Step 3: Perform the division with the adjusted numerator:

$$\frac{29 \times 1000}{8758} = \frac{29000}{8758}$$

Step 4: Perform the division as if the decimals were whole numbers:

$$\frac{29000}{8758} \approx 3.31$$

Then the correct option is A)

6. Correct option: (D)

Rounding 56.73647 to the nearest hundredth means we need to consider two decimal places.

Looking at the third decimal place (6), which is greater than 5, we round up the second decimal place (3) by 1.

So, rounding 56.73647 to the nearest hundredth gives us:

56.73647 ≈ 56.74

Then the correct option is D)

7. Correct option: (C)

To add decimal numbers, line up the decimal points and then add the digits as if they were whole numbers. Here's the calculation:

```
  78.09
+ 19.367
--------
  97.457
```

Then the correct option is C)

8. Correct option: (A)

When rounding a decimal number to the nearest tenth, we're looking at the digit immediately to the right of the tenths place. In this case, the number is 0.716.

The digit immediately to the right of the tenths place is 1. Since 1 is less than 5, we round up the tenths place. So, the tenths place becomes 7.

The remaining decimal places after the tenths place (the hundredths place) are ignored when rounding to the nearest tenth.

Therefore, 0.716 rounded to the nearest tenth is 0.7

Then the correct option is A)

9. Correct option: (A)

To determine which decimal has the least value among the options, we need to compare their values. Smaller decimals have a lower value.

Let's compare the decimals:

A) 0.0012

B) 0.0102

C) 0.012

D) 0.12

All of these decimals start with a 0 before the decimal point, so we'll move on to the digits after the decimal point.

Comparing the digits after the decimal point:
A) 0012
B) 0102
C) 012
D) 12
Among these, option A (0.0012) has the least value because it has the smallest sequence of digits after the decimal point.
Then the correct option is A)

10. Correct option: (B)

Let's calculate the sum of the numbers first and then round the result to the nearest tenth.
Sum of the numbers:
11.006 + 34 + 0.72 = 45.726
Rounded to the nearest tenth, the sum 45.726 becomes 45.7.
Then the correct option is B)

11. Correct option: (C)

Let's solve the expression step by step:
Expression: [26.2 + 38.05 - 6.05 × 5]
First, perform the multiplication: $6.05 \times 5 = 30.25$
Now substitute the result back into the expression: $26.2 + 38.05 - 30.25$
Now add and subtract: $64.25 - 30.25 = 34$
So, the value of the expression is 34.
Then the correct option is C)

12. Correct option: (D)

To find the total liters of juice in the two bottles, simply add the amount of apple juice and grape juice.
Apple juice: 1.42 liters
Grape juice: 1.89 liters
Total liters = Apple juice + Grape juice
Total liters = 1.42 + 1.89 = 3.31 liters (solve using above decimal addition method)
So, the total liters of juice in the two bottles are 3.31 liters.
Then the correct option is D)

13. Correct option: (B)

If each of the six friends paid $37.27 and the cost was evenly split among them, we can find the total cost of gas by multiplying the amount paid by one friend by the number of friends.
Total cost of gas = $\text{Cost per friend} \times \text{Number of friends}$
Total cost of gas = $\$37.27 \times 6 = \$223.62$ (solve using above decimal multiplication method)
So, the total cost of gas was $223.62.
Then the correct option is B)

14. Correct option: (B)

To find the area of a rectangle, we can use the formula: Area = Length × Breadth
Given that the length is 5.7 cm and the breadth is 3.5 cm:
Area = 5.7 cm × 3.5 cm = 19.95 cm^2 (solve using above decimal multiplication method)
Then the correct option is B)

15. Correct option: (A)

To find out how many minutes it will take Monika to walk one mile, we can set up a proportion using the given information:
Time is proportional to distance, so: $\frac{\text{Time for 1 mile}}{\text{Distance of 1 mile}} = \frac{\text{Time for 1.25 miles}}{\text{Distance of 1.25 miles}}$

Let's represent the unknown time for 1 mile as x: $\frac{x}{1} = \frac{22.4 \text{ minutes}}{1.25 \text{ miles}}$

Now, solve for x: (solve using above decimal division method) $x = \frac{22.4}{1.25} = 17.92$ minutes

So, it will take Monika 17.92 minutes to walk one mile.

Then the correct option is A)

16. Correct option: (D)

To calculate the area of the opening, which is a rectangle, we can use the formula:

Area = Length × Width

Given that the length is 11.25 inches and the width is 8.75 inches:

Area = 11.25 inches × 8.75 inches = 98.4375 square inches (solving using decimal multiplication method)

Rounded to the nearest hundredth, the area is approximately 98.44 square inches.

So, the correct answer is D) 98.44 square inches.

17. Correct option: (A)

To find out how much less the football player is earning than the baseball player, we can subtract the football player's earnings from the baseball player's earnings:

Difference = Baseball Player's Earnings - Football Player's Earnings

Difference = \$2.4 million - \$1.025 million (solve using decimal subtraction method)

Difference = \$1.375 million

So, the football player is earning \$1.375 million less per year than the baseball player.

Then the correct option is A)

18. Correct option: (B)

To find the average number of miles Alicia rode each day, we can divide the total distance by the number of days:

Average miles per day = $\frac{\text{Total miles}}{\text{Number of days}}$

Average miles per day = $\frac{25.8 \text{ miles}}{3 \text{ days}}$

Average miles per day = 8.6 miles

So, Alicia rode an average of 8.6 miles each day.

Then the correct option is B)

19. Correct option: (C)

To convert liters to quarts, we can use the conversion factor given: 1 liter = 1.06 quarts.

Given that the container holds 3.25 liters, we can calculate the equivalent in quarts:

$$\text{Quarts} = \text{Liters} \times \text{Conversion factor}$$

$$\text{Quarts} = 3.25 \text{ liters} \times 1.06 \text{ quarts/liter}$$

(Solve using decimal multiplication method)

$$\text{Quarts} = 3.445 \text{ quarts}$$

So, a 3.25-liter container will hold about 3.445 quarts of water.

Then the correct option is C)

20. Correct option: (C)

Piece B is 0.076 centimeters thicker than Piece A, and Piece A is 0.724 centimeters thick. To find the thickness of Piece B, we need to add 0.076 to 0.724:

$$\text{Thickness of Piece B} = \text{Thickness of Piece A} + 0.076$$

$$\text{Thickness of Piece B} = 0.724 + 0.076$$

$$\text{Thickness of Piece B} = 0.8 \text{ centimeters}$$

So, the thickness of Piece B is 0.8 centimeters.

Then the correct option is C)

LEVEL – II
Exercise - I

1. Correct option: (B)

To express 80 rupees and 75 paisa as a decimal value, we can divide the paisa by 100 and then add it to the rupees: $80 \text{ rupees} + \left(\frac{75 \text{ paisa}}{100}\right) = 80 + 0.75 = 80.75$

So, the answer is B) 80.75 rupees.

2. Correct option: (A)

Length of each board $= 1\frac{1}{2}$ feet $= \frac{3}{2}$ feet

Total length of the wooden log = $= 22\frac{1}{2}$ feet $= \frac{45}{2}$ feet

Now, divide the total length of the log by the length of each board:

$$\left(\frac{45}{2}\right) \div \left(\frac{3}{2}\right) = \frac{45}{2} \times \frac{2}{3} = 15$$

Number of boards that can cut from the wooden log = 15.

3. Correct option: (C)

To find the greatest 3-digit number, we need to subtract it from 5102.57.

The greatest 3-digit number is 999.

So, we should subtract 999 from 5102.57: 5102.57 - 999 = 4103.57

Therefore, we should subtract 4103.57 from 5102.57 to get the greatest 3-digit number.

The correct answer is C) 4103.57.

4. Correct option: (B)

To find out how much the sum of 33.69 and 112.55 is greater than their difference, we can simply calculate both and then find the difference between the sum and the difference.

$$\text{Sum: } 33.69 + 112.55 = 146.24$$
$$\text{Difference: } 112.55 - 33.69 = 78.86$$

Now, find the difference between the sum and the difference:

$$146.24 - 78.86 = 67.38$$

So, the sum of 33.69 and 112.55 is greater than their difference by 67.38.

The correct answer is B) 67.38.

5. Correct option: (C)

Step 1: Write down the decimal number as a fraction over a power of 10.

In this case, it's over 100 because there are two decimal places: $1.75 = \frac{175}{100}$

Step 2: Simplify the fraction by dividing both the numerator and denominator by their greatest common divisor, which in this case is 25: $\left(\frac{175}{25}\right) \div \left(\frac{100}{25}\right) = \frac{7}{4}$

So, 1.75 can be expressed as the fraction $\frac{7}{4}$.

6. Correct option: (C)

An improper fraction is a fraction where the numerator (the top number) is equal to or greater than the denominator (the bottom number). Let's check each of the options:

A) $\frac{11}{18}$ – Proper fraction $(11 < 18)$

B) $\frac{7}{9}$ – Proper fraction $(7 < 9)$

C) $\frac{35}{12}$ – Improper fraction $(35 >= 12)$

D) $\frac{17}{21}$ – Proper fraction $(17 < 21)$

So, the improper fraction is C) $\frac{35}{12}$.

7. Correct option: (B)

To compare fractions to the decimal 3.55, we can convert them to decimals and then compare:

A) $\frac{49}{21} = 2.333\ldots$

B) $\frac{31}{7} = 4.428\ldots$

C) $\frac{79}{52} = 1.519\ldots$

D) $\frac{32}{11} = 2.909\ldots$

Now, let's compare them to 3.55:

A) $2.333\ldots < 3.55$

B) $4.428\ldots > 3.55$

C) $1.519\ldots < 3.55$

D) $2.909\ldots < 3.55$

So, the fractions that are greater than 3.55 are: B) $\frac{31}{7}$

8. Correct option: (D)

To solve for x in the equation: $\frac{1}{2} + \frac{1}{x} = 2$

$$\frac{1}{x} = 2 - \frac{1}{2}$$

Now, find a common denominator: $\frac{1}{x} = \left(\frac{4}{2}\right) - \left(\frac{1}{2}\right) = \frac{3}{2}$

Now, we can take the reciprocal of both sides to solve for x: $x = \frac{2}{3}$

So, the value of x is 2/3.

9. Correct option: (A, D)

Let's convert these fractions to decimal numbers and then compare them:

A) $\frac{1}{4}$ as a decimal is 0.25, and 0.2 is less than 0.25, so $\frac{1}{4} > 0.2$ is true.

B) $\frac{1}{3}$ is approximately 0.333 …, and 0.4 is greater than 0.333 …, so $\frac{1}{3} > 0.4$ is not true.

C) $\frac{2}{5}$ as a decimal is 0.4, and $\frac{1}{2}$ is equal to 0.5, so $\frac{2}{5} > \frac{1}{2}$ is not true.

D) $\frac{3}{4}$ as a decimal is 0.75, and $\frac{6}{7}$ as a decimal is approximately 0.857 … $\frac{3}{4} < \frac{6}{7}$ is true.

So, the true statements are: A) $\frac{1}{4} > 0.2$; D) $\frac{3}{4} < \frac{6}{7}$

10. Correct option: (B, D)

Step 1: Convert the mixed numbers to improper fractions:

$$3\frac{2}{3} = \frac{(3 \times 3) + 2}{3} = \frac{9 + 2}{3} = \frac{11}{3}$$

$$2\frac{1}{2} = \frac{(2 \times 2) + 1}{2} = \frac{4 + 1}{2} = \frac{5}{2}$$

Step 2: Divide the fractions: $\left(\frac{11}{3}\right) \div \left(\frac{5}{2}\right)$

$$\frac{(11 \times 2)}{(3 \times 5)} = \frac{22}{15}$$

In mixed form; $\frac{22}{15} = 1\frac{7}{15}$

11. Correct option: (A, C, D)

To find the result of multiplying $\frac{7}{2}$ by $\frac{4}{9}$, we can multiply the numerators together to get the new numerator and the denominators together to get the new denominator.

$$\left(\frac{7}{2}\right) \times \left(\frac{4}{9}\right) = \frac{7 \times 4}{2 \times 9} = \frac{28}{18} \text{ (Option C)}$$

Now, we can simplify the fraction by finding the greatest common divisor (GCD) of the numerator and denominator and dividing both by it.

The GCD of 28 and 18 is 2. $\left(\frac{28}{2}\right) \div \left(\frac{18}{2}\right) = \frac{14}{9}$ (Option A)

In mixed form $\frac{14}{9} = 1\frac{5}{9}$ (Option D)

12. Correct option: (A)

We have to find the value of 50kg and 80g in grams.

$$1\text{g} = \frac{1}{1000}\text{kg} = 0.001\text{kg} \Rightarrow 1\text{kg} = 1000\text{g}$$

First convert 5kg to g

$$5\text{kg} = 5 \times 1000\text{g} = 5000\text{g}$$

Now we can add 80g directly as ; 5kg80g = 5080g

Then the correct option is A)

13. Correct option: (C)

To convert meters (m) to kilometers (km), we need to divide the number of meters by 1000 because there are 1000 meters in 1 kilometer.

So, to convert 7777 meters to kilometers: $7777\text{ m} \div 1000 = 7.777\text{ km}$

Therefore, 7777 meters is equal to 7.777 kilometers.

The answer is C) 7.777 km.

14. Correct option: (D)

To convert kilometers (km) to meters (m), we can use the fact that 1 kilometer is equal to 1000 meters.

So, to convert 2 kilometers and 45 meters to meters:

$$2\text{ km} = 2 \times 1000\text{ m} = 2000\text{ m}$$
$$45\text{ m} = 45\text{ m}$$

Now, add them together: $2000\text{ m} + 45\text{ m} = 2045\text{ m}$

Therefore, 2 kilometers and 45 meters is equal to 2045 meters.

The answer is D) 2045 m.

15. Correct Matching is: a-Q, b-R, c-P, d-T

Let's match the given descriptions of shapes in column II to the fractions in column I:

P) 3 circles with each circle cut into 3 parts, and 2 parts are shaded. So, the fraction is $\frac{2}{3}$.

Q) 2 circles with each circle cut into 5 parts, and 1 part is shaded. So, the fraction is $\frac{1}{5}$.

R) Two squares with each square cut into 2 parts, and one part is shaded. So, the fraction is $\frac{1}{2}$.

S) 4squares with each square cut into 4parts, and one part is shaded. So, the fraction is $\frac{1}{4}$.

T) 4 circles with each circle cut into 2 parts, and one part is shaded. So, the fraction is $\frac{1}{2}$.

Now, let's match these fractions to the provided options:

a) $2 \times \frac{1}{5}$ matches with Q) 2 circles with each circle cut into 5 parts and 1 part is shaded $\left(\frac{1}{5}\right)$.

b) $2 \times \frac{1}{2}$ matches with R) Two squares with each square cut into 2 parts with one part shaded $\left(\frac{1}{2}\right)$.

c) $3 \times \frac{2}{3}$ matches with P)3 circles with each circle cut into 3parts and 2parts are shaded $\left(\frac{2}{3}\right)$.

d) $4 \times \frac{1}{2}$ matches with T) 4 circles with each circle cut into 2 parts with one part shaded $\left(\frac{1}{2}\right)$.

So, the matches are:

a) $2 \times \frac{1}{5} - Q$

b) $2 \times \frac{1}{2} - R$

c) $3 \times \frac{2}{3} - P$

d) $4 \times \frac{1}{2}$ – T

16. Correct Answer is: 6

To find the value of $5\frac{1}{2} + \frac{1}{2}$, we can add the whole numbers and the fractions separately:

5 + 1 = 6 (whole numbers) or $5\frac{1}{2} + \frac{1}{2} = \frac{11}{2} + \frac{1}{2} = \frac{12}{2} = 6$

Now, add the results together: 5+1=6

17. Correct Answer is: 5

To find the difference between 8.5 and 3.5, we simply subtract one number from the other: $8.5 - 3.5 = 5$

So, the difference between 8.5 and 3.5 is 5.

18. Correct Answer is: 5

5000 is 5 thousand.

19. Correct Answer is: 1

To simplify the sum of these fractions, we can add their numerators while keeping the common denominator of 40:

$$\left(\frac{1}{40}\right) + \left(\frac{15}{40}\right) + \left(\frac{24}{40}\right) = \frac{1 + 15 + 24}{40}$$

Now, add the numerators: $(1 + 15 + 24) = 40$

So, the simplified sum is: $= \frac{40}{40}$

Since the numerator and denominator are the same, the fraction is equal to 1. Therefore, the answer is 1.

20. Correct Answer is: 4

To find the value of x in the equation 45/60 = 3/x, we can cross-multiply and solve for x:

$$\frac{45}{60} = \frac{3}{x}$$

First, multiply both sides of the equation by x to isolate 3: $\left(\frac{45}{60}\right) \times x = 3$

$$\left(\frac{45}{60}\right) \times x = \left(\frac{3}{4}\right) \times x$$

$$\left(\frac{3}{4}\right) \times x = 3$$

Now, to solve for x, divide both sides of the equation by $\left(\frac{3}{4}\right)$: $x = \frac{3}{\frac{3}{4}}$

When you divide by a fraction, we can multiply by its reciprocal: $x = 3 \times \left(\frac{4}{3}\right)$

Now, simplify: $x = 4$; So, the value of x is 4.

Exercise-II

1. Correct option: (A)

To find out how many bags of wheat there are, we can divide the total weight by the weight of each bag:

$$\text{Number of bags} = \frac{\text{Total weight}}{\text{Weight per bag}}$$

$$\text{Number of bags} = \frac{1743 \text{ kg}}{49.8 \text{ kg/bag}}$$

$$=> \frac{17430}{498} => 35$$

$$\text{Number of bags} = 35$$

So, there are 35 bags of wheat.

Then the correct option is A)

2. Correct option: (B)

To find out how many pieces Shikha gets when she cuts 50 meters of cloth into pieces of 1.25 meters each, we can divide the total length of cloth by the length of each piece:

$$\text{Number of pieces} = \frac{\text{Total length of cloth}}{\text{Length of each piece}}$$

$$\text{Number of pieces} = \frac{50\ \text{m}}{1.25\ \text{m/piece}}$$

$$=> \frac{5000}{125} => 40$$

$$\text{Number of pieces} = 40\ \text{pieces}$$

So, Shikha gets 40 pieces when she cuts 50 meters of cloth into pieces of 1.25 meters each. Then the correct option is B)

3. Correct option: (D)

To find the fraction of the loaf that is left, we need to subtract the total number of slices eaten from the total number of slices in the loaf and then express that as a fraction of the original number of slices.

Total slices in the loaf: 35 slices

Slices eaten: 8 + 6 + 5 + 9 = 28 slices

Slices left: 35 - 28 = 7 slices

$$\text{Fraction left} = \frac{\text{Slices left}}{\text{Total slices}}$$

$$\text{Fraction left} = \frac{7}{35}$$

$$=> \frac{7}{35} = \frac{7}{7 \times 5} = \frac{1}{5}$$

$$\text{Fraction left} = \frac{1}{5}$$

So, the fraction of the loaf that is left is $\frac{1}{5}$.

Then the correct option is D)

4. Correct option: (A)

To find out how many pages Hari has read, we can multiply the fraction of the novel she has read $\left(\frac{3}{5}\right)$ by the total number of pages in the novel (360 pages):

$$\text{Pages read} = \text{Fraction read} \times \text{Total pages}$$

$$\text{Pages read} = \left(\frac{3}{5}\right) \times 360$$

$$\text{Pages read} = 216\ \text{pages}$$

So, Hari has read 216 pages of the novel.

Then the correct option is A)

5. Correct option: (B)

To find out what part of the original 2 grams of butter is left, we need to subtract the amount of butter used from the original amount.

$$\text{Amount used for chocolate chip cookies: } \frac{1}{2}\ \text{gram}$$

$$\text{Amount used for peanut butter cookies: } \frac{1}{8}\ \text{gram}$$

$$\text{Amount used for sugar cookies: } \frac{2}{3} \text{ gram}$$

$$\text{Total amount used } = \frac{1}{2} + \frac{1}{8} + \frac{2}{3} = \frac{12}{24} + \frac{3}{24} + \frac{16}{24} = \frac{31}{24} \text{ grams}$$

$$\text{Amount left } = \text{ Original amount } - \text{ Total amount used}$$

$$\text{Amount left } = 2 \text{ grams } - \frac{31}{24} \text{ grams } = \left(\frac{48}{24}\right) - \left(\frac{31}{24}\right) \text{grams } = \frac{17}{24} \text{ grams}$$

So, the part of the original 2 grams that is left is $\frac{17}{24}$ grams.

Then the correct option is B)

6. Correct option: (B)

To find out how many cups of liquid there are in the recipe, we need to add together the measurements of water, chicken stock, and honey.

$$\text{Liquid from water: } 2\frac{3}{4} \text{ cups}$$

$$\text{Liquid from chicken stock: } 4\frac{5}{8} \text{ cups}$$

$$\text{Liquid from honey: } \frac{1}{2} \text{ cup}$$

$$\text{Total liquid } = 2\frac{3}{4} + 4\frac{5}{8} + \frac{1}{2}$$

First, convert all fractions to a common denominator, which is 8:

$$2\frac{3}{4} = \frac{11}{4}$$

$$4\frac{5}{8} = \frac{37}{8}$$

Now add them together:

$$\text{Total liquid } = \frac{11}{4} + \frac{37}{8} + \frac{1}{2}$$

$$\text{Total liquid } = \frac{22}{8} + \frac{37}{8} + \frac{4}{8}$$

$$\text{Total liquid } = \frac{63}{8}$$

Now, convert $\frac{63}{8}$ back to a mixed number: $\frac{63}{8} = 7\frac{7}{8}$

So, there are $7\frac{7}{8}$ cups of liquid in the recipe.

Then the correct option is B)

7. Correct option: (B)

To find out how many pages Swetha has read, we need to calculate $\frac{1}{6}$th of the total number of pages she needs to read.

$$\text{Total pages needed to read } = 14 + 26 + 12 + 28 = 80 \text{ pages}$$

$$\text{Pages read } = \left(\frac{1}{6}\right) \times \text{ Total pages}$$

$$\text{Pages read } = \left(\frac{1}{6}\right) \times 80$$

$$\text{Pages read } = \frac{80}{6}$$

$$\text{Pages read } = \frac{40}{3}$$

So, Swetha has read $\frac{40}{3}$ pages.

Now, convert $\frac{40}{3}$ back to a mixed number: $\frac{40}{3} = 13\frac{1}{3}$

Then the correct option is B)

8. Correct option: (D)

To solve for the value of x in the equation $\frac{23}{29} = \frac{x}{87}$, we can use cross-multiplication:

$$\frac{23}{29} = \frac{x}{87}$$

Cross-multiplying: $23 \times 87 = 29 \times x$

Solving for x:

$$x = \frac{23 \times 87}{29}$$

$$x = \frac{2001}{29}$$

$$x = 69$$

Then the correct option is D)

9. Correct option: (C, D)

Let's evaluate each option based on the given rational numbers:

$$a = -\frac{3}{4}, b = -\frac{5}{6}, c = -\frac{3}{10}, d = \frac{5}{8}$$

A) $a > b$

$-\frac{3}{4} > -\frac{5}{6}$ is true. Since $--\frac{3}{4}$ is greater than $-\frac{5}{6}$.

B) $b < d$

$-\frac{5}{6} < \frac{5}{8}$ is true. Since $-\frac{5}{6}$ is less than $\frac{5}{8}$.

C) $c > d$

$-\frac{3}{10} > \frac{5}{8}$ is false. Since $-\frac{3}{10}$ is smaller than $\frac{5}{8}$.

D) $c < b$

$-\frac{3}{10} < -\frac{5}{6}$ is false. Since $-\frac{3}{10}$ is indeed greater than $-\frac{5}{6}$.

So, the statement that is false is C) $c > d$, D) $c < b$.

Then the correct options are C), D).

10. Correct option: (A,B)

A rational number is a number that can be expressed as the quotient or fraction $\frac{p}{q}$ of two integers, where q is not equal to zero.

A) Every fraction is a rational number. This is true. Since fractions are expressed as the quotient of two integers, they fit the definition of rational numbers.

B) All integers are rational numbers. This is true. Integers can be written as fractions with a denominator of 1. For example, 5 can be expressed as $\frac{5}{1}$.

C) All rational numbers are integers. This is false. Rational numbers include fractions and decimals that are not integers. For example, $\frac{3}{4}$ and 0.25 are rational numbers but not integers.

D) Every rational number is a terminating decimal. This is false. While some rational numbers are terminating decimals (like 0.5 or 0.75), others are repeating decimals (like $\frac{1}{3}$ = 0.333...).

Then the correct options are A), B).

11. Correct option: (B,C)

Let's compare the given values: A) $A < B$; $2.3 < 2.03$ is false.

B) $A < C$; $2.3 < 2.33$ is true.

C) $B < C$; $2.03 < 2.33$ is true.

D) $C < B$; $2.33 < 2.03$ is false.

Then the correct options are B), C).

12. Correct option: (D)

From the information given above and in question we can conclude that

$$A = \frac{5}{6} \text{ (5 dotted boxes out of 6 total boxes)}$$
$$B = \frac{4}{7} \text{ (4 dotted boxes out of 7 total boxes)}$$
$$C = \frac{3}{8} \text{ (3 dotted boxes out of 8 total boxes)}$$
$$D = \frac{2}{5} \text{ (2 dotted boxes out of 5 total boxes)}$$

Then the correct option is D)

13. Correct option: (C)

Let's compare the given ratios:

$$A = \frac{5}{6}$$
$$B = \frac{4}{7}$$
$$C = \frac{3}{8}$$
$$D = \frac{2}{5}$$

Comparing the ratios:

A) A > B ; $\frac{5}{6} > \frac{4}{7}$ is true.

B) B > C ; $\frac{4}{7} > \frac{3}{8}$ is true.

C) C > A ; $\frac{3}{8} > \frac{5}{6}$ is false.

D) D > C ; $\frac{2}{5} > \frac{3}{8}$ is true.

Then the correct option is C)

14. Correct option: (C)

Based on the given ratios: $A = \frac{5}{6}$; $B = \frac{4}{7}$; $C = \frac{3}{8}$; $D = \frac{2}{5}$

Ascending order: $\frac{3}{8}, \frac{2}{5}, \frac{4}{7}, \frac{5}{6} \Rightarrow C, D, B, A$

Then the correct option is C)

Given values of A = 158.398, B= 12.4269, C = 14.1698.

Matrix Matching Type

The place value chart of a number is given below;

Thousands	Hundreds	Tens	Ones	Decimal point	Tenths	Hundredths	Thousandths
4	6	9	8	.	5	3	0

15. Correct Matching is: a-T, b-S, c-P, d-R

In the given place value chart:

Thousands (4) Hundreds (6) Tens (9) Ones (8)

Decimal Point (.)

Tenths (5) Hundredths (3) Thousandths (0)

We have the following options and their corresponding explanations:

A) The place value of 5 is **T)** $\frac{5}{10}$

Explanation: In the chart, the digit 5 is in the tenths place, which is equivalent to $\frac{5}{10}$.

B) The place value of 4 is **S)** 4000

Explanation: The digit 4 is in the thousands place, which is equivalent to $4 \times 1000 = 4000$. So, it corresponds to 4000.

C) The place value of 9 is **P)** 90

Explanation: The digit 9 is in the tens place, which is equivalent to $9 \times 10 = 90$. So, it corresponds to 90.

D) The place value of 3 is **R)** $\frac{3}{100}$

Explanation: The digit 3 is in the hundreds place, which is equivalent to $3 \times \frac{1}{100} = \frac{3}{100}$. So, it corresponds to $\frac{3}{100}$.

The correct matches are a→T; b→S; c→P; d→R.

16. Correct Answer is: 2

The number of decimal places in a decimal number refers to the number of digits after the decimal point.

In the number 12.38, there are two digits after the decimal point.

So, the number of decimal places in 12.38 is 2.

17. Correct Answer is: 4

The integral part of a decimal number refers to the digits to the left of the decimal point.

In the number 1321.28, the integral part is 1321, which consists of four digits.

So, the number of digits in the integral part of 1321.28 is 4.

18. Correct Answer is: 1

The integral part of a decimal number refers to the digits to the left of the decimal point.

In the number 8.231, the integral part is 8, which consists of one digit.

So, the number of digits in the integral part of 8.231 is 1.

19. Correct Answer is: 3

The value of the fraction $\frac{13.2}{4.4}$ can be found by dividing the numerator (13.2) by the denominator (4.4): $\frac{13.2}{4.4} => \frac{132}{44} = 3$

So, the value of $\frac{13.2}{4.4}$ is 3.

20. Correct Answer is: 6

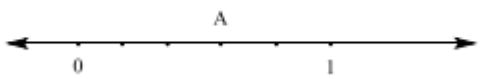
To solve for x in the equation $\frac{x}{5} = 1.2$, we can multiply both sides of the equation by 5:

$$\frac{x}{5} = 1.2$$
$$x = 1.2 \times 5$$
$$x = 6$$

So, if $\frac{x}{5} = 1.2$, then x is equal to 6.

STUDENT PRACTICE QUESTIONS

1. Correct option: (C)



From the figure we can say that the line segment is cut into 5 pieces.

From the 0, we can get point A after 3 step means A takes 3 pieces.

Then A can be represented as $\frac{3}{5}$.

2. Correct option: (B)

$\frac{2}{5}$ Simplifies to $\frac{4}{10}$; $\frac{3}{6}$ Simplifies to $\frac{1}{2}$.

To find a fraction between them, we can take their average: $\frac{\frac{4}{10}+\frac{1}{2}}{2}$

First, add the fractions in the numerator: $\frac{\frac{4}{10}+\frac{5}{10}}{2} = \frac{\frac{9}{10}}{2}$

$$\frac{\frac{9}{10}}{2} = \left(\frac{9}{10}\right) \times \left(\frac{1}{2}\right) = \frac{9}{20}$$

So, a fraction between $\frac{2}{5}$ and $\frac{3}{6}$ is $\frac{9}{20}$, which corresponds to option B) $\frac{9}{20}$.

3. Correct option: (B)

To find the fraction nearest to $\frac{3}{5}$ among the options, we can calculate the decimal value of $\frac{3}{5}$ and then compare it to the decimal values of the fractions in the options.

$\frac{3}{5}$ as a decimal is 0.6.

Now, let's calculate the decimal values of the fractions in the options:

A) $\frac{499}{600} \approx 0.83167$

B) $\frac{299}{500} = 0.598$

C) $\frac{599}{600} \approx 0.99833$

D) $\frac{399}{500} = 0.798$

The fraction that is nearest to $\frac{3}{5}$ (0.6) is option B) $\frac{299}{500}$

4. Correct option: (D)

To calculate the value of P in the expression: $870 \times 5\frac{2}{3} + 4\frac{4}{5} \div 1\frac{3}{5}$

First, let's convert the mixed numbers into improper fractions: $5\frac{2}{3} = \frac{3 \times 5 + 2}{3} = \frac{17}{3}$

$4\frac{4}{5} = \frac{5 \times 4 + 4}{5} = \frac{24}{5}$

$1\frac{3}{5} = \frac{5 \times 1 + 3}{5} = \frac{8}{5}$

1. Multiply 870 by $\frac{17}{3}$: $870 \times \left(\frac{17}{3}\right) = \frac{870 \times 17}{3} = \frac{14790}{3} = 4930$

2. Divide $\frac{24}{5}$ by $\frac{8}{5}$: $\left(\frac{24}{5}\right) \div \left(\frac{8}{5}\right) = \left(\frac{24}{5}\right) \times \left(\frac{5}{8}\right) = \frac{24 \times 5}{5 \times 8} = \frac{120}{40} = 3$

$$P = 4930 + 3 = 4933$$

So, P is equal to 4933.

Then the correct option is D)

5. Correct option: (D)

1. Convert the mixed numbers into improper fractions:

$$18\frac{3}{4}\ \text{kg} = \frac{72 + 3}{4} = \frac{75}{4}\ \text{kg}$$

$$7\frac{1}{2}\ \text{kg} = \frac{14 + 1}{2} = \frac{15}{2}\ \text{kg}$$

$$10\frac{1}{5}\ \text{kg} = \frac{50 + 1}{5} = \frac{51}{5}\ \text{kg}$$

2. Now, add these fractions together to find the total weight:

$$\text{Total weight} = \frac{75}{4}\ \text{kg} + \frac{15}{2}\ \text{kg} + \frac{51}{5}\ \text{kg}$$

$$\text{Total weight} = \left(\frac{375}{20}\right)\ \text{kg} + \left(\frac{150}{20}\right)\ \text{kg} + \left(\frac{204}{20}\right)\ \text{kg}$$

$$\text{Total weight} = \frac{375 + 150 + 204}{20}\ \text{kg}$$

$$\text{Total weight } = \frac{729}{20} \text{ kg}$$

So, the total weight carried by the porter is $\frac{729}{20}$ kg.

$$\text{Total weight } = 36\frac{9}{20} \text{ kg}$$

Then the correct option is D)

6. Correct option: (B)

The additive inverse (or opposite) of a number is a value that, when added to the original number, equals zero.

To find the additive inverse of $-\frac{8}{5}$, we simply change its sign:

$$\text{Additive inverse of } -\frac{8}{5} = -\left(-\frac{8}{5}\right) = \frac{8}{5}$$

Then the correct option is B)

7. Correct option: (A)

To find out how much balance Vineetha gets back after buying a book for Rs 18.90, a pen for Rs 8.50, and some papers for Rs. 5.05, we need to subtract the total cost of these items from the amount she gave to the shopkeeper.

$$\text{Total cost of items } = \text{Rs } 18.90 + \text{Rs } 8.50 + \text{Rs } 5.05 = \text{Rs } 32.45$$

$$\text{Amount given to the shopkeeper } = \text{Rs } 50$$

$$\text{Balance } = \text{Amount given} - \text{Total cost of items } = \text{Rs } 50 - \text{Rs } 32.45 = \text{Rs } 17.55$$

So, Vineetha gets a balance of Rs. 17.55 back.

The correct answer is A) Rs.17.55.

8. Correct option: (C)

To find the value of $\boxdot$ in the equation: $2 \boxdot = \frac{2}{5} \div 13$

To divide by 13, you can multiply by its reciprocal $\left(\frac{1}{13}\right)$: $2 \boxdot = \left(\frac{2}{5}\right) \times \left(\frac{1}{13}\right)$

Now, multiply the fractions on the right side: $2 \boxdot = \left(\frac{2}{5}\right) \times \left(\frac{1}{13}\right) = \frac{2}{65}$

$$2 \boxdot = \frac{2}{65}$$

$$\frac{2 \boxdot}{2} = \frac{\frac{2}{65}}{2}$$

$$\boxdot = \frac{\frac{2}{65}}{2} = \frac{1}{65}$$

So, the value of $\boxdot$ is $\frac{1}{65}$.

Then the correct option is C)

9. Correct option: (C)

To arrange the given sets of numbers in descending order, we need to sort them from largest to smallest:

A) 3.84, 6.70, 7.19, 3.08

Descending order: 7.19, 6.70, 3.84, 3.08

B) 0.26, 4.91, 0.462, 9.58

Descending order: 9.58, 4.91, 0.462, 0.26

C) 9.65, 8.298, 5.97, 0.23

Descending order: 9.65, 8.298, 5.97, 0.23 same as previous.

D) 2.5, 5.40, 6.2, 8.546

Descending order: 8.546, 6.2, 5.40, 2.5

Then the correct option is C)

10. Correct option: (C)

To calculate the cost of 30.5 liters of milk at the rate of Rs. 7.50 per liter, we simply multiply the number of liters by the rate per liter:

$$\text{Cost} = (\text{Number of liters}) \times (\text{Rate per liter})$$

$$\text{Cost} = 30.5 \text{ liters} \times \text{Rs. } 7.50 \text{ per liter}$$

$$\text{Cost} = \text{Rs. } 228.75$$

So, the cost of 30.5 liters of milk is Rs. 228.75.

The correct answer is C) Rs. 228.75.

11. Correct option: (D)

1. Convert the mixed numbers to improper fractions: $2\frac{1}{4} = \frac{4\times 2+1}{4} = \frac{9}{4}$

$$1\frac{1}{4} = \frac{4 \times 1 + 1}{4} = \frac{5}{4}$$

2. Add the fractions: $\frac{9}{4} + 3 + \frac{5}{4} = \frac{9}{4} + \frac{12}{4} + \frac{5}{4}$

3. Add the fractions with the same denominator: $\left(\frac{9}{4} + \frac{12}{4} + \frac{5}{4}\right) = \frac{26}{4} = \frac{13}{2} = 6\frac{1}{2}$

Therefore, sum of these given number is; $\frac{13}{2} = 6\frac{1}{2}$

Then the correct option is D)

12. Correct option: (C)

1. Convert the mixed numbers to improper fractions: $20\frac{5}{7} = \frac{140+5}{7} = \frac{145}{7}$

$$15\frac{1}{7} = \frac{105+1}{10} = \frac{106}{7}$$

2. Subtract the fractions: $\frac{145}{7} - \frac{106}{7} = \frac{39}{7}$

Therefore, subtraction of these given number is $\frac{39}{7} = 5\frac{4}{7}$

Then the correct option is C)

13. Correct option: (A)

Step 1: Convert the mixed numbers to improper fractions. $2\frac{1}{3} = \frac{7}{3}, 1\frac{5}{14} = \frac{33}{14}, 1\frac{4}{5} = \frac{9}{5}$

Step 2: Multiply the fractions using commutative property $\frac{7}{3} \times \frac{19}{14} \times \frac{9}{5} = \frac{7}{14} \times \frac{9}{3} \times \frac{19}{5} = \frac{1}{2} \times 3 \times \frac{19}{5} = \frac{57}{10}$

$$\frac{57}{10} = 5\frac{7}{10}$$

Then the correct option is A)

14. Correct option: (B)

Given $1\frac{1}{2} \div 1\frac{5}{13}$

Convert the mixed numbers to improper fractions: $\frac{3}{2} \div \frac{18}{13}$

Invert the second fraction: $\frac{3}{2} \times \frac{13}{18}$

$$\frac{3 \times 13}{2 \times 18} = \frac{39}{36} = \frac{13}{12}$$

$$\frac{13}{12} = 1\frac{1}{12}$$

Then the correct option is B)

15. Correct option: (A)

To reduce the fraction $\frac{144}{180}$ to its lowest form, we can find the greatest common divisor (GCD) of the numerator and denominator and then divide both by that GCD.

The GCD of 144 and 180 is 36.

Now, divide both the numerator and denominator by 36: $\frac{144 \div 36}{180 \div 36} = \frac{4}{5}$

Then the correct option is A)

16. Correct option: (C)

Let the total cake be represented as 1 (whole).

The part that was distributed: $\frac{5}{7}$

The remaining part of the cake: $1 - \frac{5}{7} = \frac{7}{7} - \frac{5}{7} = \frac{2}{7}$

So, $\frac{2}{7}$ of the cake is left out.

Then the correct option is C)

17. Correct option: (D)

Given fraction: => $\frac{160}{40}$

Simplify by dividing the numerator by the denominator: $160 \div 40 = 4$

So, the equivalent whole number is 4.

Then the correct option is D)

18. Correct option: (A)

Given mixed fraction: => $5\frac{1}{2}$

Convert to improper fraction: $5\frac{1}{2} = \frac{5\times2+1}{2} = \frac{11}{2}$

$$=> 5\frac{1}{2} = \frac{11}{2}$$

Then the correct option is A)

19. Correct option: (B)

To find the ascending order of fraction, it's helpful to have a common denominator for easy comparison.

In this case, the common denominator for these fractions is 12.

Let's convert the fractions and then arrange them:

Given fractions: => $\frac{2}{3}, \frac{3}{4}, \frac{1}{3}, \frac{5}{6}$

Step 1: Convert fractions to have a common denominator of 12:

$$\frac{2}{3} = \frac{8}{12}$$
$$\frac{3}{4} = \frac{9}{12}$$
$$\frac{1}{3} = \frac{4}{12}$$
$$\frac{5}{6} = \frac{10}{12}$$

Step 2: Arrange the fractions in ascending order: $\frac{4}{12}, \frac{8}{12}, \frac{9}{12}, \frac{10}{12}$ => $\frac{1}{3}, \frac{2}{3}, \frac{3}{4}, \frac{5}{6}$

Then the correct option is B)

20. Correct option: (B)

To convert hours to minutes, we multiply the number of hours by 60 (Since there are 60 minutes in 1 hour).

Given: => $7\frac{1}{6}$ hours

Step 1: Convert the mixed number to an improper fraction: $7\frac{1}{6} = \frac{7\times6+1}{6} = \frac{43}{6}$ hours

Step 2: Multiply the fraction by 60 to convert to minutes: $\left(\frac{43}{6}\right) \times 60 = 430$ minutes

Then the correct option is B)

21. Correct option: (D)

Given: => $\frac{123}{25}$

We can break down 123 into 125 and 2, so:

$$\frac{123}{25} = \frac{125 - 2}{25} = \frac{125}{25} - \frac{2}{25} = 5 - 0.08 = 4.92$$

So, the decimal equivalent of $\frac{123}{25}$ is indeed 4.92.

Then the correct option is D)

22. Correct option: (B)

Given: => $\frac{6529}{125}$

We can break down 6529 into 6250 and 279, so:

$$\frac{6529}{125} = \frac{6250 + 279}{125} = \frac{6250}{125} + \frac{279}{125} = 50 + 2.232 = 52.232$$

So, the decimal equivalent of $\frac{6529}{125}$ is indeed 52.232.

Then the correct option is B)

23. Correct option: (D)

To compare decimal numbers, we should first compare the digits to the left of the decimal point. If they are the same, then we move to the digits to the right of the decimal point.

Let's compare them:

A) 0.43 < 0.043 (False, because 0.43 is greater than 0.043)

B) 0.0043 > 0.43 (False, because 0.0043 is smaller than 0.43)

C) 0.00043 > 0.043 (False, because 0.00043 is smaller than 0.043)

D) 0.043 > 0.0043 (True, because 0.043 is greater than 0.0043)

Then the correct option is D)

24. Correct option: (A)

Let's go through each option and identify the position of the digit 9 in the thousandths place:

A) 3.0095 - The digit 9 is in the thousandths place.

B) 3.0905 - The digit 9 is in the hundredths place.

C) 3.9005 - The digit 9 is in the ones place.

D) 3.0059 - The digit 9 is in ten-thousandths place.

Then the correct option is A)

25. Correct option: (A)

To determine which decimal has the greatest value among the options, we need to compare their digits from left to right until we find a difference.

Let's compare the decimals:

A) 6.723

B) 6.0723

C) 6.7023

D) 6.7

Comparing the whole number parts (before the decimal point), they are all the same: 6.

Moving to the tenths place:

A) 7

B) 0

C) 0

D) 7

Here, option A, C and option D have a 7 in the tenths place, which is greater than 0.

Moving to the hundredths place:

A) 2

C) 0

D) 0

Here, option A has the greatest value (7) in the hundredths place.

Since we found a clear difference in the hundredths place, we can conclude that option A (6.723) has the greatest value among the given options.
Then the correct option is A)

26. Correct option: (C)

Let's solve the expression step by step: Expression: $[\frac{6}{17} \times 51 + 12 - 2.5 \times 8]$

First, perform the multiplications and divisions: $\frac{6}{17} \times 51 = 18$; $2.5 \times 8 = 20$

Now substitute the results back into the expression: $[18 + 12 - 20]$

Now add and subtract: $30 - 20 = 10 \Rightarrow \left[\frac{6}{17} \times 51 + 12 - 2.5 \times 8\right] = 10$

Then the correct option is C)

27. Correct option: (A)

Let's solve the expression step by step: Expression: $48 - [18 - \{16 - (5 - \overline{4 - 1})\}]$

1. Inside the innermost $\overline{4 - 1}$, then $(5 - \overline{4 - 1})$ solve the operations:

$$(5 - \overline{4 - 1}) = (5 - 3) = 2$$

2. Substitute the result back into the expression solves next parentheses: $\{16 - 2\} = 14$

$$48 - [18 - 14]$$

3. Solve the next set of parentheses: $[18 - 14] = 4$
4. Substitute the result back into the expression: $48 - 4 = 44$;

$$48 - [18 - \{16 - (5 - \overline{4 - 1})\}] = 44$$

Then the correct option is A)

28. Correct option: (A)

To find the rate per kg at which the apples are being sold, we can divide the total cost by the total weight in kg.
Total cost = Rs. 400
Total weight = $6\frac{1}{4}$ kg = 6.25 kg
Rate per kg = $\frac{\text{Total cost}}{\text{Total weight}}$
Rate per kg = $\frac{400}{6.25} = 64$ (solve using above decimal division method)
So, the rate per kg at which the apples are being sold is Rs. 64.
Then the correct option is A)

29. Correct option: (A)

To calculate how much money Quentin would save by waiting for the sale, subtract the sale price from the original price:
Savings = Original Price - Sale Price
Savings = $428.98 - $399.99 (solve using above decimal subtraction method)
Savings = $28.99
So, Quentin would save $28.99 by waiting until the washing machine went on sale.
Then the correct option is A)

30. Correct option: (C)

To find out how tall Govind was at the end of the year, we can add the growth of 0.6 inches to his initial height of 68.8 inches: $\text{Final height} = \text{Initial height} + \text{Growth}$

$$\text{Final height} = 68.8 \text{ inches} + 0.6 \text{ inches}$$

$$\text{Final height} = 69.4 \text{ inches}$$

So, Govind was 69.4 inches tall at the end of the year.
Then the correct option is C)

31. Correct option: (D)

To compare fractions, we can either find a common denominator for all of them or convert them to decimal equivalents.

Let's find the decimal equivalents for these fractions:

$$\frac{7}{8} = 0.875$$
$$\frac{6}{7} = 0.857\ldots$$
$$\frac{4}{5} = 0.8$$
$$\frac{5}{6} = 0.833\ldots$$

Now, comparing the decimal values, we can see that: $0.875 > 0.857\ldots > 0.833\ldots > 0.8$

So, the greatest fraction among the given options is $\frac{7}{8}$.

32. Correct option: (A, D)

A proper fraction is a fraction where the numerator (the top number) is smaller than the denominator (the bottom number). Let's check each of the options:

A) $\frac{17}{19}$ – Proper fraction $(17 < 19)$

B) $\frac{13}{12}$ – Improper fraction $(13 >= 12)$

C) $\frac{15}{14}$ – Improper fraction $(15 >= 14)$

D) $\frac{29}{31}$ – Proper fraction $(29 < 31)$

So, the proper fractions are A) $\frac{17}{19}$ and D) $\frac{29}{31}$.

33. Correct option: (A, C)

To identify equivalent fractions among the given options, we can simplify each fraction and compare them:

A) $\frac{1}{2}$

B) $\frac{2}{5}$

C) $\frac{3}{6} = \frac{1}{2}$

D) $\frac{4}{5}$

The equivalent fractions are: A) $\frac{1}{2}$;C) $\frac{3}{6} = \frac{1}{2}$

So, fractions A and C are equivalent.

34. Correct option: (A, B)

To add and subtract these fractions, we need to find a common denominator:

The common denominator for $\frac{9}{5}, \frac{1}{10}, \text{and} \frac{17}{5}$ is 10.

Now, rewrite each fraction with a denominator of 10: $\left(\frac{9}{5}\right)$ becomes $\left(\frac{9}{5}\right) \times \left(\frac{2}{2}\right) = \frac{18}{10}$

$$\left(\frac{1}{10}\right) \text{ stays as } \left(\frac{1}{10}\right)$$
$$\left(\frac{17}{5}\right) \text{ becomes } \left(\frac{17}{5}\right) \times \left(\frac{2}{2}\right) = \frac{34}{10}$$

Now, we can perform the addition and subtraction:

$$\left(\frac{18}{10}\right) + \left(\frac{1}{10}\right) - \left(\frac{34}{10}\right) = \frac{18 + 1 - 34}{10} = \frac{-15}{10} = -\frac{3}{2} = -1\frac{1}{2} = -1.5$$

Then the correct option is A), B).

35. Correct option: (A)

To find the fraction of the total test that Ramya has prepared, we need to add the number of questions she has prepared (25 multiple-choice questions) and divide it by the total number of questions (40 multiple-choice questions and 60 short-answer questions).

Total questions in the test: 40 + 60 = 100 questions

Questions prepared by Ramya: 25 multiple-choice questions.

$$\text{Fraction prepared} = \frac{\text{Questions prepared by Ramya}}{\text{Total questions}}$$

$$\text{Fraction prepared} = \frac{25}{100} => \frac{25}{25 \times 4} => \frac{1}{4}$$

$$\text{Fraction prepared} = \frac{1}{4}$$

So, Ramya has prepared $\frac{1}{4}$ of the total test.

Then the correct option is A)

36. Correct option: (A,B)

To find the average distance covered by the car in 1 hour, we can divide the total distance by the total time:

$$\text{Average distance} = \frac{\text{Total distance}}{\text{Total time}}$$

$$\text{Average distance} = \frac{89.1 \text{ km}}{2.2 \text{ hours}}$$

$$=> \frac{891}{22} \text{ km}$$

Calculating the average distance: $\text{Average distance} = 40.5 \text{ km}$

So, the average distance covered by the car in 1 hour is 40.5 kilometers.

Then the correct options are A), B).

37. Correct option: (B,C)

If A = 12.52, the number that has only 2 digits after the decimal point is:

B) 3.25

C) 13.25

Then the correct options are B), C).

38. Correct option: (C)

In this case, the denominators of both fractions are the same (both are 2). So, we can simply add the numerators: $\frac{1}{2} + \frac{3}{2} = \frac{1+3}{2} = \frac{4}{2}$

Now, simplify the fraction: $\frac{4}{2} = \frac{2}{1} = 2$

So, $\frac{1}{2} + \frac{3}{2}$ equals 2.

The correct answer is C) 2.

39. Correct option: (A)

In this case, we have two fractions with different denominators:

To add them together, we can find a common denominator.

The least common multiple (LCM) of 3 and 2 is 6. So, we can rewrite both fractions with a common denominator of 6:

$$\frac{1}{3} = \left(\frac{1}{3}\right) \times \left(\frac{2}{2}\right) = \frac{2}{6}$$

$$\frac{1}{2} = \left(\frac{1}{2}\right) \times \left(\frac{3}{3}\right) = \frac{3}{6}$$

Now, both fractions have the same denominator (6), so we can add the numerators:

$$\frac{2}{6} + \frac{3}{6} = \frac{2 + 3}{6} = \frac{5}{6}$$

40. Correct option: (B)

To determine the place value of a digit in a decimal number, we can look at its position relative to the decimal point.

In the decimal number A = 158.398:

The digit 9 is in the hundredth place (two places to the right of the decimal point).

Therefore, the place value of 9 in A is $\frac{9}{100}$.

Then the correct option is B)

41. Correct option: (D)

The decimal number B is 12.4269. The digit 6 is in the thousandths place in this decimal number.

In decimal notation, the places to the right of the decimal point are powers of 10: tenths, hundredths, thousandths, and so on.

So, the place value of 6 in the decimal represented by B (12.4269) is $\frac{6}{1000}$.

Then the correct option is D)

42. Correct option: (D)

To find out how much time you can spend on each question, you need to divide the total time ($2\frac{1}{2}$ hours) by the number of questions (50).

$$\text{Total time} = 2\frac{1}{2} \text{ hours} = 2.5 \text{ hours}$$

$$\text{Time per question} = \frac{\text{Total time}}{\text{Number of questions}}$$

$$\text{Time per question} = \frac{2.5 \text{ hours}}{50 \text{ questions}}$$

Convert hours to minutes (since we want the answer in minutes):

$$2.5 \text{ hours} = 2.5 \times 60 \text{ minutes} = 150 \text{ minutes}$$

$$\text{Time per question} = \frac{150 \text{ minutes}}{50 \text{ questions}}$$

$$\text{Time per question} = 3 \text{ minutes}$$

So, we can spend 3 minutes on each question.

Then the correct option is D)

43. Correct Matching is: a-S, b-P, c-Q, d-R

Let's match the given expressions in column I with their corresponding values in column II: a) Reciprocal of $\frac{a}{b}$ - S (Reciprocal of $\frac{a}{b}$ is $\frac{1}{\frac{a}{b}}$, which is $\frac{b}{a}$)

b) Additive inverse of $-\frac{3}{7}$ - P (The additive inverse of a fraction is a fraction with the same denominator and the negative of the numerator. In this case, it's $-\frac{3}{7}$)

c) $\frac{4}{9} + \frac{5}{9}$ - Q (Adding $\frac{4}{9}$ and $\frac{5}{9}$ results in $\frac{9}{9}$, which simplifies to 1)

d) $\frac{3}{5}$ - R (The decimal representation of $\frac{3}{5}$ is 0.6)

44. Correct Matching is: - a→ P, S; b→ P, R ; c→ Q, S; d→Q

a) 3.03, 3.30, 33.03

P) Like decimals

Explanation: All of these numbers have two digits after the decimal point, making them similar in terms of decimal format.

S) are in ascending order

Explanation: When arranged from the smallest to the largest, these numbers are in ascending order: 3.03 < 3.30 < 33.03.

b) 213.6, 64.2, 33.5

P) Like decimals
Explanation: All of these numbers have one digit after the decimal point, making them similar in terms of decimal format.
R) are in descending order
Explanation: When arranged from the largest to the smallest, these numbers are in descending order: 213.6 >64.2 > 33.5.
c) 0.1, 0.01, 0.001
Q) Unlike decimals
Explanation: All of these numbers have a decreasing number of digits after the decimal point as they move to the right, making them unlike decimals.
R) are in descending order
Explanation: When arranged from the smallest to the largest, these numbers are in descending order: 0.1 >0.01>0.001.
d) 13.48, 0.189, 9.7, 8.23
Q) Unlike decimals
Explanation: All of these numbers have a different number of digits after the decimal point as they move to the right, making them unlike decimals.
T) Cannot be composed
Explanation: These numbers do not form any clear ascending or descending pattern, so they cannot be composed into a logical order.
The Correct Matchig is a→ P, S; b→ P, R ; c→ Q, S; d→Q

45. Correct Answer is: 8
Given expression,

$$\frac{\frac{1}{3}\div\frac{1}{3}\times\frac{1}{3}}{\frac{1}{3}\div\frac{1}{3}\text{of}\frac{1}{3}}-\frac{1}{9}$$

According BODMAS
First solve; $\left(\frac{1}{3}\right)\div\left(\frac{1}{3}\right)\times\left(\frac{1}{3}\right)=\frac{1}{3}\div\frac{1}{9}=3$

$$\left(\frac{1}{3}\right)\div\left(\frac{1}{3}\times\frac{1}{3}\right)=\left(\frac{1}{3}\right)\div\left(\frac{1}{9}\right)=\left(\frac{1}{3}\right)\times\left(\frac{9}{1}\right)=3$$

Putting back into expression

$$\frac{\frac{1}{3}\div\frac{1}{3}\times\frac{1}{3}}{\frac{1}{3}\div\frac{1}{3}\text{of}\frac{1}{3}}-\frac{1}{9}=\frac{3}{3}-\frac{1}{9}=1-\frac{1}{9}=\frac{8}{9}=\frac{x}{9}=>x=8$$

So, the value of x is 8.

46. Correct Answer is: 5
Given, $3+1\frac{1}{5}+\frac{2}{5}-\frac{7}{15}$
First, convert the mixed number to an improper fraction: $1\frac{1}{5}=\frac{5\times1+1}{5}=\frac{6}{5}$
Now, add the fractions : $3+\frac{6}{5}+\frac{2}{5}-\frac{7}{15}$
Now, to combine these fractions, we need a common denominator, which is 15. Convert all fractions to have a denominator of 15:

$$\left(3\times\frac{15}{15}\right)+\left(\frac{6}{5}\times\frac{3}{3}\right)+\left(\frac{2}{5}\times\frac{3}{3}\right)-\left(\frac{7}{15}\times1\right)=\frac{45}{15}+\frac{18}{15}+\frac{6}{15}-\frac{7}{15}$$

Combine the numerators: $\frac{45+18+6-7}{15}=\frac{62}{15}$
Now we are given that, $3+\frac{6}{5}+\frac{2}{5}-\frac{7}{15}=\frac{62}{3p}$

$$=>\frac{62}{15}=\frac{62}{3p}=>3p=15=>p=5$$

So, the value of p is 5.

47. Correct Answer is: 2

To find the value of x in the equation $\frac{3}{4} = \frac{xy}{28}$, we can cross-multiply and solve for x: $\frac{3}{4} = \frac{xy}{28}$

First, multiply both sides of the equation by 28 to isolate xy: $\left(\frac{3}{4}\right) \times 28 = xy$

$$\frac{3 \times 28}{4} = \frac{84}{4} = 21$$

$$21 = xy$$

Now, we need to find the value of x.

Since $21 = xy$, x must be 2. Therefore, x = 2.

48. Correct Answer is: 3

The number of decimal places in a decimal number refers to the number of digits after the decimal point.

In the number 281.238, there are three digits after the decimal point.

So, the number of decimal places in 281.238 is 3.

49. Correct Answer is: 4

To find the decimal representation of the fraction $\frac{365}{80}$, we divide the numerator (365) by the denominator (80):$365 \div 80 = 4.5625$

In the decimal representation 4.5625, there are four digits in the decimal part (5625).

So, the number of digits in the decimal part of $\frac{365}{80}$ is 4.

50. Correct Answer is: 1

To find the decimal representation of the fraction $\frac{15}{8}$, we divide the numerator (15) by the denominator (8):$15 \div 8 = 1.875$

The given representation is x + 0.875. Since the decimal part of 1.875 is 0.875, you can see that x must be equal to 1: $x + 0.875 = 1 + 0.875 = 1.875$

So, the value of 'x' is 1.

3. Geometry

Solutions

LEVEL -I

1. Correct option: (D)

There are an infinite number of points on a line.

An infinite number of points put together closely in an orderly fashion will result in a line.

So, a line contains an infinite number of points.

Therefore, there are an infinite number of points on a line.

2. Correct option: (B)

A line segment is formed by joining any two fixed points on a line.

Line segment is a part of a line such that it starts at a particular point and ends at a different point.

So, starting point and ending points can be the exist for a line segment.

Thus, the line segment has two end points.

3. Correct option: (A)

The interiors of the figure along with its boundary is called region of the figure.

The interior of the figure together with its boundary is termed as its Region.

This is because the very definition of region is an area with distinct characteristics within a boundary.

Therefore, the interior of the figure together with its boundary is called its Region.

4. Correct option: (B)

The figure formed by two rays with the same initial points is called an angle.

The word “angle” is derived from the Latin word “angulus”, which means “corner”.

The two rays forming an angle are called the arms (or) sides of the angle.

The common end point of the two rays forming an angle is called the vertex of the angle.

5. Correct option: (A)

A closed figure is a figure with the same start and end point, and which completely encloses an area.

Some simple closed curves are made of both line segments and curved lines.

Therefore, 

is a simple closed figure.

6. Correct option: (C)

A polygon is a simple closed figure formed by more than two line segments.

- A polygon formed by three line segments is called a triangle.
- A polygon formed by four line segments is called a quadrilateral.

Therefore, a closed figure formed by three or more line segments is called as a POLYGON.

7. Correct option: (D)

The meeting point of a pair of sides is called its vertex.

Consider $\triangle ABC$.

AB, BC, and CA are the sides of this triangle, and A, B, and C are its vertices.

So, a triangle has 3 vertices.

Therefore, the number of vertices of a triangle is 3.

8. Correct option: (B)

The hour hand turns through 2 right angles starting from 11 to 5.

A complete revolution of the hour hand covers 12 hours or four right angles i.e 360∘.Which means 1 hour = 30∘.

When the hour hand goes from 2 to 8, the time difference = 11 - 5 = 6 hours i.e half of 12 hours. ⇒ 12 of 360°= 180° = Two right angles.

Therefore, the number of right angles through which the hour hand of a clock turns when it goes from 11 to 5 is 2.

9. Correct option: (C)

We know that a segment is a part of the circle enclosed by a chord and an arc of the circle.

Consider a figure:



The region enclosed by the major arc and its corresponding chord is called the major segment.

By observing the figure and according to the definition, the centre of the circle lies in the interior of the major segment.

∴ The centre of the circle always lies in the interior of the major segment.

10. Correct option: (D)

Circumference of a circle $'C' = 2\pi r$, where 'r' = radius of the circle.

Hence, $2r = d$, diameter of the circle.

$\therefore \pi \times d = C$ or $Cd = \pi$ is also referred to as the Greek alphabet,

Pi and has an approximate value = $\frac{22}{7}$ or 3.14

Therefore, the circumference of a circle is approximately equal to π times its diameter.

11. Correct option: (B).

A median of a triangle is a line segment that joins a vertex to the mid-point of the side that is opposite to that vertex.

In the figure, BD is the median that divides AC into two equal halves, that is, AD = DC.

Therefore, BD is median.

12. Correct option: (C)

A square is a regular quadrilateral.

All sides of a square are of equal length.

All angles of a square are equal in measure.

Therefore, a SQUARE is said to be regular, if its sides are equal in length and angles are equal are in measure.

13. Correct option: (B)

Coplanar simply means “lying on the same plane.” Here, “co” means “together,” and “planar” means “lying on a plane.”

In geometry, a plane is a two-dimensional, flat surface that extends infinitely in both dimensions.

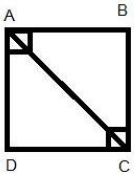
When two or more points or lines lie on the same plane or common plane, they are said to be coplanar.

Therefore, the points belong to the same plane are called CO-PLANAR POINTS.

14. Correct option: (A).

The line segments joining two non-adjacent vertices of a polygon are called diagonals.

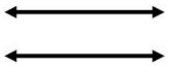
15. Correct option: (D)



In a square, each angle is 90°.
Diagonals of a square bisect the angles.
If ABCD is a square, ∠BAD will be 90°.
In the given question, AC is the diagonal then,
$\angle BAC = \frac{\angle BAD}{2} = \frac{90°}{2} = 45°$.
Hence, the value of ∠BAC is 45°.

16. Correct option: (B)

The lines which lie on the same plane and do not intersect at any point are called parallel lines.



17. Correct option: (D)

Circumference: the total boundary length, i.e., the perimeter of the circle is known as its circumference.
The boundary of the circle is called its circumference and the value of circumference is 2πr (where r is the radius of the circle).
If we cut a circle and form a line from it, then the length of the line will be the same as the circumference of the circle.
Therefore, the total length of circle is called its CIRCUMFERENCE.

18. Correct option: (A)

A line segment has a definite length and two end points. A point has a position.
There can be only one line segment can be drawn by joining the two points.



Therefore, the number of unique line segments can be drawn with the given two points are 1.

19. Correct option: (B).

The number of lines that can be drawn through 'n' non-collinear points depends on the configuration and arrangement of the points.
In general, the number of lines that can be drawn through 'n' non-collinear points is given by the formula:
$L = \frac{n(n-1)}{2}$.
This formula is based on the concept that each pair of distinct points determines a unique line. So, for 'n' points, there are n choose 2 (n_{C_2}) pairs of points, which represents the number of lines that can be drawn.
For example, if you have 4 non-collinear points, the number of lines that can be drawn would be: $L = \frac{4(4-1)}{2} = 6$.
So, in this case, 6 lines can be drawn through the 4 non-collinear points.
Therefore, Number of lines that can be drawn through 'n' non-collinear points is $\frac{n(n-1)}{2}$.

20. Correct option: (C)

Collinear points: The points which lie on the single straight line is known as collinear points.
If A, B and C are three points on a line, and B lies between A and C.

LEVEL -II

1. Correct option: (B)

The shortest distance from point A to point B is a line segment.

Distance between two points is the length of the line segment that connects the two given points. Distance between two points in coordinate geometry can be calculated by finding the length of the line segment joining the given coordinates.

Therefore, the shortest distance from point A to point B is a line segment.

2. Correct option: (A)

A line segment extending endlessly on both sides is called a STRAIGHT LINE.

3. Correct option: (B).

A line segment when extended indefinitely in one direction is called a ray.

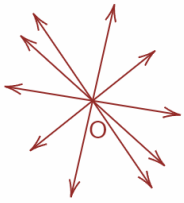
A ray is part of a line that has a fixed starting point but does not have an endpoint.

A ray can extend infinitely in one direction, meaning that a ray can go on forever in one direction.

4. Correct option: (D)

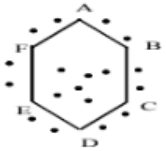
Given a point, we have to find the number of lines that can be drawn from a point.

- A point in geometry is a location.
- It has no size i.e., no width, no length and no depth.
- A point is shown by a dot.
- A line is defined as a line of points that extends infinitely in two directions.
- It has one dimension length.



An infinite number of lines can be drawn through a given point.

5. Correct option: (C)



No. of exterior points in the given figure are = 12

No. of interior points in the given figure are = 6.

The difference between the number of exterior and interior points in the given figure = 12 – 6 = 6.

Therefore, the required answer is 6.

6. Correct option: (C)

The radius of a circle is the length of the line segment from the center of a circle to a point on the circumference of the circle and diameter is a line segment from one end of the circle to the other end of the circle passing through the center of the circle. So, the radius is half the length of the diameter.

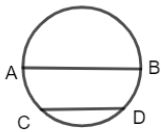
Therefore, half of a diameter of a circle is called the radius.

7. Correct option: (B)

A line segment with its endpoint lying on a circle is called the chord of a circle.

A chord of a circle is a line segment with its endpoints on the circle.

We know that the diameter is a line segment that connects the circle's extremities to its midpoint.



A chord is a line segment that has its ends on the circumference but does not traverse the center.
From the figure, AB is the diameter whereas CD is the chord of the circle.
The endpoints of the CD are on the circle.
Hence, a line segment with its endpoints on the circle is called a Chord.

8. Correct option: (D)

A central angle has an apex (vertex) that is the center O of a circle and legs (sides) that are radii that intersect the circle in two distinct points.
A circle is made up of 360 degrees. The central angle of a circle adds up to 360°.

9. Correct option: (B)

The circumference of the circle or perimeter of the circle is the measurement of the boundary of the circle.
The circumference of a circle is the perimeter of the circle.
It is the total length of the boundary of the circle.
Therefore, the circumference circle equation is $C = 2\pi r$.

10. Correct option: (C)

A right angle is always equal to 90 degrees.
Any angle greater than 90 degrees is an obtuse angle.
Therefore, the angles of measure 90° and 145° are RIGHT and OBTUSE angle.

11. Correct option: (A, C, D)

A simple curve is a curve which does not cross itself.
A curve is a continuous, smooth-flowing line that bends or changes direction at least once.
Therefore, Option A, C and D are the curves.

12. Correct option: (B)

Adjacent sides are the sides of a polygon that share a common vertex.
A polygon is made up of multiple line segments called sides.
In the given polygon ABCDEFGH, we can identify the following sides: - AB - BC - CD - DE - EF - FG – GH - HA = 8 sides.
No. of Line segments = AE- BF-CG-DH = 4 sides.
Therefore, the number of line segments in the adjacent figure is 8+4=12 sides.
In addition to these, the line segments towars O are AO, BO, CO, DO, EO, FO, GO and HO = 8 line segments.
And, AC, CE, EG anf GA line segments forms a quadrilateral. = 4 lines
Total 8+4+8+4 = 24 line segments.

13. Correct option: (B, C, D)

A quadrilateral is a closed shape and a type of polygon that has four sides, or four pairs of adjucent sides, four vertices and four angles which two pairs of opposite angels.
It is formed by joining four non-collinear points.
The sum of the interior angles of quadrilaterals is always equal to 360 degrees.
Therefore, a Quadrilateral is a polygon with four sides and a simple closed figure with four pairs of adjacent sides.

14. Correct option: (B)

The radius of the circle = 18 cm
The diameter of the circle = radius × 2
On substituting the values, we get:

The diameter of the circle = $18 \times 2 = 36$ cm.
Therefore, d = 36 cm.

15. Correct option: (C)

The radius of the circle = 16 cm
The diameter of the circle = radius × 2
On substituting the values, we get:
The diameter of the circle = $16 \times 2 = 32$ cm.
Therefore, the length of largest chord of in circle, d = 32 cm.

16. Correct option: (A)

Given: 2d = 64 cm

$d = \frac{64}{2} = 32$

d = 32 cm

$\text{Radius} = \frac{d}{2}$

Radius= $\frac{32}{2} = 16$.

r = 16 cm.

Therefore, radius is 16 cm.

17. Correct Matching is: a → R, b → S, c → T, d → Q

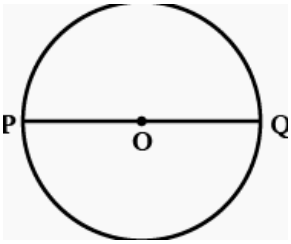
a) A ray starts from a point and extends infinitely in one direction. So, it has one endpoint.
Therefore, the Ray has only one endpoint.

b) A triangle is a closed, 2-dimensional shape with 3 sides, 3 angles, and 3 vertices. A triangle is also a polygon.
Therefore, triangle has 3 vertices.

c) Every circle has infinite no. of radii. This because number of points on the circumference of circle is infinite.
Therefore, number of radii in a circle is infinite.

d) A semicircle is a half-circle, formed by cutting a whole circle along a diameter line, as shown above.

Here, PQ is a diameter of a circle.



Any diameter of a circle cuts it into two equal semicircles.
∴ There are 2 semicircles in a circle.

18. Correct Answer is: 1

One and only one line segment can be drawn to pass through two given points.
Therefore, the number of line segments obtained when two points are given is 1.

19. Correct Answer is: 6

In geometry, a hexagon can be defined as a closed two-dimensional polygon with six sides. Hexagon has 6 vertices and 6 angles also. Hexa means six and gonia means angles.

20. Correct Answer is: 4

A square and a rectangle each have 4 right (90 degree) angles.
Both square and rectangle have 4 sides and 4 angles, but the only difference is square is a rectangle with all sides equal.
Therefore, number of right angles in a square is 4.

STUDENT PRACTICE QUESTIONS

1. Correct option: (A)
We know that a line segment is made up of two points.
So, there is exactly one line that can be drawn through any two distinct points.
Hence, the answer is one.

2. Correct option: (D).
As we know, a circle is a collection of infinite points having equal distance from a fixed point.
Also, diameter is a line segment passing through the center joining two points on the circumference.
So, we can draw an infinite number of diameters in a circle.
Therefore, a circle can have an infinite number of diameters.

3. Correct option: (D)
The angle subtended at the centre of circle is of 360°, and it is called as a central angle.
Hence, the answer is 360°.

4. Correct option: (A).
Every individual flat surface of a solid is called its face. Solids have more than one face.
The given figure is a cube.
Cube is a 3D solid object bounded by six square faces with three meeting at each vertex.
Hence a cube has 6 faces.
Therefore, the number of faces a solid cube have 6.

5. Correct option: (B)
By angle sum property, the sum of interior angles of a quadrilateral is 360°.
If ABCD is a quadrilateral, then $\Rightarrow \angle A + \angle B + \angle C + \angle D = 360°$
$\Rightarrow x° + (x + 10)° + (x - 20)° + 2x° = 360°$

$\Rightarrow x° + x° + x° + 2x° + (10 - 20)° = 360°$

$\Rightarrow 5x° = 360° + 10° = 370°$

$\Rightarrow x° = 74°.$

So, the angles of the quadrilateral are, $x° = 74°$,

$(x + 10)° = (74° + 10°) = 84°$,

$(x - 20)° = (74 - 20)° = 54°$,

$(2x)° = (2 \times 74)° = 148°.$

Hence, the smallest angle of the quadrilateral is 54°.

6. Correct option: (B)
ABCD is a quadrilateral, then $\Rightarrow \angle A + \angle B + \angle C + \angle D = 360°$
$\Rightarrow 60° + 120° + \angle C + \angle D = 360°$

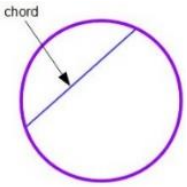
$\Rightarrow 180° + \angle C + \angle D = 360°$
$\Rightarrow \angle C + \angle D = 360° - 180° = 180°.$
$\Rightarrow \angle C + \angle D = 180°.$

Therefore, $\angle C + \angle D = 180° = 2$ right angles.

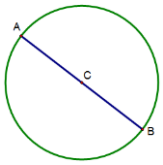
7. Correct option: (C)
For a circle, **Diameter** $= 2 \times$ **Radius**
This means Diameter and Radius are proportional to each other.
In a circle diameter is double of radius, so if the radius increases by 4 times, then the diameter will increase by 8 times.
Therefore, if the radius increases by 4 times, then the diameter will increase by 8 times.

8. Correct option: (B)

A chord is a line segment whose ends lie on the ends of a circle.



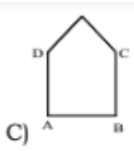
When a chord passes through the center of the circle, it is called diameter of the circle.



Diameter is the longest chord which passes through the centre of the circle.
Therefore, A chord which passes through the center of circle is called Diameter.

9. Correct option: (C)

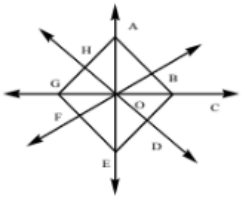
A convex polygon is defined as a polygon with all its interior angles less than 180°.
This means that all the vertices of the polygon will point outwards, away from the interior of the shape.
Think of it as a 'bulging' polygon. Note that a triangle (3-gon) is always convex.
A convex polygon is a closed shape, where none of the vertices are pointed inward.
When a line crosses the convex polygon, it intersects at most the two sides of the polygon.
Any polygon such as pentagon, hexagon, etc, whose vertices are pointed outside the center of the polygon, are convex.
Therefore, the convex polygon is.



C) Since all its interior angles less than 180°.

10. Correct option: (B)

The corresponding figure is as follows:



Four concurrent lines are given as follows: AE, BF, CG and DH.
Therefore, the number of lines passing through O in the given figure is 4.

11. Correct option: (C, D)

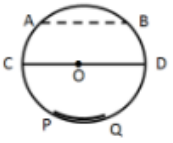
By observing the given figure point, A and point B lies within or inside the figure representing the interior points.
Therefore, points A and B are the interior points in the adjacent figure.

12. Correct option: (A, C)

A simple plane closed figure bounded by three line segments is called a triangle.

13. Correct option: (C)

A circle is the set of all points (or locus of all points) in the plane that lie at a fixed distance from a fixed point.
The fixed point is called the “center” of the circle.
From the given figure, the fixed point is O.
Therefore, centre of the circle is pointing O.



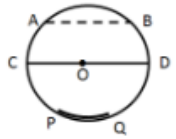
14. Correct option: (B)

A segment of a straight line joining two points on a circle is called a chord.

A chord that passes through the center of the circle is called a diameter.

A chord that passes through the center of a circle is called a diameter and is the longest chord.

From the given figure, the longest chord is CD.



Therefore, the longest chord of the circle is CD.

15. Correct Matching is: a → PS, b → R, c → SP, d → Q

a) Parallelogram is a quadrilateral in which opposite sides are equal and parallel. Rhombus is a quadrilateral in which opposite sides are parallel and all of its sides are equal. Square is quadrilateral in which all the fourse sides are of equal length. So adjucent sides are equl insquare.

Hence, if adjacent sides of a parallelogram are equal, then it is a rhombus.

b) We know that the angles of a rectangle and square are right angles.

c) Property of square: Diagonals of a square cure perpendicular bisectors of each other.

Hence, when the diagonals of a quadrilateral are perpendicular to each other, then it is called square. In case of Rhomus also, the diagonals are perpendicular.

d) A rhombus is a parallelogram in which adjacent sides are equal but the angles are not equal. In case of square, the adjucent sides are equal and the angles are 90°.

Therefore, Adjacent sides are equal, and the angle is 90 degree is a square.

16. Correct Answer is: 3

Let three collinear points be A,B,C

They can represent three line segments namely, AB,BC,AC.

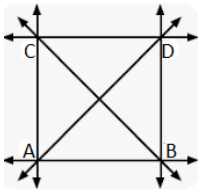
Thus namely 3 line segments are possible with three collinear points.

Therefore, the number of line segments obtained when three points are given is 3.

17. Correct Answer is: 6

A line can be drawn with every pair of two points.

So, we can draw the number of possible lines as follows:



Therefore, 6 lines can be drawn.

18. Correct Answer is: 4

At the point of intersection of the diagonals of a rhombus, there are four right angles.

A rhombus is a quadrilateral with all sides of equal length. When the diagonals of a rhombus intersect, they bisect each other at right angles. This property holds true for all rhombi, regardless of their size or orientation.

Therefore, at the point where the diagonals of a rhombus intersect, there will always be four right angles formed.

19. Correct Answer is: 2

The diagonal of a rhombus is a line segment that joins any two non-adjacent vertices of a rhombus. A rhombus has two diagonals that bisect each other at right angles, thus, they form 4 right-angled triangles which are congruent.

Therefore, number of diagonals in a rhombus is 2.

20. Correct Answer is: 0

A parallelogram is a quadrilateral in which opposite sides are parallel and equal in length.
When the diagonals of a parallelogram intersect, they divide each other into two equal halves.
However, the angles formed at the point of intersection are not necessarily right angles unless the parallelogram is a rectangle.
In a rectangle, all angles are right angles, including the angles formed by the intersection of its diagonals.
However, in a general parallelogram, the angles formed at the point of intersection of the diagonals are not right angles unless additional conditions are met.

Lines and Angles

LEVEL -I

1. Correct option: (B)

An angle which is measuring less than 90 degrees is called an acute angle.
This angle is smaller than the right angle (which is equal to 90 degrees).
For example, $\angle 30°, \angle 45°, \angle 60°, \angle 75°, \angle 33°, \angle 55°, \angle 85°$, etc. are all acute angles.
Therefore, $60°$ is an acute angle.

2. Correct option: (D)

A reflex angle is an angle that is more than 180 degrees and less than 360 degrees.
We know, reflex angle must be between 180° and 360°.
It can't be equal to 180° or 360° as these are straight and complete angle, respectively.
Here, only 192° is greater than 180° and less than 360°.

3. Correct option: (C)

An angle which is measuring less than 90 degrees is called an acute angle.
This angle is smaller than the right angle (which is equal to 90 degrees).



4. Correct option: (C).

A complete angle is also called a full angle, a round angle, or a perigon.
We know that if two rays meet at a common point (vertex), then an angle is formed.
It is measured in units like “degrees” or “radians.”
If the amount of turn around a point is equal to 360 degrees then the angle formed is a complete angle.
Therefore, complete angle means 360°.

5. Correct option: (C)

x, y are complementary angles.
The sum of two angles is equal to 90 degrees.
$\angle x + \angle y = 90^0$
Given $\angle x = 70^0$
$70^0 + \angle y = 90^0$
$\angle y = 90^0 - 70^0$
$\angle y = 20^0$; Therefore, $\angle y = 20^0$.

6. Correct option: (A)

x, y, z are complementary angles.
The sum of two angles is equal to 90 degrees.
$\angle x + \angle y + \angle z = 90^0$

Given $\angle x + \angle y = 60^0$
$60^0 + \angle z = 90^0$
$\angle z = 90^0 - 60^0$
$\angle z = 30^0$; Therefore, $\angle z = 30^0$.

7. Correct option: (B)

When the sum of two angles is 180°, then the angles are known as supplementary angles. In other words, if two angles add up, to form a straight angle, then those angles are referred to as supplementary angles.

8. Correct option: (A)

x, y, z are supplementary angles.
The sum of two angles is equal to 180 degrees.
$\angle x + \angle y = 180^0$
Given $\angle x = 110^0$
$110^0 + \angle y = 180^0$
$\angle y = 180^0 - 110^0$
$\angle y = 70^0$; Therefore, $\angle y = 70^0$.

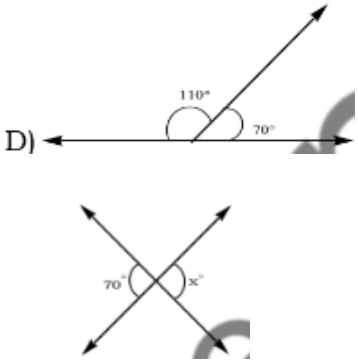
9. Correct option: (B)

The figure is a straight line. The sum of angles is equal to 180 degrees.
$65^0 + 70^0 + x^0 = 180^0$
$135^0 + x^0 = 180^0$
$x^0 = 180^0 - 135^0$
$x^0 = 45^0$; Therefore, the value of x is 45^0.

10. Correct option: (D)

A pair of angles formed on a straight line are said to be a linear pair. The sum of angles of a linear pair is always 180 degrees.
Here, $110^0 + 70^0 = 180^0$.

11. Correct option: (C)



The angle labelled x and the angle with value 70° are vertically opposite to one another and they are equal to one another.
Therefore, the value of x = 70°.

12. Correct option: (D)

Since lines l , m, and n intersect at O.
$\angle a = \angle c$ [Vertically opposite angles]
$\Rightarrow \angle c = 38°$ [$\angle a = 38°$]
$\angle a + \angle b = 180°$ [Linear pair of angles]
$\Rightarrow 38° + \angle b = 180°$
$\Rightarrow \angle b = 142°$
Since $\angle b$ and $\angle d$ are vertically opposite angles.
$\Rightarrow \angle d = \angle b$
$\Rightarrow \angle d = 142°$ [$\angle b = 142°$]
Hence, $\angle b = 142°, \angle c = 38°, \angle d = 142°$.

13. Correct option: (D)
From the figure, $2y^0, y^0, 3y^0$ form a linear pair.
Therefore, their sum must be supplementary.
$2y^0 + y^0 + 3y^0 = 180^0$.
$6y^0 = 180^0$.
$y^0 = 30^0$. ; Therefore, the value is $y^0 = 30^0$.
14. Correct option: (B)
If two straight lines intersect one another, the four angles so formed are together equal to four right angles i.e., 360°.
If two straight lines intersect one another, the vertically opposite angles are equal.
15. Correct option: (C)
Corresponding angles are the angles which are formed in matching corners or corresponding corners with the transversal when two parallel lines are intersected by any other line (i.e. the transversal).
The corresponding angles formed by two parallel lines are always equal.
The angle labelled x and the angle with value 60° are corresponding angles and they are equal to one another.
Therefore, the value of x = 60°.
16. Correct option: (C)
The co-interior angle, also known as the consecutive interior angle, if the transversal intersects the two parallel lines, each pair of co-interior angles sums up to 180 degrees (supplementary angles).
From the figure, $2x^0, x^0 + 30^0$ form a linear pair.
Therefore, their sum must be supplementary.
$2x^0 + x^0 + 30^0 = 180^0$.
$3x^0 + 30^0 = 180^0$.
$3x^0 = 180^0 - 30^0$.
$3x^0 = 150^0$
$x^0 = 50^0$. ; Therefore, the value is $x^0 = 50^0$.
17. Correct option: (D)
The angle labelled A and the angle with value 110° forms a linear pair.
$\angle A + 110^0 = 180^0$.
$\angle A = 180^0 - 110^0$.
$\angle A = 70^0$
It is given that, **l ‖ m** and t is transversal.
We know that corresponding angles are equal.
∠B is the corresponding angle of the angle near to 70°.
Therefore, 70° and ∠B are in a straight line.
So, **B** + 70° = 180°
∠B = 180° − 70° = 110°
Therefore, the value of ∠B is 110°.
Now, ∠B − ∠A =110°−70° = 40°. ; Therefore, ∠B − ∠A = 40°.
18. Correct option: (A)
The alternate exterior angles in the given figure is observed.
Since, we know that alternate exterior angles are equal.
$110° = 2y^0 - 50°$
$110° + 50° = 2y^0$
$160° = 2y^0$
$y^0 = 80°$; Therefore, the value of y is 80°.

19. Correct option: (B)
It is given that, **l ∥ m** and **p ∥ q** t is transversal.
We know that corresponding angles are equal.
Since A and x are corresponding angles, then A= x.
x is the corresponding angle of the angle near to 110°.
Therefore, 110° and x are in a straight line.
So, **y** + 110° = 180°
y = 180° − 110° = 70° ; Therefore, the value of y is 70°.

20. Correct option: (A)
Corresponding angles are the angles which are formed in matching corners or corresponding corners with the transversal when two parallel lines and the transversal.
Also, if a transversal intersects two parallel lines, then each pair of corresponding angles is equal.
Therefore, the pairs of corresponding angles are ∠b = ∠f, ∠d = ∠h, ∠a = ∠e, ∠c = ∠g.
Therefore, the corresponding angles are b, f.

LEVEL -II

1. Correct option: (C)
In the above figure, we can see that a transversal line intersects the parallel lines 'l' and 'm'.
The angle labelled ∠e and the angle with value ∠a = 65° form a corresponding angle.
Since, ∠a and ∠e, are on the left-hand side of the transversal in the corresponding position.
Therefore, ∠a = ∠e = 65°.

2. Correct option: (D)
Angle 6 is an obtuse angle because it is more than 90 degrees while it is less than 180 degrees.
Therefore, the segment which has obtuse angle is 6.

3. Correct option: (C)
Let two angles be 3x and 5x
Sum of supplementary angles is 180°

$$\therefore 3x + 5x = 180°$$
$$8x = 180°$$
$$x = \frac{180^0}{8} = 22.5° = 22\frac{1}{2}°$$

So, one angle = $3x = 3 \times 22.5° = 67.5° = 67\frac{1}{2}°$
Another angle = $5x = 5 \times 22.5° = 112.5° = 112\frac{1}{2}°$
Larger of two angle is $112\frac{1}{2}°$.

4. Correct option: (C)
The figure formed by two rays with the same initial points is called an angle.
An angle is formed by two rays or two line segments that begin at the same point or share the same endpoint.

5. Correct option: (B)
From the figure
∠AOB + ∠AOC = ∠BOC
∠BOC + 90° = 134°
∠BOC = 134° − 90° = 44°
Therefore, ∠BOC = 44°.

6. Correct option: (D)

From the figure

$\angle AOB - \angle BOC = \angle AOC$

Since $\angle AOB$ is a straight angle which is equal to 180°, $\angle BOC$ is an obtuse angle whose value is given as 156°.

Subtract $\angle BOC$ from $\angle AOB$ to get $\angle AOC$.

$\angle AOC = 180° - 156° = 24°$; Therefore, $\angle AOC = 180° - 156°$.

7. Correct option: (D)

Vertical angles are pairs of opposite angles made by two intersecting lines.

In the above figure,

We have the following pairs of vertical angles:

· ∠a and ∠d

· ∠b and ∠c

· ∠e and ∠h

· ∠f and ∠g

Hence, we can conclude that 4 pairs of vertical angles are formed when a transversal cuts two parallel lines.

8. Correct option: (D)

Transversal is defined as a line that intersects two lines in the same plane at two distinct points.

Observe from the figure that the line f intersects lines a and e at two distinct points.

Observe from the figure that the line a intersects lines b and f at two distinct points.

Therefore, line f and line a is the transversal.

9. Correct option: (C)

Let AB be a line parallel to PQ and TS and meeting at R bisecting b.

Let each bisected angle be x.

Then, $a + x = 180°$ (consecutive angles) .....(1)

$x = 180° - a$

also, $c + x = 180°$(consecutive angles).....(2)

$x = 180° - c$

On adding equation (1) and (2)--

$x + x = 180° - a + 180° - c$

$b = 360° - a - c \ldots\ldots\ldots (2x = b)$

$a + b + c = 360°$.

Therefore, $a + b + c = 360°$.

10. Correct option: (A)

A reflex angle is an angle that measures more than 180 degrees and less than 360 degrees.

Therefore, 356° is reflex angle.

11. Correct option: (C, D)

An obtuse angle is a type of angle that is always larger than 90° but less than 180°.

In other words, it lies between 90° and 180°.

120° and 140° lies between 90° and 180°.

Therefore, 120° and 140° is an obtuse angle.

12. Correct option: (A, C, D)

From the given options, a triangle, square, and rhombus are all formed by line segments.

A) Triangle: A triangle is a polygon with three sides, and each side is a line segment connecting two vertices.

C) Square: A square is a quadrilateral with four equal sides, and each side is a line segment connecting two vertices.

D) Rhombus: A rhombus is also a quadrilateral with four equal sides, and each side is a line segment connecting two vertices.

13. Correct option: (A, B)

An angle which is measuring less than 90 degrees is called an acute angle.
This angle is smaller than the right angle (which is equal to 90 degrees).



14. Correct option: (A)

All the parts or points which lie inside the arms or inside the angle are called points in the interior of the angle.
Points that lie in the interior: a, c, j.

15. Correct option: (C).

The points that lie on ∠POQ boundary: g, i, l.

16. Correct option: (D).

The point where the rays meet is called the vertex.
In the given figure PO and OQ are two rays meeting at a common end point Y.
The point O where the angle is formed, is the vertex of the angle.

17. Correct Matching is: $a \rightarrow S, b \rightarrow R, c \rightarrow T, d \rightarrow P$.

a) A reflex angle is an angle that is more than 180 degrees and less than 360 degrees.
Therefore, 220 degree is a Reflex angle.

b) An angle whose measure is greater than a right angle but less than a straight angle is called an obtuse angle.
Therefore, 110 degree is an Obtuse angle.

c) An angle which is measuring less than 90 degrees is called an acute angle.
Therefore, 60 degrees is called an acute angle.

d) A straight angle is an angle equal to 180 degrees. It is called straight because it appears as a straight line.
Therefore, 180 degrees is called a straight angle.

18. Correct Answer is: 3

A triangle is a polygon made up of three line segments.
A triangle has three vertices.
A triangle has three angles.

19. Correct Answer is: 6.

A hexagon is a two-dimensional shape with six sides and six angles.
That also means that hexagons are polygons.
Hexagon is a polygon, a closed figure bounded by sides that are joined together. The number of sides is 6.

20. Correct Answer is: 4.

Each diagonal divides the rhombus into two congruent triangles.
Since the diagonals of a rhombus are perpendicular bisectors of each other, these triangles are also congruent right triangles.
When the diagonals of a rhombus intersect at the vertex, they form four congruent right triangles. Each of these triangles has a right angle at the vertex.
Therefore, at the point of intersection of the diagonals in a rhombus, there can be a maximum of four right angles.

STUDENT PRACTICE QUESTIONS

1. Correct option: (D)
An angle whose measure is greater than a right angle but less than a straight angle is called an obtuse angle.
We know 102° is greater than 90°.
So, it is an obtuse angle as angles greater than 90° and less than 180° are referred to as obtuse angles.

2. Correct option: (B)
A reflex angle is an angle that is more than 180 degrees and less than 360 degrees.
We know, reflex angle must be between 180° and 360°.
It can't be equal to 180° or 360° as these are straight and complete angle, respectively.
Here, only 180° is equal to 180° which is not greater than 180° and less than 360°.
Thus, 180° is not a reflex angle.

3. Correct option: (B)
A reflex angle is an angle that is more than 180 degrees and less than 360 degrees.
We know, reflex angle must be between 180° and 360°.
It can't be equal to 180° or 360° as these are straight and complete angle, respectively.



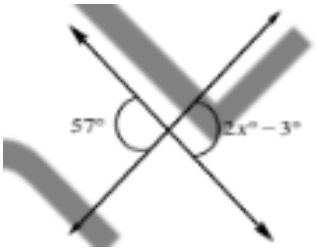
Therefore, it is represented by

4. Correct option: (A).
When the sum of two angles is equal to 90 degrees, they are called complementary angles.
For example, 30 degrees and 60 degrees are complementary angles.

5. Correct option: (B)
Find angles using the sum of supplementary angles.
Let, **x, x** be two angles.
Now, $x + x = 180°$ [Sum of supplementary angles =180∘]
$\Rightarrow 2x = 180°$
$x = 90°$; Therefore, the angle is 90°.

6. Correct option: (B)
The sum of angles of a linear pair is always equal to 180°.
Such angles are also known as supplementary angles.
The adjacent angles are the angles which have a common vertex.

7. Correct option: (B)



The angle labelled $(2x - 3)°$ and the angle with value 57° are vertically opposite to one another and they are equal to one another.
So, $(2x - 3)° = 57°$
$2x = 57° + 3°$; $2x = 60°$
$x = 30°$; Therefore, the value of $x = 30°$.

8. Correct option: (D)
It is given that, **l ∥ m** and t is transversal.
We know that corresponding angles are equal.
y is the corresponding angle of the angle near to 110°.
Therefore, 110° and y are in a straight line.
So, $\mathbf{y} + 110° = 180°$
$y = 180° - 110° = 70°$; Therefore, the value of y is 70°.
9. Correct option: (B)
The co-interior angle, also known as the consecutive interior angle, if the transversal intersects the two parallel lines, each pair of co-interior angles sums up to 180 degrees (supplementary angles).
From the figure, $x^0 - 40^0, 3x^0 - 20^0$ form a linear pair.
Therefore, their sum must be supplementary.
$x^0 - 40^0 + 3x^0 - 20^0 = 180^0$.
$4x^0 - 60^0 = 180^0$.
$4x^0 = 180^0 + 60^0$.
$4x^0 = 240^0$
$x^0 = 60^0$. ; Therefore, the value is $x^0 = 60^0$.
10. Correct option: (B)
A pair of angles formed on a straight line are said to be a linear pair. The sum of angles of a linear pair is always 180 degrees.
$\angle a + \angle b = 180°$; Therefore, the sum is 180°.
11. Correct option: (A, C)
In a traditional analog clock, there are two times where the angle between the hour and minute hands is exactly 90 degrees. These times are:
A) 3'0 clock
C) 9'0 clock
At 3 o'clock and 9 o'clock, the hour hand and the minute hand form a right angle, which measures 90 degrees.
Therefore, at 3 o'clock and 9 o'clock which measures 90 degrees.
12. Correct option: (A, D)
A straight angle is defined as an angle that is equal to 180 degrees.

- The reason it is called a straight angle is because it appears as a straight line.
- It is exactly half of a revolution, i.e., it is half of a complete angle.
- It can also be formed by joining two right angles, i.e., 90° + 90° = 180°.

Therefore, straight angle means two right angles and 180°.
13. Correct option: (C).
Another name of ∠4 from the given figure can be written ∠CDA or ∠ADC.
14. Correct option: (A).
Another name of ∠2 from the given figure can be written ∠BAC or ∠CAB.
15. Correct Matching is: $\mathbf{a \rightarrow R, b \rightarrow P, c \rightarrow S, d \rightarrow Q}$.
a) An equilateral triangle is a triangle in which all three sides have the same length.
Therefore, equilateral triangle is all sides are equal.
b) An isosceles triangle is a triangle that has any two sides equal in length and angles opposite to equal sides are equal in measure.
Therefore, isosceles triangle is two sides are equal.
c) A scalene triangle is a triangle in which all three sides are in different lengths, and all three angles are of different measures.

Therefore, scalene triangle is no two sides are equal.

d) The hypotenuse is always the longest side in a right triangle because it is opposite of the largest angle, the ninety degree angle.

Therefore, Largest side in right triangle is Hypotenuse.

16. Correct Answer is: 1

Three or more points in a plane are said to be collinear if they all lie on the same line and this line is called the line of collinearity for the given points.

We can draw only one line passing through three collinear points.

17. Correct Answer is: 4.

Complete angle = 360°.

Right angle =90°.

So, four right angles are required to form a complete angle. [$90° \times 4 = 360°$]

18. Correct Answer is: 5.

In the given figure, the line segments are AB, BC, AC, AD, and CD.

Therefore, the number of line segments drawn in the figure is 5.

19. Correct Answer is: 1.

A triangle can have at most one right angle, or an angle that has a measure of 90°. When a triangle has a right angle, it is called a right triangle.

In general, all triangles are either a right triangle, or they are not a right triangle.

Therefore, the number of possible right angles in a triangle is 1.

20. Correct Answer is: 0

Square is a quadrilateral with four equal sides and angles.

It's also a regular quadrilateral as both its sides and angles are equal.

∴ A square has four right angles. But, zero straight angles.

4. Ratio and Proportion

Solutions

LEVEL-I

1. Correct option: (D) 10/11

Simplest form of 500:650 is

$$\frac{500}{650} = \frac{50}{65} = \frac{10}{11}$$

2. Correct option: (D) 520

Rs. 1300 is shared between Venkat and Kiran in ratio 2:3

Venkat's share is 2

Total is $2 + 3 = 5$

Then $\frac{2}{5} \times 1300 = 520$

3. Correct option: (A) 200

Let the number be a and b

Two numbers are in ratio 3:8, then numbers are 3x and 8x

Sum of two number is 440

$$a + b = 440$$
$$3x + 8x = 440$$
$$11x = 440$$
$$x = 40$$

a is; $a = 3(x) = 3(40) = 120$

b is; $b = 8x = 8(40) = 320$

difference is; $320 - 120 = 200$

4. Correct option: (B)

Ratio of led and copper is 3:7

Led weight is 60.9gm

So, value of 3 ratio of led is $3x = 60.9$

$$x = 20.3$$

Weight of copper is $7(20.3) = 142.1\text{gm}$

5. Correct option: (C) 45

Side of triangle are in ratio of 2:3:5

Perimeter = 90cm

$$P = 2x + 3x + 5x$$
$$90 = 10x$$
$$x = 9$$

Sides are

$2x = 2 \times 9 = 18$

$3x = 3 \times 9 = 27$

$5x = 5 \times 9 = 45$

Longest side is 45cm

6. Correct option: (C) 5/12

a) $\frac{7}{6} = 1.166$

b) $\frac{4}{9} = 0.444$

c) $\frac{5}{12} = 0.416$

d) $\frac{3}{4} = 0.75$

$\frac{5}{12}$ is smallest ratio

7. Correct option: (B) 6:3:2

Given, $a = 2b = 3c$

Take LCM, $\frac{a}{6} = \frac{b}{3} = \frac{c}{2}$

Ratio is 6:3:2

8. Correct option: (D) 15:10:6

Given

$$\frac{1}{p} : \frac{1}{q} : \frac{1}{r} = 2 : 3 : 5$$

So

$\frac{1}{p} = 2$, $p = \frac{1}{2}$

$\frac{1}{q} = 3$, $q = \frac{1}{3}$

$\frac{1}{r} = 5$, $r = \frac{1}{5}$

$$p : q : r = \frac{1}{2} : \frac{1}{3} : \frac{1}{5}$$

$$p : q : r = 15 : 10 : 6$$

9. Correct option: (B) 252

Ratio of male and female is 2:3

There are 168 males

$$2x = 168$$

$$x = 84$$

Number of females

$$3x = 3 \times 84 = 252$$

10. Correct option: (D) 408

Rs. 1428 is divided between Haneesh and Hemanth in ratio 2:5

Total ratio is 2 + 5 = 7

Haneesh shares is

$$\frac{2}{7} \times 1428 = 408$$

11. Correct option: (B) 9

Given

12, x, 8, 6 are in ratio

Then

$$\frac{12}{8} = \frac{x}{6}$$

$$72 = 8x$$

$$x = \frac{72}{8}$$

$$x = 9$$

12. Correct option: (A) 2

Let the number be x, which is to be added to numbers 16, 2, 25 and 4 to be in proportion

$$\frac{16 + x}{25 + x} = \frac{2 + x}{4 + x}$$

$$64 + 16x + 4x + x^2 = 50 + 25x + 2x + x^2$$

$$20x + 64 = 27x + 50$$

$$64 - 50 = 27x - 20x$$

$$7x = 14$$

$$x = 2$$

13. Correct option: (D) 256/3

$$\frac{16}{18} = \frac{x}{96}$$
$$x = \frac{16 \times 96}{18}$$
$$x = \frac{8 \times 96}{9}$$
$$x = \frac{8 \times 32}{3}$$
$$x = \frac{256}{3}$$

14. Correct option: (C) 21,875

Income and saving ratio are 7:4
Savings of one year = Rs. 12500
Let income be x

$$\frac{x}{12500} = \frac{7}{4}$$
$$4x = 87500$$
$$x = 21875$$

15. Correct option: (A) 84

$$\frac{72}{186} = \frac{x}{217}$$
$$x = \frac{72 \times 217}{186}$$
$$x = \frac{24 \times 217}{62}$$
$$x = \frac{12 \times 217}{31}$$
$$x = 12 \times 7$$
$$x = 84$$

16. Correct option: (C) 14

Given, Rahul and Rohit age ratio is 7:5
Sum of their age = 84

$$7x + 5x = 84$$
$$12x = 84$$
$$x = 7$$

Rahul age= $7 \times 7 = 49$yrs
Rohit age= $5 \times 7 = 35$yrs
Difference in their age is $49 - 35 = 14$

17. Correct option: (B) 5

Let the number to be subtracted be x

$$\frac{35 - x}{40 - x} = \frac{6}{7}$$
$$245 - 7x = 240 - 6x$$
$$245 - 240 = 7x - 6x$$
$$x = 5$$

18. Correct option: (D) 28

Given that 7, 14, x is in continued fraction $7 : 14 : : 14 : x$

$$\frac{7}{14} = \frac{14}{x}$$
$$\frac{1}{2} = \frac{14}{x}$$
$$x = 28$$

19. Correct option: (C) 12

Mean proportion between 4 and 36

$$= \sqrt{4 \times 36}$$
$$= \sqrt{144}$$
$$= 12$$

20. Correct option: (B) 12

8, x, 18 are in continued fraction

Then

$$8 : x : : x : 18$$
$$\frac{8}{x} = \frac{x}{18}$$
$$x^2 = 144$$
$$x = \sqrt{144}$$
$$x = 12$$

LEVEL-II

1. Correct option: (C) 40

Value of x is 9 times the value of y, $x = 9y$

Mean proportion of x and y is 12

$$\sqrt{xy} = 12$$
$$\sqrt{(9y)y} = 12$$
$$3y = 12$$
$$y = 4$$

x is; $x = 9 \times 4 = 36$

Now $x + y = 36 + 4 = 40$

2. Correct option: (B) 42

Shiva and Rohan age ratio is 6:7

Difference of age is 7; $7x - 6x = 7$

$$x = 7$$

Shiva's age is $= 6 \times 7 = 42$

3. Correct option: (C) 350

Three numbers are in ratio 4:3:5

Sum of numbers = 840

$$4x + 3x + 5x = 840$$
$$12x = 840$$
$$x = 70$$

$4x = 4 \times 70 = 280$

$3x = 3 \times 70 = 210$

$5x = 5 \times 70 = 350$

Highest number is 350

4. Correct option: (D) 16

Total students = 80

2 out of every 10 students fails

Ratio is $\frac{2}{10} = \frac{1}{5}$

Number of pass members $\frac{1}{5} \times 80 = 16$

5. Correct option: (A) 5

Given $\frac{x}{y} = \frac{7}{5}$

Then $\frac{5x+4y}{3x-2y}$

Divide by y

$$\frac{5\left(\frac{x}{y}\right)+4}{3\left(\frac{x}{y}\right)-2}$$

$$= \frac{5\left(\frac{7}{5}\right)+4}{3\left(\frac{7}{5}\right)-2}$$

$$= \frac{7+4}{\frac{21}{5}-2}$$

$$= \frac{11}{\left(\frac{21-10}{5}\right)}$$

$$= \frac{55}{11}$$

$$= 5$$

6. Correct option: (A) 20

Cost of 12 pencils = Rs. 24

Cost of one pencil; $\frac{24}{12} = 2$ Rs

Cost of 10 pencils; $2 \times 10 = 20$ Rs

7. Correct option: (D) 1/4

In a day, there are 24 hours

Ratio is

$$\frac{6 \text{ hours}}{24 \text{ hours}} = \frac{1}{4}$$

8. Correct option: (A) Rs. 416

39 packets of 12 pencils costs Rs. 374.40

Cost of one packet pencil

$$= \frac{374.40}{39 \times 12}$$

$$= \frac{3744}{39 \times 12 \times 10}$$

$$= \frac{1248}{13 \times 12 \times 10}$$

$$= \frac{104}{13 \times 10}$$

$$= \frac{52}{13 \times 5}$$

$$= \frac{52}{65}$$

$$= 0.8$$

Cost of 52 packets of 10 pencils

$$0.8 \times 52 \times 10$$

$$= 416 \text{Rs.}$$

9. Correct option: (A) 2:1

Given; 10% of $m = 20\%$ of n

$$\frac{10}{100} \times m = \frac{20}{100} \times n$$
$$\frac{m}{10} = \frac{n}{5}$$
$$\frac{m}{n} = \frac{10}{5}$$
$$\frac{m}{n} = \frac{2}{1}$$

10. Correct option: (D) 6

AB= 14cm

Point X divides in ratio 3:4

Length of AX $\frac{3}{7} \times 14 = 6\text{cm}$

11. Correct options: (A), (B) and (C)

a) $\frac{8}{18} = \frac{\left(\frac{8}{2}\right)}{\left(\frac{18}{2}\right)} = \frac{4}{9}$

b) $\frac{20}{45} = \frac{\left(\frac{20}{5}\right)}{\left(\frac{45}{5}\right)} = \frac{4}{9}$

c) $\frac{12}{27} = \frac{\left(\frac{12}{3}\right)}{\left(\frac{27}{3}\right)} = \frac{4}{9}$

d) $\frac{12}{20} = \frac{\left(\frac{12}{4}\right)}{\left(\frac{20}{4}\right)} = \frac{3}{5}$

Equivalent ratios of 4:9 are 8:18, 20:45 and 12:27

12. Correct option: (A) 18/19

Given; $\frac{a}{b} = \frac{3}{4}$

Then ; $\frac{2a+3b}{5a+b} = \frac{\left(\frac{2a+3b}{b}\right)}{\left(\frac{5a+b}{b}\right)}$

$$= \frac{\frac{2a}{b} + 3}{\frac{5a}{b} + 1}$$

$$= \frac{2\left(\frac{3}{4}\right)+3}{5\left(\frac{3}{4}\right)+1} = \frac{\frac{6}{4}+3}{\frac{15}{4}+1}$$

$$= \frac{\left(\frac{6+12}{4}\right)}{\left(\frac{15+4}{4}\right)}$$

$$= \frac{18}{19}$$

13. Correct options: (A) and (B)

a) If 3, 5, 12, 20 are in proportion $\frac{3}{12} = \frac{5}{20}$

$\frac{3}{12} = \frac{1}{4}$ and $\frac{5}{20} = \frac{1}{4}$

Hence are proportional.

b) If 3, 6, 15, 30 are in proportional $\frac{3}{15} = \frac{6}{30}$

$\frac{3}{15} = \frac{1}{5}$ and $\frac{6}{30} = \frac{1}{5}$

Hence are proportional

c) If 3, 9, 5, 15 are proportional $\frac{3}{5} = \frac{5}{15}$

Which are not equal in ratio
Hence are not proportional

d) If 5, 9, 6, 11 are proportional $\frac{5}{6} = \frac{9}{11}$

Which are not equal in ratio
Hence are not proportional

14. Correct option: (B) 10

Given 25, x, 4 are in continued proportion.

$$25 \times 4 = (x) \times (x)$$
$$x^2 = 100$$
$$x = \sqrt{100}$$
$$x = 10$$

15. Correct option: (C) 14

Mean proportion between 4, 49 $\sqrt{4 \times 49}$

$$= 2 \times 7$$
$$= 14$$

16. Correct option: (D) 45

Given mean proportion of x and y is 18

x is four times value of y, x = 4y $\sqrt{xy} = 18$

$$\sqrt{(4y)y} = 18$$
$$\sqrt{4y^2} = 18$$
$$2y = 18$$
$$y = 9$$

x is , $x = 4 \times 9 = 36$

x + y $= 36 + 9 = 45$

17. Correct matching is: $(\mathbf{a}) \rightarrow (\mathbf{S}); (\mathbf{b}) \rightarrow (\mathbf{R}); (\mathbf{c}) \rightarrow (\mathbf{P}); (\mathbf{d}) \rightarrow (\mathbf{Q})$

a) Simplest form of 16:36 is; $\frac{16}{36} = \frac{\left(\frac{16}{4}\right)}{\left(\frac{36}{4}\right)} = \frac{4}{9}$

b) $\frac{12}{20} = \frac{\left(\frac{12}{4}\right)}{\left(\frac{20}{4}\right)} = \frac{3}{5}$

c) Given; $\frac{p}{q} = \frac{2}{5}$

Then

$$\frac{25p + 14q}{5p + 7q} = \frac{\left(\frac{25p + 14q}{q}\right)}{\left(\frac{5p + 7q}{q}\right)}$$
$$= \frac{25\left(\frac{p}{q}\right) + 14}{5\left(\frac{p}{q}\right) + 7}$$
$$= \frac{25\left(\frac{2}{5}\right) + 14}{5\left(\frac{2}{5}\right) + 7}$$
$$= \frac{10 + 14}{2 + 7}$$
$$= \frac{24}{9}$$
$$= \frac{8}{3}$$

d) Given $\frac{a}{b} = \frac{3}{5}$

$$\frac{5a}{3b} = \frac{5\left(\frac{a}{b}\right)}{3\left(\frac{b}{b}\right)}$$
$$= \frac{5\left(\frac{3}{5}\right)}{3}$$
$$= \frac{3}{3}$$
$$= \frac{1}{1}$$

18. Correct Answer is: 2

Given

$$\frac{14}{21} = \frac{x}{3}$$
$$x = \frac{14 \times 3}{21}$$
$$x = \frac{14}{7}$$
$$x = 2$$

19. Correct Answer is: 30

15, x, 60 are in continued fraction

$$\frac{15}{x} = \frac{x}{60}$$
$$x^2 = 15 \times 60$$
$$x = \sqrt{900}$$
$$x = 30$$

20. Correct Answer is: 4

Consequent in a ratio is second term

Hence, 4 is consequent of 7:4

STUDENT PRACTICE QUESTIONS

1. Correct option: (B) 4:7

Venkat and Thirupathi take Rs. 600 and Rs. 1050 as pocket expenses

Ratio of their expenses are

$$\frac{600}{1050}$$

Simplest form

$$\frac{600}{1050} = \frac{60}{105} = \frac{12}{21} = \frac{4}{7}$$

2. Correct option: (D) 72

Bag contains Rs 216 in form of 1 rupee, 50 paise and 10 paise

Let the number of coins be x

One rupee ratio is 2x

50 paise ratio is 3x

10 paise ratio is 4x

Then

$$2x + 3x + 4x = 216$$
$$9x = 216$$
$$x = \frac{216}{9}$$
$$x = 24$$

Number of 50 paise coins are $3 \times 24 = 72$

3. Correct option: (D) 6/10

a) $\frac{6}{15} = 0.4$

b) $\frac{9}{21} = 0.428$

c) $\frac{4}{16} = 0.25$

d) $\frac{6}{10} = 0.6$

$\frac{6}{10}$ is biggest ratio

4. Correct option: (C) 27:64

Given

$\frac{x}{y} = \frac{9}{16}$ and $\frac{y}{z} = \frac{3}{4}$

Then $y = \frac{3z}{4}$

Put this value of y in $\frac{x}{y}$

$$\frac{x}{y} = \frac{9}{16}$$
$$\frac{x}{\left(\frac{3z}{4}\right)} = \frac{9}{16}$$
$$\frac{4x}{3z} = \frac{9}{16}$$
$$\frac{x}{z} = \frac{27}{64}$$

5. Correct option: (C) 9

Let the number added be x

$$\frac{3+x}{5+x} = \frac{6}{7}$$
$$7(3+x) = 6(5+x)$$
$$21 + 7x = 30 + 6x$$
$$7x - 6x = 30 - 21$$
$$x = 9$$

6. Correct option: (C) 41

164, 8, x, 2 are in proportion;

$$\frac{164}{x} = \frac{8}{2}$$
$$8x = 328$$
$$x = \frac{328}{8}$$
$$x = 41$$

7. Correct option: (B) 15

$$\frac{9}{x} = \frac{12}{20}$$
$$12x = 180$$
$$x = 15$$

8. Correct option: (D)

Let the fourth term of proportion be x

$$\frac{25}{19} = \frac{50}{x}$$
$$x = \frac{50 \times 19}{25}$$
$$x = 2 \times 19$$
$$x = 38$$

9. Correct option: (B) 3344

Ratio of boys and girls is 8:3

Number of girls = 912

Let boys' number be x

$$\frac{x}{912} = \frac{8}{3}$$

$$x = \frac{912 \times 8}{3}$$

$$x = 304 \times 8$$

$$x = 2432$$

Total strength of school is $= 2432 + 912 = 3344$

10. Correct option: (A) 243

Mean proportion between 3 and x is 27

$$\sqrt{3 \times (x)} = 27$$

Squaring both sides

$$3x = (27)^2$$

$$x = \frac{27 \times 27}{3}$$

$$x = 9 \times 27$$

$$x = 243$$

11. Correct options: (A), (C) and (D)

$\frac{9}{12}$ can be written as $\frac{3}{4}$

a) $\frac{27}{36} = \frac{\left(\frac{27}{9}\right)}{\left(\frac{36}{9}\right)} = \frac{3}{4}$

b) $\frac{18}{48} = \frac{\left(\frac{18}{6}\right)}{\left(\frac{48}{6}\right)} = \frac{3}{8}$

c) $\frac{3}{4}$

d) $\frac{18}{24} = \frac{\left(\frac{18}{6}\right)}{\left(\frac{24}{6}\right)} = \frac{3}{4}$

12. Correct options: (B), (C) and (D)

a) $\frac{6}{2} < \frac{2}{5}$

cross- multiply; $30 < 18$, which is incorrect.

b) $\frac{5}{12} > \frac{3}{8}$

cross-multiply; $40 > 36$ Which is correct.

c) $\frac{7}{6} > \frac{4}{9}$

cross-multiply, $63 > 24$

Which is correct.

d) $\frac{21}{9} > \frac{23}{12}$

cross-multiply, $252 > 207$

Which is correct

13. Correct option: (D) $\sqrt{21}$

$$7 : x = x : 3$$

$$7 \times 3 = (x) \times (x)$$

$$x^2 = 21$$

$$x = \sqrt{21}$$

14. Correct option: (A) 3:7

Inverse ratio of 7:3 is $\frac{3}{7}$

15. Correct matching is: $(\mathbf{a}) \rightarrow (\mathbf{R}); (\mathbf{b}) \rightarrow (\mathbf{P}); (\mathbf{c}) \rightarrow (\mathbf{S}); (\mathbf{d}) \rightarrow (\mathbf{T})$

a) Given $\frac{4}{3} = \frac{x}{9}$

$$36 = 3x$$

$$x = 12$$

b) Given $\frac{8}{24} = \frac{x}{12}$

$$\frac{1}{3} = \frac{x}{12}$$

$$3x = 12$$

$$x = 4$$

c) Given $\frac{4}{x} = \frac{20}{35}$

$$\frac{4}{x} = \frac{4}{7}$$

$$4x = 7 \times 4$$

$$x = 7$$

d) Given $\frac{12}{20} = \frac{x}{5}$

$$\frac{3}{5} = \frac{x}{5}$$

$$x = 3$$

16. Correct Answer is: 5

$$\frac{45}{108} = \frac{x}{12}$$

$$x = \frac{45 \times 12}{108}$$

$$x = \frac{45}{9}$$

$$x = 5$$

17. Correct Answer is: Rs. 2

Cost of dozen bananas is Rs. 24

In a dozen there are 12 bananas

So, cost of a banana is $\frac{24}{12} = 2$

18. Correct Answer is: Rs. 8

5kg potato costs Rs 20

1kg potato costs $\frac{20}{5} = 4$ Rs

Then 2kg potato cost $2 \times 4 = 8$ Rs

19. Correct Answer is: 8

Antecedent in a ratio is first term

Hence, 8 is antecedent of 8:5

20. Correct Answer is: 6

Half dozen is 6 items.

5. Data Handling
Solutions

LEVEL-I

1. Correct option: (C) 10

Given; Tally chart of favorite seasons of people.

Season	**Tally Marks**
Winter	5
Spring	2
Summer	6
Fall	4

Number of people who choose to fall or summer are, $= 6 + 4 = 10$

2. Correct option: (C) 4

Given

Average of a, a+2, a+4, a+12, a+7 is 9

$$\frac{a + a + 2 + a + 4 + a + 12 + a + 7}{5} = 9$$

$$5a + 25 = 45$$

$$5a = 20$$

$$a = 4$$

3. Correct option: (D) 360°

In a pie chart, the sum of angles of all its components is 360°

4. Correct option: (C) value of component

The height of a bar is proportional to the value of the component, in a bar graph.

5. Correct option: (B) 4326

Number of candidates appeared in 8 years are given

12464, 14731, 13213, 14219, 15613, 11287 and 14162

The range of data is calculated by finding the difference between the maximum and minimum value.

Maximum candidate number = 15613

Minimum candidate number =11287

Range of the number of candidates appeared is $= 15613 - 11287 = 4326$

6. Correct option: (A) 40

Given

Size of the bat	10	15	20	30	40	50
Cost	80	120	150	250	400	600

The range of data is calculated by finding the difference between the maximum and minimum value.

Maximum value of size = 50

Minimum value of size = 10

Range of bat size $= 50 - 10 = 40$

7. Correct option: (B) 120

Given Central angle = 72°

Value of component = 24

Now sector angle $= \frac{\text{value of component}}{\text{total value of all compomponents}} \times 360°$

Let total value of all components be x

$$72 = \frac{24}{x} \times 360$$
$$x = \frac{24}{72} \times 360$$
$$x = \frac{1}{3} \times 360$$
$$x = 120$$

8. Correct option: (C) 7

Given, 5 cm height represents 40 units value in a bar.

Then 1cm $= \frac{40}{5} = 8$ units

Now to find 56 units value in cm $= \frac{56}{7} = 7$cm

9. Correct option: (B) 13

Given, Range of 10 observations = 15

Highest score = 28

Let lowest score be x

Now, The range of data is calculated by finding the difference between the maximum and minimum value;

$$15 = 28 - x$$
$$x = 28 - 15$$
$$x = 13$$

10. Correct option: (D) 20

Given Central angle = 72°

Let x be the value of component.

Value of all components in percentage is 100

Then,

$$\text{sector angle} = \frac{\text{value of component}}{\text{total value of all compomponents}} \times 360°$$
$$72 = \frac{x}{100} \times 360$$
$$x = \frac{72}{360} \times 100$$
$$= 20\%$$

11. Correct option: (C) 250

Given, Bae graph represents 1cm = 2m

Value of 875 units is 1.75cm $1\text{cm} = \frac{875}{1.75}$

$1\text{cm} = 500\text{units}$

Now

Equating value of 1 cm, $2\text{m} = 500$

$1\text{m} = 250$

12. Correct option: (B) 11250

Given, A picture represents 375 units.

30 pictures are represented by $= 375 \times 30 = 11250$ units

13. Correct option: (C) 5

Tallies are marked in a bunch of five.

14. Correct option: (B) 9

Student	**Number of Oranges**
Ben	9
Tim	12
Pam	18
John	4

Now the difference between number of oranges purchased by Pam and Ben is
$= 18 - 9 = 9$

15. Correct option: (C)

Given data of motorcycles parked in a week.

Day	**Number of Motorcycles parked**
Sun	23
Mon	13
Tue	10
Wed	12
Thu	16
Fri	17
Sat	20

Total number of motorcycles parked in a week$= 23 + 13 + 10 + 12 + 16 + 17 + 20 = 111$

16. Correct option: (B) Observation

Each numerical figure in data is called an Observation.

17. Correct option: (B) Fantasy

Given a survey.

Book/Magazine	**Number of votes**
Fashion Magazine	4
Drama	3
Fantasy	6
Sports fiction	5
Biographies	2
Historical Fiction	4

Maximum number of people prefer to read Fantasy books.

18. Correct option: (B) 48

Given, Number of bananas eaten by a monkey in 4 days Total = 12
Then multiplied by 4; $12 \times 4 = 48$

19. Correct option: (D) 198

Given data of flowers sold.

Flowers	**Number of flowers**
Rose	63
Tulip	45
Orchid	54
Lily	36

Total numbers of flowers sold $= 63 + 45 + 54 + 36 = 198$

20. Correct option: (D) 2006

Given a data of number of trees planted during 4 consecutive years.
2003 = 400 trees
2004 = 550 trees
2005 = 500 trees
2006 = 600 trees
So, the maximum number of trees are planted in the year 2006.

LEVEL – II

1. Correct option: (B) Class Mark

The mid value of class interval is called class mark.

2. Correct option: (C) Rs. 0.48

Given, Price of fuel over different states

Rs. 1.79, Rs. 1.61, Rs. 1.96, Rs. 1.84, Rs. 1.75, Rs. 2.09

The range of data is calculated by finding the difference between the maximum and minimum value.

Maximum value = Rs. 2.09

Minimum value = Rs. 1.61

So, range is= $2.09 - 1.61 = 0.48$

3. Correct option: (D) 3

In observation 11, 4, 12, 9, 8, 11, 12, 4, 12, 8, 8

Frequency of 12 is 3

4. Correct option: (D) 80

Given Number of videos rented in 4 months.

Month	**Videos rented in last 4 months**
January	60
April	40
July	20
October	80

Number of videos rented in October is 80

5. Correct option: (B) 52 units

Given, 1cm = 10 units

5.2cm is $= 5.2 \times 10 = 52$ units

6. Correct option: (D) 26

Given data of favorite activities of students in a class.

Sports = 7

Dance = 5

Painting = 6

Drama = 8

So, total students in a class are $= 7 + 5 + 6 + 8 = 26$

7. Correct option: (A) 10

Given data of favorite sports of students of grade 3.

Badminton = 25 students

Tennis = 30 students

Soccer = 20 students

Baseball = 35

Difference between soccer and tennis students is $= 30 - 20 = 10$

8. Correct option: (B) 60

Given, Library has 300 books.

Novels percentage is 25%

Magazine percentage is 35%

Journals percentage is 20%

Total $= 25 + 35 + 20 = 80\%$

Remaining percentage is $= 100 - 80 = 20\%$ which is of newsletters.

Then number of newsletters is $= \frac{20}{100} \times 300 = 60$

9. Correct option: (A) x+4

Mean $= \frac{\text{number of observation}}{\text{total observations}}$

So, mean of x, x+2, x+4, x+6, x+8

$= \frac{x+x+2+x+4+x+6+x+8}{5} = \frac{5x+20}{5} = x + 4$

10. Correct option: (D) 8

Given, 1cm = 250 populations

Length of rectangle which represents 2000 population is $= \frac{2000}{250} = 8$

11. Correct option: (A) pam

Student	**Number of Oranges**
Ben	9
Tim	12
Pam	18
John	4

Pam purchased more than 15 oranges.

12. Correct option: (D) Village D

Given Number of people died = 3000

Village A died people = 30%

Village B died people = 35%

Village C died people = 20%

Village D died people = 15%

So, village D has victims less than 20%

13. Correct option: (ABCD)

14. Correct option: (C)

15. Correct option: (B) 27

Given a set of observations.

7, 21, 11, 5, 9, 20, 32, 28

The range of data is calculated by finding the difference between the maximum and minimum value.

Maximum value = 32

Minimum value = 5

So, range is$= 32 - 5 = 27$

16. Correct option: (B) 900

Total students enrolled in school from 1990 to 1993 is $= 100 + 250 + 200 + 350 = 900$

17. Correct option: (A) 2:5

Ratio of students enrolled in 1990 to 1991 is $= \frac{100}{250} = \frac{2}{5}$

18. Correct option: (B) 20%

Percentage decrease in enrollment from 1991 to 1992 is $\frac{250-200}{250} \times 100$

$= \frac{50}{250} \times 100 = 20\%$

19. Correct option: (A) 75%

The percentage increase in students enrolled in 1993 from 1992 is $\frac{350-200}{200} \times 100$

$= \frac{150}{100} \times 100 = 75\%$

20. Correct option: (B) 225

Average of students enrolled in school from 1990 to 1993 is $\frac{100+250+200+350}{4} = \frac{900}{4} = 225$

STUDENT PRACTICE QUESTIONS

1. Correct option: (C) 35 and 24

Given; Marks of mathematics and statistics.

Marks in mathematics	25	40	30	35	21	45	23	33	10	29
Marks in Statistics	30	39	23	42	20	40	25	30	18	19

The range of data is calculated by finding the difference between the maximum and minimum value.

Now, first consider mathematics.

Maximum marks in mathematics = 45

Minimum marks in mathematics = 10

So, range of mathematics marks is $= 45 - 10 = 35$

Consider for Statistics

Maximum marks in Statistics = 42

Minimum marks in Statistics = 18

So, range of Statistics marks is $= 42 - 18 = 24$

2. Correct option: (C) 17:20

Given

Age -Group	**Number of people who go to work**	**Number of people who play**
10-13	1	17
13-16	2	23
16-19	6	11
19-21	11	5
22-24	15	3
25-28	20	2

Now the ratio of people who play between 10-13 to number of people who go to work between 25-28 is $= \frac{17}{20}$

3. Correct option: (D) 3

In observation 11, 4, 12, 9, 8, 11, 12, 4, 12, 8, 8

The frequency of 12 is 3

4. Correct option: (B) 4750

Given, one picture represents 1000 units.

4 pictures represent $= 4 \times 1000 = 4000$

$\frac{3}{4}$ of picture represent $\frac{3}{4} \times 1000 = 750$

Total number of units produced are $= 4000 + 750 = 4750$

5. Correct option: (A) 108

Given, Let total value of component be x

One component is 30% the value of total components $= \frac{30x}{100} = \frac{3x}{10}$

$$\text{sector angle} = \frac{\text{value of component}}{\text{total value of all compomponents}} \times 360°$$

$$= \frac{\left(\frac{3x}{10}\right)}{x} \times 360$$

$$= \frac{3}{10} \times 360$$

$= 3 \times 36 = 108°$

6. Correct option: (B) 126

Given, The ratio of salary of person contributed to three welfare societies.

Total salary share = 20

Share of A = 5

Share of B = 8

Then share of C is $= 20 - (5 + 8) = 20 - 13 = 7$

Now $\text{sector angle} = \frac{\text{value of component}}{\text{total value of all compomponents}} \times 360°$

$$= \frac{7}{20} \times 360$$

$= 7 \times 18 = 126°$

7. Correct option: (D) 20%

Given, Sector angle = 72°

Let x be the value of component.

The value of all components in percentage is 100.

Then $\text{sector angle} = \frac{\text{value of component}}{\text{total value of all compomponents}} \times 360°$

$$72 = \frac{x}{100} \times 360$$

$x = \frac{72}{360} \times 100 = 20\%$

8. Correct option: (C) 1/4

Given a data of shoes types and brands

Types of Brands	**Number of children wearing them**
Lace-up shoes	3
Ballerina shoes	4
Sneakers	3
Keds	5

Fraction of children wearing lace-up to rest shoes is $= \frac{3}{4+3+5}$

$$= \frac{3}{12} = \frac{1}{4}$$

9. Correct option: (D) Steel and Coal

Given a of revenue earned by a country.

Oil = \$100 Steel = \$150 Coal = \$200 Gold = \$50

Now, combining steel and coal amount will result to \$350

10. Correct option: (B)

Given a data of number of vehicles passing through road

Total number of vehicles passing through the crossing during day is

$$= 50 + 400 + 250 + 250 + 150 + 100$$

$$= 1200$$

11. Correct option: (C) July 22 – July 29

Greater number of deaths in between two consecutive weeks is July 22 to July 29

12. Correct option: (D) 5

More than 145 deaths are registered in 8 July week, 22 July week, 29 July week, 5 August week and 12 August week

So, there are 5 weeks who registered more than 145 deaths.

13. Correct option: (C) 5

Less than 165 deaths are in 1 July week, 15 July week, 22 July week, 29 July week and 5 August week

So, there are 5 weeks who registered less than 165 deaths.

6. Algebra

Solutions

LEVEL – I

1. Correct option: (C)

The expression for the statement "x multiplied by 6 and added to 13" is:

$$6x + 13$$

Then the correct option is C)

2. Correct option: (A)

The statement "One-fifth of the product of 'x' and 'y' is subtracted from P" can be expressed as:

$$P - \left(\frac{1}{5}\right)xy$$

In this expression:

- 'x' and 'y' represent two variables.
- The product of 'x' and 'y' is calculated as xy.
- One-fifth of that product is $\left(\frac{1}{5}\right)xy$.
- Finally, that value is subtracted from P to obtain the expression.

Then the correct option is A)

3. Correct option: (B)

In the expression $5x^2y + 2x + 7$ the numerical term is the term that does not have any variable (x or y) attached to it.

So, the numerical term in the given expression is: B) 7

4. Correct option: (C)

The false statement is:

C) A literal coefficient is a constant.

A literal coefficient is not always a constant; it always a variable.

5. Correct option: (B)

Given expression:

$$12x + 25y - 25 + 25xy + 7xy^2 - 5x^2y - 5xy + 10x^2y^2 + 6xy$$

Expression after grouping the like terms

$$10x^2y^2 - 5x^2y + 7xy^2 + 26xy + 12x + 25y - 25$$

Then the numerical coefficient of maximum number of like terms in the expression is 26.

Therefore the correct option is B)

6. Correct option: (C)

An expression that contains two unlike terms is called a Binomial.

So, the correct answer is: C) Binomial

7. Correct option: (C)

A Binomial expression consists of two unlike terms. Let's analyze the given options:

A) $x^2 + (2x)x \Rightarrow 3x^2$ - This is a Monomial because it has only one term.

B) $4xy + (2x)(2y) \Rightarrow 8xy$ - This is a Monomial because it has only one term.

C) $2x^2 - 4$ - This is a Binomial because it has two terms ($2x^2$ and -4).

D) $(3x)y - x^2 - 3yx \Rightarrow -x^2$ - This is a Monomial because it has only one term.

So, the correct answer is: C) $2x^2 - 4$

8. Correct option: (B)

A Trinomial expression consists of three unlike terms. Let's analyze the given options:

A) $(2x^2)y - 4x + (2x)(2xy) \Rightarrow -4x$ - This is a Monomial because it has only one term.

B) $x^2y - 3x^2 - (2x)(2y) \Rightarrow$ - This is a Trinomial because it has three unlike terms.

C) $-2xy + xy + y^2 => y^2 - xy$ - This is a Binomial because it has two unlike terms only.
D) $(2x)y^2 - y^2(2xy) + (-y)(2xy) => -y^2(2xy)$ - This is a Monomial because it has only one term.
So, the correct answer is: B) $x^2y - 3x^2 - (2x)(2y)$

9. Correct option: (A)

To find the degree of the polynomial, add up the exponents of each term and select the highest sum.
The degree of the algebraic expression $4x^2 + 3xyz + y^2$ is indeed 3 because the highest power of variable's add up (in this case, the variable 3xyz) is 3.
Then the correct option is A)

10. Correct option: (C)

To find the degree of the polynomial, add up the exponents of each term and select the highest sum.
For the given expression $pq - 6p^2q^2 + q^2p + 9$
$pq: 1 + 1 = 2$
$-6p^2q^2: 2 + 2 = 4$
$q^2p: 2 + 1 = 3$
$9: 0 + 0 = 0$
Therefore the highest degree of the expression is 4.
Then the correct option is C)

11. Correct option: (A)

To find the sum of the expressions
$4xy^2 - 7x^2y, 12x^2y - 6xy^2, -3x^2y + 6xy^2$

$$4xy^2 - 7x^2y + 12x^2y - 6xy^2, +(-3x^2y) + 6xy^2$$

Now, combine like terms by adding the coefficients (the numbers in front of the variables) that have the same variables:

$$(4 - 6 + 6)xy^2 + (12 - 3 - 7)x^2y = 4xy^2 + 2x^2y$$

So, the sum of the expressions is $4xy^2 + 2x^2y$.
Then the correct option is A)

12. Correct option: (B)

To find the sum of the expressions
$6x^2 - 4x, 2x - 3x^2 + 4, 4x^2 - 5x + 3$

$$6x^2 - 4x + 2x - 3x^2 + 4 + 4x^2 - 5x + 3$$

Now, combine like terms by adding the coefficients (the numbers in front of the variables) that have the same variables:

$$(6 - 3 + 4)x^2 + (-4 + 2 - 5)x + (4 + 3) = 7x^2 - 7x + 7$$

So, the sum of the expressions is $7x^2 - 7x + 7$.
Then the correct option is B)

13. Correct option: (B)

To subtract $-5x^2y$ from $6xy - x^2y$, we can follow these steps:
1. Start with the expression: $6xy - x^2y$
2. Subtract $-5x^2y$ from it: $6xy - x^2y - (-5x^2y)$
3. To subtract a negative number, we can change it to addition:

$$6xy - x^2y + 5x^2y$$

4. Now, we can combine like terms by adding the coefficients that have the same variables:

$$(6xy + 5x^2y) - x^2y => 6xy + 4x^2y$$

Then the correct option is B)

14. Correct option: (C)

To find the sum of the expressions we can simply add them together term by term:

$$\left(\frac{7}{2}\right)x^3 - \left(\frac{3}{2}\right)x^2 + 5 + \left(\frac{7}{2}\right)x^2 - \left(\frac{3}{2}\right)x^3 + \frac{1}{3} + \left(\frac{3}{2}\right)x^3 - \left(\frac{1}{2}\right)x^2 + \frac{2}{3}$$

Now, let's combine like terms:

$$\left[\left(\frac{7}{2}\right)x^3 - \left(\frac{3}{2}\right)x^3 + \left(\frac{3}{2}\right)x^3\right] + \left[-\left(\frac{3}{2}\right)x^2 + \left(\frac{7}{2}\right)x^2 - \left(\frac{1}{2}\right)x^2\right] + \left[5 + \frac{1}{3} + \frac{2}{3}\right]$$

Combine the coefficients for each power of x and the constants:

$$\frac{7}{2}x^3 + \left(-\left(\frac{3}{2}\right) + \frac{7}{2} - \frac{1}{2}\right)x^2 + \left(5 + \frac{1}{3} + \frac{2}{3}\right)$$

$$=> \left(\frac{7}{2}\right)x^3 + \left(\frac{3}{2}\right)x^2 + 6$$

Then the correct option is C)

15. Correct option: (A)

From the question we can write as:

$$\left(\frac{1}{3} - \frac{5}{3}y^2\right) - \left(-\frac{7}{3}y^2 + \frac{1}{2}y\right)$$

To subtract a negative expression, change it to addition:

$$\left(\frac{1}{3} - \frac{5}{3}y^2\right) + \left(\frac{7}{3}y^2 - \frac{1}{2}y\right)$$

Now, combine like terms by adding or subtracting the coefficients that have the same variable (y) or constant terms:

$$\left(\frac{7}{3}y^2 - \frac{5}{3}y^2\right) - \frac{1}{2}y + \frac{1}{3} = \frac{2}{3}y^2 - \frac{1}{2}y + \frac{1}{3}$$

Then the correct option is A)

16. Correct option: (B)

To find the perimeter of a triangle with side lengths of $3x, 4x, \text{and } 6x$, simply add up the lengths of all three sides:

$$\text{Perimeter} = 3x + 4x + 6x$$

Now, combine like terms by adding the coefficients (the numbers in front of x):

$$\text{Perimeter} = (3 + 4 + 6)x$$

$$\text{Perimeter} = 13x$$

So, the perimeter of the triangle is 13x.

Then the correct option is B)

17. Correct option: (A)

The product of xy and x^2y is calculated by multiplying the two expressions together:

$$xy \times x^2y = (xy) \times (x^2y)$$

Now, use the properties of exponents to simplify:

$$(xy) \times (x^2y) = x^1 \times y^1 \times x^2 \times y^1$$

When multiplying variables with the same base (in this case, x, y), we add their exponents. So, combining the exponents:

$$x^{1+2} \times y^{1+1} = x^3 \times y^2$$

Therefore, the product of xy and x^2y is x^3y^2.

Then the correct option is A)

18. Correct option: (B)

To find the product of $4x, -2xy, 4x^3, -5x^2y^2, \text{and } -2xy^2$, we simply multiply all of these terms together:

$$(4x) \times (-2xy) \times (4x^3) \times (-5x^2y^2) \times (-2xy^2)$$

Now, let's multiply the coefficients and the variables separately:

Coefficients: $(4) \times (-2) \times (4) \times (-5) \times (-2) = -320$
Variables:

$$x \times xy \times x^3 \times x^2y^2 \times xy^2 = x^8y^5$$

Putting it all together:

$$=> -320x^8y^5$$

Then the correct option is B)

19. Correct option: (A)

To find the area of a rectangle, we simply multiply its length and breadth.
In this case, the length of the rectangle is $3x - 2y^2$, and the breadth is $4x^2 - 5y$. So, the area (A) of the rectangle is:

$$A = (\text{Length}) \times (\text{Breadth})$$
$$A = (3x - 2y^2) \times (4x^2 - 5y)$$

Now, let's multiply these two expressions together using the distributive property:

$$A = (3x \times 4x^2) + \big(3x \times (-5y)\big) + (-2y^2 \times 4x^2) + (-2y^2 \times (-5y))$$

Now, simplify each term:

$$A = 12x^3 - 15xy - 8x^2y^2 + 10y^3$$

Then the area of rectangle is $A = 12x^3 - 15xy - 8x^2y^2 + 10y^3$
Therefore the correct option is A)

20. Correct option: (C)

Let's denote the length of the cuboid as L, the breadth as B, and the height as H.
According to the given information:
1. The length (L) is 2 times its breadth (B), so $L = 2B$.
2. The height (H) is one-third of the length (L), so $H = \left(\frac{1}{3}\right)L$.

Now, we mentioned that the breadth is x cm, so B = x.
We need to find the volume (V) of the cuboid, which is given by the formula:

$$V = \text{Length} \times \text{Breadth} \times \text{Height}$$

Substitute the values we found for L, B, and H:

$$V = (2x) \times (x) \times \left(\frac{1}{3}\right)(2x)$$

Now, simplify the expression:

$$V = (2x) \times (x) \times \left(\frac{2}{3x}\right) = \frac{4}{3}x^3$$

So, the volume of the cuboid is $\left(\frac{4}{3}\right)x^3$ cubic centimeters.
Therefore the correct option is C)

LEVEL-II

1. Correct option: (A)

Firstly the price for 1 dozen of apple is x, and y dozens of apples is needed.
After discount the price for 1 dozen is x-4 and (y-6) dozens is needed now.
The total amount to pay is the product of the price per dozen of apples and the number of dozens of apples.

$$\text{Amount to pay} = (x - 4) \times (y - 6)$$

Therefore the correct option is A)

2. Correct option: (B)

The area of a rectangle is given by the formula:

$$\text{Area} = \text{Length} \times \text{Breadth}$$

If the length is increased by 6 units and the breadth is decreased by 9 units, the new length (l') and new breadth (b') will be:

$$\text{New Length } (l') = l + 6$$
$$\text{New Breadth } (b') = b - 9$$

Now, we can calculate the area of the new rectangle:

$$\text{New Area} = l' \times b'$$
$$\text{New Area} = (l + 6) \times (b - 9) = lb + 6b - 9l - 54$$

This is the area of the rectangle after the changes in length and breadth have been made.
Therefore the correct option is B)

3. Correct option: (D)

To subtract the product of $3x^2(2x - 3y)$ from the product of $6x(x^2 + 4y)$, follow these steps:

1. Calculate the product of $3x^2(2x - 3y)$:
$3x^2(2x - 3y) = 6x^3 - 9x^2y$
2. Calculate the product of $6x(x^2 + 4y)$:
$6x(x^2 + 4y) = 6x^3 + 24xy$
3. Now, subtract the result from step 1 from the result from step 2:
$$(6x^3 + 24xy) - (6x^3 - 9x^2y)$$
4. Distribute the negative sign into the second expression inside the parentheses:
$$6x^3 + 24xy - 6x^3 + 9x^2y$$
5. Now, combine like terms:
$(6x^3 - 6x^3) + (24xy + 9x^2y)$

So, the result of subtracting the product of $3x^2(2x - 3y)$ from the product of $6x(x^2 + 4y)$ is $24xy + 9x^2y$.
Therefore, the correct option is D)

4. Correct option: (C)

$$\text{Total Sum: } 6y + 13x - 7x^2$$

Given Expressions: $(3x^2 - 4x + 5y) + (3x - 10x^2 - 8y)$
Now, subtract the sum of the two given expressions from the total sum:

$$\text{Third Expression} = \text{Total Sum} - \text{Sum of Given Expressions}$$
$$\text{Third Expression} = (6y + 13x - 7x^2) - [(3x^2 - 4x + 5y) + (3x - 10x^2 - 8y)]$$

Now, distribute the negative sign inside the brackets:

$$\text{Third Expression} = 6y + 13x - 7x^2 - 3x^2 + 4x - 5y - 3x + 10x^2 + 8y$$

$$\text{Third Expression} = (6y - 5y + 8y) + (13x - 3x + 4x) + (-7x^2 - 3x^2 + 10x^2)$$
$$\text{Third Expression} = 9y + 14x$$

So, the third expression is $9y + 14x$
Therefore, the correct option is C)

5. Correct option: (A)

The perimeter of a rectangle is given by the formula:

$$\text{Perimeter} = 2(\text{Length} + \text{Breadth})$$

In this case, the length of the rectangle is $6x^2 - 4y^2$, and the breadth is $2x^2 - 8xy$.
So, the perimeter is:

$$\text{Perimeter} = 2(6x^2 - 4y^2 + 2x^2 - 8xy)$$

Now, let's simplify the expression inside the parentheses:

$$\text{Perimeter} = 2(8x^2 - 4y^2 - 8xy)$$
$$\text{Perimeter} = 16x^2 - 8y^2 - 16xy$$

So, the perimeter of the rectangle is $16x^2 - 8y^2 - 16xy$.
Then the correct option is A)

6. Correct option: (C)

If the population of a certain species of insects is currently $\frac{x^2}{2}$, and it becomes 3y times itself after one week, we can calculate the population after one week:

$$\text{Population after one week} = (3y) \times \left(\frac{x^2}{2}\right)$$

Now, to find the population after 5 weeks, we need to keep multiplying it by 3y for each week:

$$\text{Population after 5 weeks} = (3y)^5 \times \left(\frac{x^2}{2}\right)$$

To simplify, raise 3y to the 5th power:

$$\text{Population after 5 weeks} = 3^5 \times y^5 \times \left(\frac{x^2}{2}\right)$$

$$\text{Population after 5 weeks} = 243 \times y^5 \times \left(\frac{x^2}{2}\right)$$

Then the correct option is C)

7. Correct option: (C)

To subtract $y^2 - 7 - 2y$ from $8 - y + 2y^2$, we need to perform the subtraction operation term by term:

$(8 - y + 2y^2) - (y^2 - 7 - 2y)$

Now, distribute the negative sign into the second set of parentheses:

$$8 - y + 2y^2 - y^2 + 7 + 2y$$

Next, combine like terms:

$$(2y^2 - y^2) + (-y + 2y) + (8 + 7)$$

Simplify each set of parentheses:

$$y^2 + y + 15$$

So, the result of subtracting $y^2 - 7 - 2y$ from $8 - y + 2y^2$ is $y^2 + y + 15$.

Then the correct option is C)

8. Correct option: (C)

To find the value of the quadratic expression $ax^2 + bx + c$ at $x = -\frac{b}{a}$, we can simply substitute $-\frac{b}{a}$ for x in the expression and calculate:

$$ax^2 + bx + c \text{ at } x = -\frac{b}{a}:$$

$$= a\left(-\frac{b}{a}\right)^2 + b\left(-\frac{b}{a}\right) + c$$

$$= a\left(\frac{b^2}{a^2}\right) - \left(\frac{b^2}{a}\right) + c$$

Now, simplify:

$$= \frac{b^2}{a} - \frac{b^2}{a} + c$$

$$= \left(\frac{b^2}{a} - \frac{b^2}{a}\right) + c$$

$$= 0 + c$$

So, the value of the quadratic expression $ax^2 + bx + c$ at $x = -\frac{b}{a}$ is simply c.

Then the correct option is C)

9. Correct option: (C)

Given $a = 0, b = 1,$ and $c = 2$, we can calculate the expression $a^2 + b + c$ as follows:

$$a^2 + b + c = (0)^2 + 1 + 2$$

Now, simplify:

$$a^2 + b + c = 0 + 1 + 2$$

$$a^2 + b + c = 3$$

So, when a = 0, b = 1, and c = 2, the value of $a^2 + b + c$ is 3.

Then the correct option is C)

10. Correct option: (D)

The sum of $x + 1$ and the reciprocal of $(x - 1)$ can be expressed as:

$$(x + 1) + \frac{1}{x - 1}$$

To simplify this expression further, we can find a common denominator:

$$\frac{[(x + 1)(x - 1) + 1]}{x - 1}$$

Now, expand the numerator:

$$\frac{x^2 - 1 + 1}{x - 1}$$

Simplify the numerator:

$$\frac{x^2}{x - 1}$$

So, the sum of x + 1 and the reciprocal of (x - 1) is $\frac{x^2}{x-1}$.

None of the option matches with this.

Then the correct option is D)

11. Correct option: (B, C)

The statement can be represented as an algebraic expression as follows:

1. The sum of p and q is: p + q.
2. The difference between p and q is: p - q.
3. Taking away 5 from their difference: (p - q) - 5.

Now, combine the two parts:

The sum of p and q is multiplied by taking away 5 from their difference: $(p + q)((p - q) - 5)$

$$(p + q)\big((p - q) - 5\big) = (p + q)(p - q) - 5(p + q) = p^2 - q^2 - 5(p + q)$$

Then the correct option is B), C).

12. Correct option: (A, B)

To find the total number of persons in the large hall, we can multiply the number of rows by the number of benches in each row and then multiply that by the number of persons each bench can accommodate.

Given:

$$\text{Number of rows of benches} = 4a^2b$$

$$\text{Number of benches in each row} = 16a^3b^3$$

$$\text{Number of persons each bench can accommodate} = 3ab^2$$

Total number of persons

$$= (\text{Number of rows}) \times (\text{Number of benches per row}) \times (\text{Number of persons per bench})$$

$$\text{Total number of persons} = (4a^2b) \times (16a^3b^3) \times (3ab^2)$$

Now, multiply the coefficients and combine the variables with the same base:

$$\text{Total number of persons} = (4 \times 16 \times 3) \times (a^2 \times a^3 \times a) \times (b \times b^3 \times b^2)$$

$$\text{Total number of persons} = 192a^6b^6 = 2^6 \times 3a^6b^6$$

Then the correct option is A), B).

13. Correct option: (B, D)

To find the value of 2A - B, first, we'll find the expressions for 2A and then subtract B from it.

Given:

$$A = x^2 + 2x + 2$$

$$B = 5x^2 - x + 4$$

Now, let's calculate 2A:

$$2A = 2(x^2 + 2x + 2)$$

Now, distribute the 2 into each term inside the parentheses:

$$2A = 2x^2 + 4x + 4$$

Now, subtract B from 2A:

$$2A - B = (2x^2 + 4x + 4) - (5x^2 - x + 4)$$

Now, distribute the negative sign into the second expression inside the parentheses:

$$2A - B = 2x^2 + 4x + 4 - 5x^2 + x - 4$$

$$2A - B = (2x^2 - 5x^2) + (4x + x) + (4 - 4)$$

Simplify each set of parentheses:

$$2A - B = -3x^2 + 5x + 0$$

So, the value of 2A - B is $-3x^2 + 5x$.

Then the correct option is B), D).

14. Correct option: (C)

To find the perimeter of a triangle with sides A, B, and C, we simply need to add the lengths of these sides together.

Given:

$$A = 5x^2y + 2xy^2 - 4xy$$

$$B = 3xy^2 + 4x^2y - 2$$

$$C = 7x^2y - 5$$

Now, add the lengths of these sides to find the perimeter:

$$\text{Perimeter} = A + B + C$$

$$P = (5x^2y + 2xy^2 - 4xy) + (3xy^2 + 4x^2y - 2) + (7x^2y - 5)$$

$$P = 5x^2y + 2xy^2 - 4xy + 3xy^2 + 4x^2y - 2 + 7x^2y - 5$$

$$P = (5x^2y + 4x^2y + 7x^2y) + (2xy^2 + 3xy^2 - 4xy) - 7$$

$$P = 16x^2y + 5xy^2 - 4xy - 7$$

So, the perimeter of the triangle with sides A, B, and C is $16x^2y + 5xy^2 - 4xy - 7$.

Then the correct option is C)

15. Correct option: (B)

To find the value of 2B + C - A, simply substitute the given expressions for A, B, and C into the expression and simplify: $2B + C - A$

$$= 2(3xy^2 + 4x^2y - 2) + (7x^2y - 5) - (5x^2y + 2xy^2 - 4xy)$$

Now, distribute the coefficients:

$$= 6xy^2 + 8x^2y - 4 + 7x^2y - 5 - 5x^2y - 2xy^2 + 4xy$$

$$= (6xy^2 - 2xy^2) + (8x^2y + 7x^2y - 5x^2y) + (-4 - 5 + 4xy)$$

Now, simplify each set of parentheses:

$$2B + C - A = 4xy^2 + 10x^2y - 9 + 4xy$$

So, the value of $2B + C - A$ is $4xy^2 + 10x^2y + 4xy - 9$.

Then the correct option is B)

16. Correct option: (D)

The additive inverse of a number or expression is what we add to it to get zero.

$$A + B - C$$

The additive inverse would be:

$$-(A + B - C)$$

Now, let's substitute the expressions for A, B, and C:

$$-(5x^2y + 2xy^2 - 4xy + 3xy^2 + 4x^2y - 2 - 7x^2y + 5)$$

Now, distribute the negative sign to each term inside the parentheses:

$$-5x^2y - 2xy^2 + 4xy - 3xy^2 - 4x^2y + 2 + 7x^2y - 5$$

Now, combine like terms:

$$(-5x^2y - 4x^2y + 7x\text{^}2y) + (-2xy^2 - 3xy^2) + (4xy + 2 - 5)$$

Combine the like terms further:

$$-2x^2y - 5xy^2 + 4xy - 3$$

So, the additive inverse of A + B - C is

$$-2x^2y - 5xy^2 + 4xy - 3.$$

Then the correct option is D)

17. Correct Matching is: a-T, b-S,c-P,d-R

a) $(x^2 + 2)^2$

$(x^2 + 2)^2 = x^4 + 4 + 4x^2$ Highest power of x = 4

Then the degree of the above expression is 4.-→(T)

b) $(x^2 + 2)(x + 3)$

$(x^2 + 2)(x + 3) = x^3 + 2x + 3x^2 + 6$ Highest power of x = 3

Then the degree of the above expression is 3.-→(S)

c) 2

Degree of a constant term is always 0.

Then the degree of the above expression is 0.-→P)

d) $(x + 3)(x - 3)$

$(x + 3)(x - 3) = x^2 - 9$ Highest power of x = 2

Then the degree of the above expression is 2.-→(R)

18. Correct Answer is: (2)

The degree of a polynomial expression is determined by the highest power of the variable in the expression.

In the expression $(x + 2)(x^n + 1)$, we have two factors: (x+2) and $(x^n + 1)$.

The degree of the first factor (x+2) is 1 because the highest power of x is 1.

The degree of the second factor $(x^n + 1)$ is n because it contains the term x^n , which has the highest power of x.

To find the degree of the entire expression, we add the degrees of the factors:

$$\text{Degree of expression} = \text{Degree of } (x + 2) + \text{Degree of } (x^n + 1) = n + 1$$

We are given that degree of the expression is 3.

$$n + 1 = 3 \Rightarrow n = 2$$

Then the value of n is 2.

19. Correct Answer is: (5)

Let's substitute the values of a, b, c, and d into the expression and calculate:

$$(cd - ab) - (bd - ac)$$

Substitute the values: $(4 \times 3 - 1 \times 2) - (2 \times 4 - 1 \times 3)$

Now, perform the multiplications and subtractions inside the parentheses:

$$(12 - 2) - (8 - 3)$$

Now, continue simplifying: $(10) - (5) = 5$

So, the value of $(cd - ab) - (bd - ac)$ when $a = 1, b = 2, c = 3,$ and $d = 4$ is 5.

20. Correct Answer is: (2)

To find the value of x in the equation $7x - 5 = 3x + 3$, we need to isolate x on one side of the equation.

$$7x - 3x - 5 = 3x - 3x + 3$$
$$4x - 5 = 3$$
$$4x - 5 + 5 = 3 + 5$$
$$4x = 8$$
$$\frac{4x}{4} = \frac{8}{4}$$
$$x = 2$$

So, the value of x is 2.

STUDENT PRACTICE QUESTIONS

1. Correct option: (A)

The correct statement for the expression "4x + 9" is:

A) 4 is multiplied by x and added to 9

2. Correct option: (D)

To find the number of terms in the expression:

$$6x^2y + 2xy + 5xy^2 + 9x + 6y + 20x^2y^2 + (-2x^3) - 7$$

Count all the individual terms:

1. $6x^2y$
2. $2xy$
3. $5xy^2$
4. $9x$
5. $6y$
6. $20x^2y^2$
7. $-2x^3$
8. -7

There are a total of 8 terms in the given expression.

3. Correct option: (A)

An expression that contains three like terms is called a Trinomial.

So, the correct answer is: A) Trinomial

4. Correct option: (D)

A Monomial expression consists of three unlike terms. Let's analyze the given options:

D) $(2x)y^2 - y^2(2xy) + (-y)(2xy) => -y^2(2xy)$ - This is a Monomial because it has only one term.

So, the correct answer is: (D)

5. Correct option: (C)

To find the sum of the expressions $3a^2b, -4a^2b,$ and $9a^2b$

$$3a^2b + (-4a^2b) + 9a^2b$$

Now, combine like terms by adding the coefficients (the numbers in front of the variables) that have the same variables (a^2b):

$$(3 - 4 + 9)a^2b$$
$$= (8)a^2b$$

So, the sum of the expressions is $8a^2b$.

Then the correct option is C)

6. Correct option: (D)

To subtract $3a - 2b$ from $5b - 6a$, we can directly do subtraction:

$$5b - 6a - (3a - 2b) = 5b - 6a - 3a + 2b = 7b - 9a$$

Then the correct option is D)

7. Correct option: (B)

To find out what should be taken away from $3x^2 - 4y^2 + 5xy + 20$ to get $-x^2 - y^2 + 6xy + 20$, we need to calculate the difference between the two expressions:

$$(3x^2 - 4y^2 + 5xy + 20) - (-x^2 - y^2 + 6xy + 20)$$

Now, distribute the negative sign across the terms inside the second set of parentheses:

$$3x^2 - 4y^2 + 5xy + 20 + x^2 + y^2 - 6xy - 20$$

Next, combine like terms by adding or subtracting the coefficients that have the same variables:

$$(3x^2 + x^2) + (-4y^2 + y^2) + (5xy - 6xy) + (20 - 20)$$

Now, perform the arithmetic operations within each set of parentheses:

$$4x^2 - 3y^2 - xy$$

So, to get from $3x^2 - 4y^2 + 5xy + 20$ to $-x^2 - y^2 + 6xy + 20$, you need to subtract $4x^2 - 3y^2 - xy$.

Then the correct option is B)

8. Correct option: (C)

To find the product of $4xy$ and $-6xyz$, we simply multiply the two expressions together:

$$(4xy) \times (-6xyz)$$

Now, apply the rules of multiplication:

$$4 \times (-6) \times x \times x \times y \times z \times y$$

Multiply the coefficients:

$$-24 \times x \times x \times y \times z \times y$$

$$-24 \times x^2 \times y^2 \times z = -24x^2y^2z$$

Then the correct option is C)

9. Correct option: (D)

To find the perimeter of a rectangle, we add up the lengths of all four sides.

In this case, the length of the rectangle is $6x$ and the breadth (width) is $4y + 3$. A rectangle has two pairs of equal sides, so we have:

1. Two sides of length $6x$.
2. Two sides of length $4y + 3$.

Now, calculate the perimeter:

$$\text{Perimeter} = 2(\text{Length} + \text{Breadth})$$

$$\text{Perimeter} = 2(6x + 4y + 3)$$

$$\text{Perimeter} = 12x + 8y + 6$$

So, the perimeter of the rectangle is $12x + 8y + 6$.

Then the correct option is D)

10. Correct option: (B)

To find the total distance Rahul moves in $6y$ steps, we can simply multiply the distance he covers in one step ($8x$) by the number of steps ($6y$).

$$\text{Distance moved} = (\text{Distance per step}) \times (\text{Number of steps})$$

$$\text{Distance moved} = (8x) \times (6y)$$

Now, multiply the terms together:

Distance moved = $48xy$

So, Rahul covers a distance of $48xy$ centimeters in $6y$ steps.

Therefore the correct option is B)

11. Correct option: (B, D)

To determine what should be subtracted from $a^2 + b^2 + 2ab$ to get $-4ab + 2b^2$, we can set up the subtraction as follows:

$$(a^2 + b^2 + 2ab) - (\text{What should be subtracted}) = -4ab + 2b^2$$

Now, let's find out what should be subtracted:

$$\text{What should be subtracted} = (a^2 + b^2 + 2ab) - (-4ab + 2b^2)$$

Now, subtract the expressions inside the parentheses:

$$= a^2 + b^2 + 2ab + 4ab - 2b^2$$

Combine like terms: $= (a^2 + 6ab - b^2) => a^2 + b(-b + 6a)$

Then the correct option is B), D).

12. Correct option: (A, B)

Like terms are terms that have the same variables raised to the same powers. Let's identify the like terms in each set:

$$\text{A) } 7xy \text{ and } \frac{8}{7}xy$$

$$\text{Like terms: } 7xy \text{ and } \frac{8}{7}xy$$

$$\text{B) } 9x^3y^2 \text{ and } -7x^3y^2$$

$$\text{Like terms: } 9x^3y^2 \text{ and } -7x^3y^2$$

$$\text{C) } 71\,xy^2 \text{ and } \frac{8}{3}x$$

- These terms do not have the same variables raised to the same powers, so they are not like terms.

$$\text{D) } 3x^2y \text{ and } 8xy$$

- These terms do not have the same variables raised to the same powers, so they are not like terms.

So, the like terms in each set are as follows:

$$\text{A) } 7xy \text{ and } \frac{8}{7}xy$$

$$\text{B) } 9x^3y^2 \text{ and } -7x^3y^2$$

Then the correct options are A), B).

13. Correct option: (A)

A) $2x^2y$ is a monomial because it contains only one term.

B) x + y is a binomial because it contains two terms.

C) $x + y^2$ is also a binomial because it contains two terms.

Then the correct option is A)

14. Correct option: (D)

A binomial is an algebraic expression containing exactly two terms. Let's identify the binomial from the options:

A) 8abc - This is a monomial because it contains only one term.

B) 2 + x + y - This is not a binomial; it's a trinomial because it contains three terms (2, x, and y).

C) x^2y - This is a monomial because it contains only one term.

D) $x + y^2$ - This is a binomial because it contains two terms.

So, the binomial among the given options is D) $x + y^2$.

15. Correct Matching is: a-S, b-R, c-Q, d-P

a) $(x + y)^2 - (x - y)^2 =$

Solving the expression

$$x^2 + y^2 + 2xy - (x^2 + y^2 - 2xy) = 4xy$$

Then we can say that a) -→(S)

b) $(ax + b)(cx + d) =$

Solving the expression

$$acx^2 + cbx + adx + bd = cax^2 + (ad + bc)x + bd$$

Then we can say that b) -→(R)

c) $\left(x-\frac{1}{xy}\right)\left(xy+\frac{1}{x}\right)$

Solving the expression

$$\left(x^2y-1+1-\frac{1}{x^2y}\right)=x^2y-\frac{1}{x^2y}$$

Then we can say that c) -→(Q)

d) $(ax+d)(cx+b)=$

Solving the expression

$$acx^2+cdx+abx+bd=cax^2+(ab+cd)x+bd$$

Then we can say that d) -→(P)

16. Correct Answer is: (0)

Given the values of A, B, and C as follows:

$$A=-2x+3y-5z$$
$$B=-4x-9y-3z$$
$$C=6x+6y+8z$$

We want to find the value of A + B + C, which is the sum of these three expressions:

$$A+B+C=(-2x+3y-5z)+(-4x-9y-3z)+(6x+6y+8z)$$

Now, simply combine like terms by adding the coefficients of the same variables:

$$=(-2x-4x+6x)+(3y-9y+6y)+(-5z-3z+8z)$$

Now, perform the additions:

$$A+B+C=0+0+0=0$$

So, the value of A + B + C is 0.

17. Correct Answer is: (2)

To find the coefficient of x^2 in the expression $(3y+x^2)(xy+2)$, we need to perform polynomial multiplication. The term x^2 can be obtained from multiplying x^2 in the first factor and the constant 2 in the second factor.

So, multiply these two terms: $x^2\times 2=2x^2$

The coefficient of x^2 in the expression is 2.

18. Correct Answer is: (6)

To find the number of terms in the expression $(x+xy-2)(x+y)$, we can use the distributive property to multiply the two polynomial:

$$(x+xy-2)(x+y)$$

When you multiply each term in the first binomial by each term in the second binomial, we will generate several terms:

$$x^2+x^2y-2x+xy+xy^2-2y$$

Now, simplify each term:

$$x^2+xy+x^2y+xy^2-2x-2y$$

Now, count the number of terms:

There are 6 terms in the expression $(x+xy-2)(x+y)$.

19. Correct Answer is: (8)

To solve for x in the equation 3x + 9 = 33, we'll need to isolate x.

First, subtract 9 from both sides of the equation:

$$3x+9-9=33-9$$

This simplifies to: $3x=24$

Now, divide both sides by 3 to solve for x:

$$\frac{3x}{3}=\frac{24}{3}$$
$$x=8$$

So, the solution to the equation is x = 8.

20. Correct Answer is: (8)

To find the length of one side of a square when we know its area, we can use the formula for the area of a square:

$$\text{Area} = \text{side}^2$$
$$64 = \text{side}^2$$

Now, to solve for the side length, we can take the square root of both sides:

$$\sqrt{64} = \sqrt{\text{side}^2}$$
$$8 = \text{side}$$

So, the length of one side of the square is 8 cm.

Algebraic Expressions and Equations

LEVEL-I

1. Correct option: (B)

Given expression

$$(x^6 - 7x^5 + 4x^4) \div x^2$$

To divide the expression we can apply the rules of exponents. When we divide two terms with the same base we subtract the exponents.

Divide each term separately:

$$\left(\frac{x^6}{x^2} - \frac{7x^5}{x^2} + \frac{4x^4}{x^2}\right) = x^4 - 7x^3 + 4x^2$$

So, the result of $(x^6 - 7x^5 + 4x\text{^}4) \div x^2$ is:

$$x^4 - 7x^3 + 4x^2$$

Then the correct option is B)

2. Correct option: (A)

To divide the expression abx^2 by $(-abx)$, we can simplify as follows:

$$abx^2 \div (-abx) = \frac{abx^2}{-abx}$$

Now, divide both the numerator and the denominator by $-abx$

$$\frac{abx^2}{-abx} = -\frac{x \times abx}{abx} = -x$$

So, the result of $abx^2 \div (-abx)$ is $-x$.

Then the correct option is A)

3. Correct option: (B)

To divide the expression $16p^6q^7r^8$ by $-4p^2q^3r^4$ we can simplify as follows:

$$\frac{16p^6q^7r^8}{-4p^2q^3r^4} = -\frac{(4p^4q^4r^4)(4p^2q^3r^4)}{4p^2q^3r^4} = -4p^4q^4r^4$$

So, the result of $16p^6q^7r^8 \div (-4p^2q^3r^4)$ is

$$-4p^4q^4r^4$$

Then the correct option is B)

4. Correct option: (B)

Given expression

$$\left(-\frac{5}{2}x^2 + \frac{5}{3}xy + \frac{10}{3}x\right) \div \left(\frac{5}{6}x\right)$$

Find the reciprocal of $\left(\frac{5}{6}x\right)$, which is $\left(\frac{6}{5x}\right)$.

Now, multiply the expression by the reciprocal:

$$\left(-\frac{5}{2}x^2 + \frac{5}{3}xy + \frac{10}{3}x\right) \times \left(\frac{6}{5x}\right)$$

Now, perform the multiplication for each term:

$$1.\left(-\frac{5}{2}x^2\right) \times \left(\frac{6}{5x}\right) = \left(-\frac{5}{2}\right) \times \left(\frac{6}{5}\right)x = -3x$$

$$2.\left(\frac{5}{3}xy\right) \times \left(\frac{6}{5x}\right) = 2y$$

$$3.\left(\frac{10}{3}x\right) \times \left(\frac{6}{5x}\right) = 4$$

So, the result of $\left(-\frac{5}{2}x^2 + \frac{5}{3}xy + \frac{10}{3}x\right) \div \left(\frac{5}{6}x\right)$ is:

$$-3x + 2y + 4$$

Then the correct option is B)

5. Correct option: (D)

Given expression

$$\left(\frac{1}{4}a^2x - \frac{1}{10}abx - \frac{3}{8}acx\right) \div \left(\frac{3}{8}ax\right)$$

Find the reciprocal of $\left(\frac{3}{8}ax\right)$, which is $\left(\frac{8}{3ax}\right)$.

Now, multiply the expression by the reciprocal:

$$\left(\frac{1}{4}a^2x - \frac{1}{10}abx - \frac{3}{8}acx\right) \times \left(\frac{8}{3ax}\right)$$

Now, perform the multiplication for each term:

$$\frac{1}{4}a^2x \times \frac{8}{3ax} - \frac{1}{10}abx \times \frac{8}{3ax} - \frac{3}{8}acx \times \frac{8}{3ax} = \frac{2}{3}a - \frac{4}{15}b - c$$

So, the result of

$$\left(\frac{1}{4}a^2x - \frac{1}{10}abx - \frac{3}{8}acx\right) \div \left(\frac{3}{8}ax\right)$$

is:

$$\frac{2}{3}a - \frac{4}{15}b - c$$

Then the correct option is D)

6. Correct option: (C)

To simplify the expression $(-48x^6 - 6x^3) \div (-12x^2)$, we can divide each term in the numerator by the denominator.

1. Divide $-48x^6$ by $-12x^2$:

$$\frac{-48x^6}{-12x^2} = 4x^{6-2} = 4x^4$$

2. Divide $-6x^3$ by $-12x^2$:

$$\frac{-6x^3}{-12x^2} = \left(\frac{1}{2}\right)x^{3-2} = \left(\frac{1}{2}\right)x$$

So, the simplified expression is:

$$4x^4 + \left(\frac{1}{2}\right)x$$

Then the correct option is C)

7. Correct option: (A)

To simplify the expression $(x^2 - 11x + 30) \div (x - 6)$, we can use factorization.

We can also simplify the expression by factoring the numerator:

$x^2 - 11x + 30$ can be factored as $(x - 5)(x - 6)$.

Now, we can cancel out the common factor $(x - 6)$ in the numerator and denominator:

$$\frac{[(x - 5)(x - 6)]}{x - 6} = x - 5$$

So, the simplified expression is $x - 5$.

Then the correct option is A)

8. Correct option: (C)

To simplify the expression $(12a^2 + 10a - 2) \div (6a - 1)$, We can use polynomial long division

Polynomial Long Division Perform polynomial long division by dividing each term of the numerator by the denominator:

$$\begin{array}{r} 2a + 2 \\ \hline 6a - 1 \;|\; 12a^2 + 10a - 2 \\ -(12a^2 - 2a) \\ \hline 12a - 2 \\ -(12a - 2) \\ \hline 0 \end{array}$$

So, the result of the division is 2a + 2.

Then the correct option is C)

9. Correct option: (A)

In the context of multinomials, the term "prime" is not commonly used as it is for integers. The concept of primality primarily applies to integers.

In the context of integers, a number is considered prime if it is divisible by only 1 and itself (option A). However, this concept doesn't directly apply to multinomials, which are algebraic expressions with multiple terms involving variables and coefficients. Multinomials are not typically classified as prime or non-prime.

10. Correct option: (A)

The product of two polynomials, $p(x)$ and $q(x)$, can be expressed in terms of their (HCF) and least common multiple (LCM) as follows:

$$p(x) \times q(x) = [HCF(p(x), q(x))] \times [LCM(p(x), q(x))]$$

This equation expresses the product of two polynomials in terms of their HCF and LCM. This is a useful relation when dealing with polynomial factorization and simplification.

Then the correct option is A)

11. Correct option: (A)

To find the highest common factor (HCF) of two polynomials, we need to determine the highest power of the variable that is common to both polynomials.

In this case, we have the polynomials x^4 and $8x^3$.

The highest power of x that is common to both polynomials is x^3.

Therefore, the HCF of x^4 and $8x^3$ is x^3.

Then the correct option is A)

12. Correct option: (C)

1. Factorize each expression: $22h^2t = 2 \times 11 \times h \times h \times t$

$$33ht^2 = 3 \times 11 \times h \times t \times t$$

2. Now, identify the common factors:

The common factors among these expressions are 11, h, and t.

3. Calculate the HCF by multiplying these common factors:

$$HCF = 11 \times h \times t = 11ht$$

Then the correct option is C)

13. Correct option: (D)

To find the three numbers in the ratio 2:3:5 with a highest common factor (HCF) of 6, we can follow these steps:

1. Write the numbers in the form of 2x, 3x, and 5x, where x is a common multiplier.

2. Find the HCF of these numbers $\text{HCF}(2x, 3x, 5x) = x$
Equate it with the given HCF as $x = 6$
Then the numbers are $2x = 12, 3x = 18, 5x = 30$
So, the three numbers are 12, 18, and 30, which corresponds to option D) 12, 18, and 30.

14. Correct option: (B)

If three monomials are in the ratio x:y:z and their highest common factor (HCF) is xyz, then we can express the monomials as follows:

$$\text{First monomial: HCF (xyz)} \times x = x^2yz$$
$$\text{Second monomial: HCF (xyz)} \times y = xy^2z$$
$$\text{Third monomial: HCF (xyz)} \times z = xyz^2$$

So, the three monomials are $x^2yz, xy^2z,$ and xyz^2.
Then the correct option is B)

15. Correct option: (C)

To find the least common multiple (LCM) of the two expressions abc and cd, we'll need to factorize each expression and then determine the LCM based on their prime factors.
1. Factorize the expressions: $abc = a \times b \times c, cd = c \times d$
2. Determine the LCM by taking the highest power of each prime factor:

$$\text{LCM} = a \times b \times c \times d$$

So, the LCM of abc and cd is abcd.
Then the correct option is C)

16. Correct option: (D)

1. Factorize each expression:

$3x^2y^2z^2$: Prime factors include 3, x, y, and z.
$4xy^2z^4$: Prime factors include 2, x, y, and z.

2. Now, identify the prime factors and their respective highest powers:

– LCM will have the prime factors 2, 3, x, y, and z.
– The highest power of 2 is $2^2 = 4$.
– The highest power of 3 is $3^1 = 3$.
– The highest power of x is x^2.
– The highest power of y is y^2.
– The highest power of z is z^4.

3. Calculate the LCM by multiplying these highest powers:

$$\text{LCM} = 4 \times 3 \times x^2 \times y^2 \times z^4$$

Then the correct option is D)

17. Correct option: (C)

1. Factorize each expression:

$7p^3q^4$: Prime factors include 7, p^3, and q^4.
$8pq^5$: Prime factors include 2^3, p, and q^5.
$2p^6q^2$: Prime factors include 2, p^6, and q^2.

2. Now, identify the prime factors and their respective highest powers:

– LCM will have the prime factors $2^3, 7, p^6$, and q^5.
– The highest power of 2 is 2^3.
– The highest power of 7 is $7^1 = 7$.
– The highest power ofp is p^6.
– The highest power of q is q^5.

3. Calculate the LCM by multiplying these highest powers:

$$\text{LCM} = 2^3 \times 7 \times p^6 \times q^5 = 56p^6q^5$$

Then the correct option is C)

18. Correct option: (B)

1. Expand each expression: $(x+y)(x-y)$

$$(x+y)^2 = (x+y)(x+y)$$
$$(x-y)^2 = (x-y)(x-y)$$

2. Identify the prime factors and their respective highest powers:

$$\text{highest power of the } (x+y) \text{is } 2$$
$$\text{highest power of the } (x-y) \text{is } 2$$

3. Calculate the LCM by multiplying these highest powers: $LCM = (x-y)^2(x+y)^2$

Then the correct option is B)

19. Correct option: (B)

To find the least common multiple (LCM) and highest common factor (HCF) of the two expressions $A = (x+y)x$ and $B = (x+y)y$, we can start by analyzing their factors and expressions:

1. $A = x + y$

$$\text{Factors: } x, (x+y)$$

2. $B = (x+y)y$

$$\text{Factors: } (x+y), y$$

Now, let's find the LCM and HCF:

To find the LCM, we need to take the highest power of each factor that appears in both expressions.

$$LCM = (x+y)xy$$

To find the HCF, we need to take the lowest power of each common factor. In this case, the common factors are $x+y$.

$$HCF = x+y$$

So, the LCM of A and B is $(x+y)xy$, and the HCF of A and B is $x+y$.

Then the correct option is B)

20. Correct option: (A)

We can use the relationship between LCM, HCF, and the product of two numbers to find the unknown monomial: $LCM * HCF = \text{Product of the two numbers}$

Plugging in the given values: $(12a^2b^4c) \times (a^2b^2) = (12a^2b^2c) \times \text{Monomial}_2$

Now, simplify:

$$12a^2b^4c \times \frac{a^2b^2}{12a^2b^2c} = \text{Monomial}_2 = a^2b^4$$

Then the correct option is A)

LEVEL-II

1. Correct option: (C)

To find the highest common factor (HCF) of the two expressions we can compare their prime factors and take the lowest power of each common factor.

First, let's factorize both expressions:

$$A = 2^7 \times 3 \times x^4 \times y^5 \times z^3$$
$$B = 2^3 \times 3^3 \times x^2 \times y^3 \times z^3$$

Now, identify the common factors and their lowest powers:

$$\text{Common factors: } 2, 3, x, y, z$$
$$HCF = 2^3 \times 3 \times x^2 \times y^3 \times z^3 = 24x^2y^3z^3$$

So, the highest common factor (HCF) of $A = 384x^4y^5z^3$ and $B = 216x^2y^3z^3$ is $24x^2y^3z^3$.

Then the correct option is C)

2. Correct option: (B)

To factor the expression $2x(x+y) + 3(x+y)$, we can factor out the common binomial factor $(x + y)$:

$$2x(x + y) + 3(x + y) = (x + y)(2x + 3)$$

So, the binomial factor of the expression is $(x + y)$.

Then the correct option is B)

3. Correct option: (A)

To find the monomials B and C when the highest common factor (HCF) is given as $3^4x^2a^3b^7$, we need to express B and C in a way that their HCF matches the given value.

Let's break down the HCF into its prime factors:

$$\text{HCF} = 3^4 \times x^2 \times a^3 \times b^7$$

Now, let's look at the answer choices:

$$\text{A) } 324x^2a^5b^9, 405x^4a^3b^7$$
$$\text{B) } 324x^2a^5b^9, 405x^4ab^4$$
$$\text{C) } 324x^2a^2b^3, 405x^4ab^4$$
$$\text{D) } 324x^2a^2b^3, 405x^4ab^4$$

We can see that option A has the correct prime factors to match the HCF:

$$B = 324x^2a^5b^9$$
$$C = 405x\text{^}4a^3b^7$$

Then the correct option is A)

4. Correct option: (B)

To find the least common multiple (LCM) and highest common factor (HCF) of two numbers or expressions, we can compare their prime factors.

Let's start by factoring both A and B:

$$A = 2^3 \times 3^2 \times x^2 \times y^2 \times z^2$$
$$B = 2^4 \times 3 \times x \times y \times z$$

Now, let's calculate the LCM and HCF:

LCM (Least Common Multiple):

To find the LCM, we need to take the highest power of each prime factor that appears in both A and B:

For 2, the highest power is 2^4.
For 3, the highest power is 3^2.
For x, the highest power is x^2.
For y, the highest power is y^2.
For z, the highest power is z^2.

So, the LCM is: $\text{LCM} = 2^4 3^2 x^2 y^2 z^2$

HCF (Highest Common Factor):

To find the HCF, we need to take the lowest power of each common prime factor:

For 2, the lowest power is 2^3.
For 3, the lowest power is 3.
For x, the lowest power is x.
For y, the lowest power is y.
For z, the lowest power is z.

So, the HCF is: $\text{HCF} = 2^3 \times 3 \times x \times y \times z$

Now, you have the LCM and HCF:

$$\text{LCM} = 2^4 \times 3^2 \times x^2 \times y^2 \times z^2$$
$$\text{HCF} = 2^3 \times 3 \times x \times y \times z$$

Then the correct option is B)

5. Correct option: (C)

To find the product of two polynomials when their least common multiple (LCM) and highest common factor (HCF) are given, we can use the following relationship:

$$\text{Product of polynomials} = \text{LCM} \times \text{HCF}$$

In this case, the LCM is given as $6x^2y^2$, and the HCF is given as $2xy^2$.

Now, multiply the LCM and HCF to find the product of the polynomials:

$$\text{Product} = (6x^2y^2) \times (2xy^2)$$

Multiply the coefficients and combine the like terms:

$$\text{Product} = (6 \times 2) \times (x^2 \times x) \times (y^2 \times y^2)$$

$$\text{Product} = 12x^3y^4$$

So, the product of the two polynomials is $12x^3y^4$.

Then the correct option is C)

6. Correct option: (C)

To find the product of the expressions $(m+2)(m+5)(m+3)$, we can use the distributive property to multiply them step by step:

1. First, multiply the first two expressions:

$$(m+2)(m+5) = m(m) + m(5) + 2(m) + 2(5) = m^2 + 5m + 2m + 10 = m^2 + 7m + 10$$

2. Now, multiply the result from step 1 by the third expression (m+3):

$$(m^2 + 7m + 10)(m+3) = m^2(m) + m^2(3) + 7m(m) + 7m(3) + 10(m) + 10(3)$$

$$= m^3 + 3m^2 + 7m^2 + 21m + 10m + 30$$

$$= m^3 + 10m^2 + 31m + 30$$

So, the product of the expressions $(m+2)(m+5)(m+3)$ is equal to:

$$m^3 + 10m^2 + 31m + 30$$

Then the correct option is C)

7. Correct option: (A)

To find the product of the expressions $(2x^2 + 3x + 4)((2x^2)^2 + (3x)^2 + (4)^2 - 6x^3 - 12x - 8x^2)$, we can simplify the inner expression first:

After solving the inner expression

$$4x^4 + 9x^2 + 16 - 6x^3 - 12x - 8x^2$$

Next, let's combine like terms in the inner expression:

$$4x^4 - 6x^3 + x^2 - 12x + 16$$

Now, let's multiply this simplified inner expression by $(2x^2 + 3x + 4)$:

$$(2x^2 + 3x + 4)(4x^4 - 6x^3 + x^2 - 12x + 16)$$

To find the product, we can use the distributive property to multiply each term in the first expression by each term in the second expression and then combine like terms:

First, distribute $(2x^2 + 3x + 4)$ to each term in the second expression:

$$(2x^2)(4x^4) + (2x^2)(-6x^3) + (2x^2)(x^2) + (2x^2)(-12x) + (2x^2)(16) +$$
$$(3x)(4x^4) + (3x)(-6x^3) + (3x)(x^2) + (3x)(-12x) + (3x)(16) +$$
$$(4)(4x^4) + (4)(-6x^3) + (4)(x^2) + (4)(-12x) + (4)(16)$$

Now, calculate each of these products:

$$8x^6 - 12x^5 + 2x^4 - 24x^3 + 32x^2 +$$
$$12x^5 - 18x^4 + 3x^3 - 36x^2 + 48x +$$
$$16x^4 - 24x^3 + 4x^2 - 48x + 64$$

Now, combine like terms:

$$8x^6 - 45x^3 + 64$$

Now, simplify further:

So, the product of the expressions is $8x^6 - 45x^3 + 64$

Then the correct option is A)

8. Correct option: (A)

To simplify the expression $\frac{a^9 - b^9}{a^3 - b^3}$, we can use the formula for the difference of cubes, which states that:

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2).$$

So, the expression becomes:

$$\frac{a^9 - b^9}{a^3 - b^3} = \frac{[(a^3)^3 - (b^3)^3]}{a^3 - b^3}$$

Now, notice that we have a difference of cubes in the numerator, which we can factor:

$$(a^3)^3 - (b^3)^3 = (a^3 - b^3)(a^6 + a^3b^3 + b^6).$$

So, the expression becomes:

$$\frac{[(a^3 - b^3)(a^6 + a^3b^3 + b^6)]}{[a^3 - b^3]}$$

Now, we can cancel the common factor $(a^3 - b^3)$ from the numerator and the denominator:

$$\frac{[(a^3 - b^3)(a^6 + a^3b^3 + b^6)]}{[a^3 - b^3]} = (a^6 + a^3b^3 + c^6)$$

So, the simplified expression is:

$$a^6 + a^3b^3 + b^6.$$

Then the correct option is A)

9. Correct option: (C)

To find the highest common factor (HCF) of two numbers or expressions, we can compare their prime factorizations and take the lowest power of each common prime factor.

Let's factorize both A and B:

$$A = 384x^4y^5z^3$$

$$B = 256x^2y^3z^3$$

Now, let's find the HCF:

- For the numbers 384 and 256, we can find their common factors:

$$384 = 2^7 \times 3$$

$$256 = 2^8$$

So, the common factor for both is 2^7.

For x, the common factor is x^2 (the lowest power).

For y, the common factor is y^3 (the lowest power).

For z, the common factor is z^3 (the lowest power).

Now, multiply all these common factors together:

$$HCF = 2^7 \times x^2 \times y^3 \times z^3$$

So, the HCF of A and B is $2^7 \times x^2 \times y^3 \times z^3$.

Then the correct option is C)

10. Correct option: (C)

1. Factorize each expression:

$22h^2t = 2 \times 11 \times h \times h \times t$

$$33ht^2 = 3 \times 11 \times h \times t \times t$$

2. Now, identify the common factors:

The common factors among these expressions are 11, h, and t.

3. Calculate the HCF by multiplying these common factors:

$$HCF = 11 \times h \times t = 11ht$$

Then the correct option is C)

11. Correct option: (A, C)

Let's analyze the given expressions:

$$1.\ 14x^2y^2$$
$$2.\ 21x^2y^2$$
$$3.\ 35x^4y^5z$$

Numerical Coefficients:

- The numerical coefficients are 14, 21, and 35.

Literals (variables and their exponents):

- The literals for the first two expressions are x^2y^2.
- The literal for the third expression is x^4y^5z.

Now, let's consider each statement:

A) The HCF of numerical coefficients is 7.

- This statement is true because the numerical coefficients 14, 21, and 35 have a common factor of 7.

B) The HCF of numerical coefficients is 35.

- This statement is not true. The highest common factor (HCF) of the numerical coefficients is 7, not 35.

C) The HCF of literals is x^2y^2.

- This statement is true because the first two expressions have the common literal factor x^2y^2.

D) The HCF of literals is x^4y^5z.

- This statement is not true. The HCF of literals for the given expressions is x^2y^2, not x^4y^5z.

So, the correct statements are A) The HCF of numerical coefficients is 7 and C) The HCF of literals is x^2y^2.

12. Correct option: (C, D)

Let's analyze the given expressions:

$$1.\ 12x^2y^3z$$
$$2.\ 6xy^2z^4$$
$$3.\ 8x^4yz^3$$

Numerical Coefficients: - The numerical coefficients are 12, 6, and 8.

Literals (variables and their exponents):

- The literals for the first expression are x^2y^3z.
- The literals for the second expression are xy^2z^4.
- The literals for the third expression are x^4yz^3.

Now, let's consider each statement:

A) The HCF of numerical coefficients is 6.

- This statement is false because the highest common factor (HCF) of 12, 6, and 8 is indeed 2.

B) The LCM of numerical coefficients is 36.

- This statement is not true. The least common multiple (LCM) of the numerical coefficients 12, 6, and 8 is 24, not 36.

C) The HCF of literals is xyz.

- This statement is true because the HCF of the literals $x^2y^3z, xy^2z^4,$ and x^4yz^3 is xyz.

D) The LCM of literals is $x^4y^3z^4$.

- This statement is true because the LCM of the literals $x^2y^3z, xy^2z^4,$ and x^4yz^3 is $x^4y^3z^4$.

So, the correct statements are C) The HCF of literals is xyz, and D) The LCM of literals is $x^4y^3z^4$.

13. Correct option: (A, C)

To determine what expression divides evenly into the given expression $a^2 + b^2 + 2b$, we can factor the expression further to find its divisors.

$a^2 + b^2 + 2b$ Can be factored as follows: $a^2 + b^2 + 2b = (a + b)^2$

Now $a^2 + b^2 + 2b = (a + b)^2$

We can say that

$$\frac{(a+b)^2}{a+b} = (a+b), \frac{(a+b)^2}{(a+b)^2} = 1$$

Therefore, the expression $a^2 + b^2 + 2ab$ is divide by $(a + b)^2$, $(a + b)$.

14. Correct option: (A)

To find the highest common factor (HCF) of the two polynomials $g(x) = 15x^3y^3z$ and $f(x) = 25xy^3z^2$, we can use the relationship between the LCM (Least Common Multiple), HCF, and the product of the two polynomials:

$$\text{LCM} \times \text{HCF} = \text{Product of the two polynomials}$$

Given that the LCM is $75x^3y^3z^2$, we can rearrange the equation to find the HCF:

$$\text{HCF} = \frac{\text{Product of the two polynomials}}{\text{LCM}}$$

First, calculate the product of the two polynomials:

$$\text{Product} = g(x) \times f(x) = (15x^3y^3z) \times (25xy^3z^2) = 375x^4y^6z^3$$

Now, divide the product by the LCM to find the HCF:

$$\text{HCF} = (375x^4y^6z^3)/(75x^3y^3z^2)$$

Now, simplify:

$$\text{HCF} = 5x^{4-3} \times y^{6-3} \times z^{3-2} = 5xy^3z$$

So, the highest common factor (HCF) of $g(x) = 15x^3y^3z$ and $f(x) = 25xy^3z^2$ is $5xy^3z$.

Then the correct option is A)

15. Correct option: (B)

We know that

$$\text{LCM} \times \text{HCF} = \text{Product of the two monomials}$$

In this case, the HCF is given as $5p^2q^3$. The two monomials are:

$$1.\ 15p^3q^4$$
$$2.\ 20m^2p^2q^3$$

Now, we can use the formula to find the LCM:

$$\text{LCM} \times \text{HCF} = (15p^3q^4) \times (20m^2p^2q^3)$$

Now, divide both sides by the HCF to find the LCM:

$$\text{LCM} = ((15p^3q^4) \times (20m^2p^2q^3))/(5p^2q^3)$$

Now, simplify:

$$\text{LCM} = \frac{300m^2p^5q^7}{5p^2q^3} = 60m^2p^3q^4$$

So, the least common multiple (LCM) of $15p^3q^4$ and $20m^2p^2q^3$, given their HCF is $5p^2q^3$, is $60m^2p^3q^4$.

Then the correct option is B)

16. Correct option: (A)

We know that $\text{LCM} \times \text{HCF} = \text{Product of the two monomials}$

In this case, we have: $\text{LCM} = 36k^3m^2n^5$

$$\text{HCF} = 216k^3m^3n^5$$
$$\text{One of the monomials} = 72k^2m^2n^4$$

Now, we can use the formula to find the other monomial:

$$\text{Other monomial} = \frac{36k^3m^2n^5 \times 216k^3m^3n^5}{72k^2m^2n^4}$$

Cancel common factors: $= 108k^4m^3n^6$
So, the other monomial is $= 108k^4m^3n^6$
Then the correct option is A)

Given $A = 2x^2 - 3x + 5, B = 5x + 2, C = x^2 + 3x - 7$

17. Correct Matching is: a-Q,b-T,c-S,d-P

a)

Find the LCM by taking the highest power of each factor that appears in both expressions:
- The LCM of $x^3 \times y^2$ and $-5 \times y^2$ is: $LCM = x^3 \times y^2 \times (5)$
So, the LCM of x^3y^2 and $-5y^2$ is $5x^3y^2$.
a) -→(Q)

b)

The least common multiple (LCM) of $4x^3, 6y^2,$ and $8z^2$ is:

$$LCM(4x^3, 6y^2, 8z^2) = LCM(2^2x^3, 2 \times 3y^2, 2^3z^2) = 2^3 \times 3x^3y^2z^2$$

So, the LCM is $24x^3y^2z^2$.
b) -→(T)

c)

The least common multiple (LCM) of $2x^3y^3, 10y^2x^3,$ and $18xy$ is:

$$LCM(2x^3y^3, 10y^2x^3, 18xy) = LCM(2x^3y^3, 2 \times 5y^2x^3, 2 \times 3^2xy) = 2 \times 5 \times 3^2x^3y^3$$

So, the LCM is $90x^3y^3$.
c) -→(S)

d)

The least common multiple (LCM) of $25x^3, 15x^2,$ and $45x^3$ is:

$$LCM(5^2x^3, 3 \times 5x^2, 3^2 \times 5x^3) = 225x^3$$

So, the LCM is $225x^3$.
d) -→(P)

18. Correct Answer is: (6)

To find the least common multiple (LCM) of $2x^2, 3x^4,$ and $6x^8$ when x = 1, simply substitute x = 1 into each term and calculate:

$$LCM(2x^2, 3x^4, 6x^8) = LCM(2(1)^2, 3(1)^4, 6(1)^8) = LCM(2, 3, 6)$$

Now, find the LCM of the numbers 2, 3, and 6:

$$LCM(2, 3, 6) = 6$$

So, the LCM of $2x^2, 3x^4,$ and $6x^8$ when x = 1 is 6.

19. Correct Answer is: (3)

To find the highest common factor (HCF) of $3x^2, 9y^3,$ and $3z^4$, we can identify the common factors for each term and take the lowest power of those common factors:

only common factor is 3

$$HCF(3x^2, 9y^3, 3z^4) = 3$$

So, the HCF is 3.

20. Correct Answer is: (7)

If we want to find the quotient of $(3x^2 - 2x - 1)$ when divided by (x - 1) and x = 2, we can substitute the value of x into the expression: $\frac{3x^2 - 2x - 1}{x - 1}$
Now, substitute x = 2:

$$\frac{3(2)^2 - 2(2) - 1}{2 - 1}$$

$$\frac{3(4) - 2(2) - 1}{1}$$

Simplify the numerator: $\frac{12 - 4 - 1}{1} = 7$
The quotient is 7.

STUDENT PRACTICE QUESTIONS

1. Correct option: (A)

Given expression $(10x^7 - 8x^6 + 3x^4) \div x^3$

Divide each term separately:

$$\frac{10x^7}{x^3} - \frac{8x^6}{x^3} + \frac{3x^4}{x^3} = 10x^4 - 8x^3 + 3x$$

So, the result of $(10x^7 - 8x^6 + 3x^4) \div x^3$ is:

$$10x^4 - 8x^3 + 3x$$

Then the correct option is A)

2. Correct option: (A)

To simplify the expression $(3x^2 - x - 14) \div (x + 2)$ using factorization, we can first factor the numerator and then cancel out common factors:

1. Factor the numerator $(3x^2 - x - 14)$:
   The quadratic expression $3x^2 - x - 14$ factors as $(3x - 7)(x + 2)$.
2. Now, rewrite the expression with the factored numerator:
   $[(3x - 7)(x + 2)] \div (x + 2)$
3. Cancel out the common factor $(x + 2)$ in both the numerator and denominator:

$$\frac{(3x - 7)(x + 2)}{x + 2} = 3x - 7$$

So, the simplified expression is $3x - 7$

Then the correct option is A)

3. Correct option: (A)

To simplify the expression $(15a^2 + 17ax - 4x^2) \div (3a + 4x)$ using factorization, we can first factor the numerator and then cancel out common factors:

1. Factor the numerator $(15a^2 + 17ax - 4x^2)$:
   The quadratic expression $15a^2 + 17ax - 4x^2$ factors as $(3a + 4x)(5a - x)$.
2. Now, rewrite the expression with the factored numerator:
   $$[(3a + 4x)(5a - x)] \div (3a + 4x)$$
3. Cancel out the common factor $(3a + 4x)$ in both the numerator and denominator:

$$\frac{(3a + 4x)(5a - x)}{3a + 4x} = 5a - x$$

So, the simplified expression is $5a - x$.

Then the correct option is A)

4. Correct option: (D)

To find the highest common factor (HCF) of the two polynomials pq and qrs, we can identify the common factors in each term and then determine the HCF:

1. Break down each polynomial into its prime factors:
   $$pq = p \times q$$
   $$qrs = q \times r \times s$$
2. Now, identify the common factors:
   The only common factor between pq and qrs is q.

Then the correct option is D)

5. Correct option: (B)

1. Factorize each expression:
   $$4x^2y = 2 \times 2 \times x \times x \times y$$
   $$6xy = 2 \times 3 \times x \times y$$
   $$8x^2y^2 = 2 \times 2 \times 2 \times x \times x \times y \times y$$

2. Now, identify the common factors:
   The common factors among these expressions are 2, x, and y.
3. Calculate the HCF by multiplying these common factors:

$$\text{HCF} = 2 \times x \times y = 2xy$$

Then the correct option is B)

6. Correct option: (D)

If three monomials are in the ratio $xy : yz : zx$ and their highest common factor (HCF) is xyz, we can express the monomials as follows:

1. First monomial: $\text{HCF}(xyz) \times (xy) = xyz \times xy = x^2y^2z$
2. Second monomial: $\text{HCF}(xyz) \times (yz) = xyz \times yz = xy^2z^2$
3. Third monomial: $\text{HCF}(xyz) \times (zx) = xyz \times zx = x^2yz^2$

Then the correct option is D)

7. Correct option: (C)

To find the least common multiple (LCM) of the three expressions $xy, yz,$ and xz, we can determine the LCM based on their prime factors.

1. Factorize each expression:

xy: Prime factors include x and y.
yz: Prime factors include y and z.
xz: Prime factors include x and z.

2. Calculate the LCM by multiplying these prime factors:

$$\text{LCM} = x \times y \times z$$

Then the correct option is C)

8. Correct option: (A)

1. Factorize each expression:

$120x^2y^3$: Prime factors include $2, 3, 5, x^2,$ and y^3.
$36x^6y^8$: Prime factors include $2, 3, x^6,$ and y^8.

2. Now, identify the prime factors and their respective highest powers:
   – LCM will have the prime factors $2, 3, 5, x^6,$ and y^8.
   – The highest power of 2 is $2^3 = 8$.
   – The highest power of 3 is $3^1 = 3$.
   – The highest power of 5 is $5^1 = 5$.
   – The highest power of x is x^6.
   – The highest power of y is y^8.
3. Calculate the LCM by multiplying these highest powers:
   $\text{LCM} = 8 \times 3 \times 5 \times x^6 \times y^8 = 120x^6y^8$

Then the correct option is A)

9. Correct option: (A)

1. Factorize each expression:

a^2bc: Prime factors include a^2, b, c.
ab^2c: Prime factors include a, b^2, c.
abc^2: Prime factors include a, b, c^2.

2. Now, identify the prime factors and their respective highest powers:
   – LCM will have the prime factors $a^2, b^2,$ and c^2.
   – The highest power of a is a^2.
   – The highest power of b is b^2.
   – The highest power of c is c^2.
3. Calculate the LCM by multiplying these highest powers:

$$\text{LCM} = a^2 \times b^2 \times c^2$$

Then the correct option is A)

10. Correct option: (C)

Given that the least common multiple (LCM) of two monomials is $8x^2y^2$ and the highest common factor (HCF) is 2xy, and we already have one of the monomials as $4xy^2$, let's find the second monomial.

The LCM can be expressed as the product of the two monomials divided by their HCF:

$$\text{LCM} = \frac{\text{Monomial}_1 \times \text{Monomial}_2}{\text{HCF}}$$

Plugging in the values:

$$8x^2y^2 = \frac{4xy^2 \times \text{Monomial}_2}{2xy}$$

$$\text{Monomial}_2 = 8x^2y^2 \times \frac{2xy}{4xy^2} = 4x^2y$$

So, the second monomial is $4x^2y$.

Then the correct option is C)

11. Correct option: (A, B, C)

To determine what expression divides evenly into the given expression $a^2 - b^2$, we can factor the expression further to find its divisors.

$a^2 - b^2$ Can be factored as follows:

$$a^2 - b^2 = (a + b)(a - b)$$

Now we can say that

$$=> \frac{(a + b)(a - b)}{a + b} = (a - b), \frac{(a + b)(a - b)}{a - b} = (a + b)$$

$$\frac{(a + b)(a - b)}{(a + b)(a - b)} = 1$$

Therefore, the expression $a^2 - b^2$ is divide by

$(a + b), (a - b), (a + b)(a - b)$.

12. Correct option: (A, B, C ,D)

To determine what expression divides evenly into the given expression $a^4 - b^4$, we can factor the expression further to find its divisors.

$a^4 - b^4$ Can be factored as follows:

$$a^4 - b^4 = (a^2 + b^2)(a^2 - b^2)$$

Further we can factor this as

$$(a^2 + b^2)(a^2 - b^2) = (a^2 + b^2)(a - b)(a + b)$$

Now we can say that

$$=> \frac{(a^2 + b^2)(a + b)(a - b)}{a + b} = (a^2 + b^2)(a - b)$$

$$\frac{(a^2 + b^2)(a + b)(a - b)}{a - b} = (a^2 + b^2)(a + b)$$

$$\frac{(a^2 + b^2)(a + b)(a - b)}{(a + b)(a - b)} = (a^2 + b^2)$$

$$\frac{(a^2 + b^2)(a + b)(a - b)}{(a^2 + b^2)} = (a + b)(a - b)$$

Therefore, the expression $a^4 - b^4$ is divide by

$(a + b), (a - b), (a + b)(a - b), (a^2 + b^2)$

13. Correct option: (B)

To find the product of A and B, we need to multiply the expressions A and B:

$$A \times B = (2x^2 - 3x + 5) \times (5x + 2)$$

To calculate the product, we can use the distributive property and multiply each term in A by each term in B: $A \times B = (2x^2 \times 5x) + (2x^2 \times 2) + (-3x \times 5x) + (-3x \times 2) + 25x + 10$

Now, simplify each term: $A \times B = 10x^3 + 4x^2 - 15x^2 - 6x + 25x + 10$

Combine like terms: $A \times B = 10x^3 - 11x^2 + 19x + 10$

So, the product of A and B is $10x^3 - 11x^2 + 19x + 10$

Then the correct option is B)

14. Correct option: (C)

To find the product of B and C, we can simply multiply the two polynomials together:

$$B \times C = (5x + 2)(x^2 + 3x - 7)$$

Use the distributive property to perform the multiplication:

$$= 5x \times (x^2 + 3x - 7) + 2 \times (x^2 + 3x - 7)$$
$$= 5x^3 + 15x^2 - 35x + 2x^2 + 6x - 14$$

Simplify each term: $= 5x^3 + (15x^2 + 2x^2) + (-35x + 6x) - 14$

$$= 5x^3 + 17x^2 - 29x - 14$$

So, the product of B and C is $5x^3 + 17x^2 - 29x - 14$.

Then the correct option is C)

15. Correct Matching is: a-S,b-T,c-Q,d-P

a)

To find the highest common factor (HCF) of x^3y^2 and $-5y^2$, follow these steps:

1. Identify the common factors in both expressions.

$$\text{common factor}(y)$$

2. Determine the HCF by taking the lowest power of each common factor.

$$\text{HCF}(x^3y^2, -5y^2) = y^2$$

The HCF of x^3y^2 and $-5y^2$ is y^2.

a) -→(S)

b)

The highest common factor (HCF) of $4x^3, 6y^2,$ and $8z$ is: $\text{HCF}(4x^3, 6y^2, 8z) = 2$

So, the HCF is 2.

b) -→(T)

c)

The highest common factor (HCF) of $2x^3y^3, 10x^3y^2,$ and $14xy$ is:

$$\text{HCF}(2x^3y^3, 10x^3y^2, 14xy) = 2xy$$

So, the HCF is 2xy.

c) -→(Q)

d)

The highest common factor (HCF) of $5x^3, -15x^2,$ and $45x$ is: $\text{HCF}(5x^3, -15x^2, 45x) = 5x$

So, the HCF is 5x.

d) -→(P)

16. Correct Answer is: (4)

To find the least common multiple (LCM) of $abc, 2a^2b,$ and $4a^3b^2c$ when a = 1, b = 1, and c = 1, simply substitute these values into the expressions and calculate the LCM:

$$\text{LCM}(abc, 2a^2b, 4a^3b^2c) = \text{LCM}(1, 2 \times 1, 4 \times 1)$$

Now, calculate the LCM:

$$\text{LCM}(1, 2, 4) = 4$$

So, the LCM of $abc, 2a^2b,$ and $4a^3b^2c$ when a = 1, b = 1, and c = 1 is 4.

17. Correct Answer is: (5)

To find the highest common factor (HCF) of $35x^2y, 5xy^3,$ and $25x^2y^2$ when x = 1 and y = 1, simply substitute these values into the expressions and calculate the HCF:

$$\text{HCF}(35x^2y, 5xy^3, 25x^2y^2) = \text{HCF}(35,5,25)$$

Now, calculate the HCF: $\text{HCF}(35, 5, 25) = 5$

So, the HCF of $35x^2y, 5xy^3,$ and $25x^2y^2$ when x = 1 and y = 1 is 5.

18. Correct Answer is: (9)

To divide $153x^3y^2$ by $17x^3y^2$, we can simply cancel out the common factors in the numerator and denominator:

$$\frac{153x^3y^2}{17x^3y^2}$$

Now, cancel out the common factors:

$$\frac{153x^3y^2}{17x^3y^2} = \frac{153}{17} = 9$$

So, the result of dividing $153x^3y^2$ by $17x^3y^2$ is 9.

19. Correct Answer is: (0)

First, find the product of the expressions $(1 + x)(1 - x)(1 + x^2)$:

Now, use the difference of squares formula to simplify the first two factors:

$$(1 + x)(1 - x) = 1^2 - x^2 = 1 - x^2$$

So, the expression becomes:

$$(1 - x^2)(1 + x^2)$$

Now, apply the difference of squares formula again:

$$(1 - x^2)(1 + x^2) = 1^2 - x^4 = 1 - x^4$$

Now, we have the simplified expression:

$$1 - x^4$$

Now, put x = 1 into the simplified expression:

$$1 - (1)^4 = 1 - 1 = 0$$

So, when we put x = 1 into the expression $(1 + x)(1 - x)(1 + x^2)$, we get the result 0.

20. Correct Answer is: (3)

To divide $x(3x^2 - 108)$ by $(x^3 - 36x)$, follow these steps:

1. Factor both the numerator and denominator.

Numerator: $x(3x^2 - 108) = 3x^3 - 108x$

Denominator: $x^3 - 36x = x(x^2 - 36) = x(x - 6)(x + 6)$

2. Rewrite the division problem with the factored expressions:

$$\frac{3x^3 - 108x}{[x(x - 6)(x + 6)]}$$

3. Now, divide the numerator by the denominator. We can cancel common factors:

$$\frac{3x^3 - 108x}{[x(x - 6)(x + 6)]} = \frac{3x(x-6)(x+6)}{x(x-6)(x+6)} = 3$$

So, $\frac{x(3x^2 - 108)}{x^3 - 36x}$ simplifies to 3.

7. Exponents and Powers

Solutions

LEVEL – I

1. Correct option: (B) 45

$5 \times 3^2 = 5 \times 3 \times 3 = 5 \times 9 = 45$

2. Correct option: (A) 1600

$4^3 \times 5^2 = 4 \times 4 \times 4 \times 5 \times 5 = 64 \times 25 = 1600$

3. Correct option: (D) 11664

$(-3)^2 \times (6)^4 = (-3) \times (-3) \times 6 \times 6 \times 6 \times 6 = 9 \times 1296$

$= 11664$

4. Correct option: (C) -1

$(-1)^{2005} \times (-1)^{2016} = (-1) \times 1 = -1$

5. Correct option: (B) $-\frac{128}{675}$

Cube of $-\frac{2}{3} = \frac{-2}{3} \times \frac{-2}{3} \times \frac{-2}{3} = \frac{-8}{27}$

Square of $\frac{-4}{5} = \frac{-4}{5} \times \frac{-4}{5} = \frac{16}{25}$

Product of cube of $\frac{-2}{3}$ and square of $\frac{-4}{5}$ is

$$= \frac{-8}{27} \times \frac{16}{25}$$
$$= -\frac{128}{675}$$

6. Correct option: (B) $\frac{25}{16}$

$(-3)^2 \times \left(\frac{-5}{12}\right)^2 = 9 \times \frac{25}{144} = \frac{25}{16}$

7. Correct option: (B) $2^3 \times 5^2$

Exponential form of $2 \times 2 \times 2 \times 5 \times 5 = 2^3 \times 5^2$

8. Correct option: (A) $2^4 \times 3^2 \times 5^2$

Exponential form of 3600 is

$$= 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 5$$
$$= 2^4 \times 3^2 \times 5^2$$

9. Correct option: (D) $\left(\frac{-1}{3}\right)^5$

Exponential form of $\frac{-1}{243}$ is

$$= \frac{-1}{3} \times \frac{-1}{3} \times \frac{-1}{3} \times \frac{-1}{3} \times \frac{-1}{3}$$
$$= \left(\frac{-1}{3}\right)^5$$

10. Correct option: (B) 3^{14}

$3^2 \times 3^4 \times 3^8 = 3^{2+4+8} = 3^{14}$

11. Correct option: (C) 5^3

$\{(5^2)^3 \times 5^4\} \div 5^7 = \frac{5^6 \times 5^4}{5^7}$

$$= \frac{5^{10}}{5^7}$$
$$= 5^{10-7}$$
$$= 5^3$$

12. Correct option: (D) 1

$\frac{3^5 \times 10^5 \times 25}{5^7 \times 6^5} = \frac{3^5 \times 2^5 \times 5^5 \times 5^2}{5^7 \times 2^5 \times 3^5} = 1$

13. Correct option: (B) $3^6 \times 2^6$

Prime factors of 729 are $3 \times 3 \times 3 \times 3 \times 3 \times 3$ and 64 are $2 \times 2 \times 2 \times 2 \times 2 \times 2$

Expression of 729×64 is given as $3^6 \times 2^6$

14. Correct option: (C) 3

Given $5^{n-2} \times 3^{2n-3} = 135$

$$5^{n-2} \times 3^{2n-3} = 3 \times 3 \times 3 \times 5$$

$$5^{n-2} \times 3^{2n-3} = 5^1 \times 3^3$$

Equating powers since base are equal

$n - 2 = 1$ or $2n - 3 = 3$

So, $n = 3$

15. Correct option: (C) $\frac{729}{64}$

Given $\frac{p}{q} = \left(\frac{2}{3}\right)^2 \div \left(\frac{6}{7}\right)^0$

$$\frac{p}{q} = \frac{4}{9} \div 1$$

$$\frac{p}{q} = \frac{4}{9}$$

Now, $\frac{q}{p} = \frac{9}{4}$

$$\left(\frac{q}{p}\right)^3 = \left(\frac{9}{4}\right)^3 = \frac{729}{64}$$

16. Correct option: (A) 2

Given

$$7^{2n+1} \div 49 = 7^3$$

$$\frac{7^{2n}7^1}{49} = 7^3$$

$$\frac{7^{2n}}{7} = 7^3$$

$$7^{2n} = 7^4$$

Equating powers, since base is same

$2n = 4$

$n = 2$

17. Correct option: (D) $\frac{3}{5}$

$$\frac{10 \times 5^{n+1} + 25 \times 5^n}{3 \times 5^{n+2} + 10 \times 5^{n+1}} = \frac{10 \times (5^n \times 5^1) + 25 \times 5^n}{3 \times (5^n \times 5^2) + 10 \times 5^n \times 5^1}$$

$$= \frac{10 \times 5 \times 5^n + 25 \times 5^n}{3 \times 5^n \times 25 + 10 \times 5 \times 5^n}$$

$$= \frac{5^n(50 + 25)}{5^n(75 + 50)}$$

$$= \frac{75}{125}$$

$$= \frac{3}{5}$$

18. Correct option: (D) 3.8012×10^7

Standard form of 3,80,12,000 is 3.8012×10^7

19. Correct option: (C) 1.4×10^9

The diameter of sun 1,400,000,000 m in standard form is 1.4×10^9

20. Correct option: (A) 3.628×10^{-4}

Standard form of 0.0003628 is 3.628×10^{-4}

LEVEL – II

1. Correct option: (A) $2^3 \times 5^2 \times x^3$

Exponential form of $2 \times 2 \times 2 \times 5 \times 5 \times x \times x \times x$ is

$$2^3 \times 5^2 \times x^3$$

2. Correct option: (D) 625

The value of 5^4 is

$$= 5 \times 5 \times 5 \times 5$$
$$= 625$$

3. Correct option: (B) $100a^7$

$5 \times (2a)^2 \times 5a^5 =$

$$5 \times 4a^2 \times 5a^5$$
$$= 100a^7$$

4. Correct option: (C) 1

$$(-1)^{2018} \times (-1)^{2019} \times (-1)^{2020} \times (-1)^{2021}$$

Even power = positive number

Odd power = negative number

$$= 1 \times (-1) \times 1 \times (-1)$$
$$= 1$$

5. Correct option: (D) 0

$$(-1)^1 + (-1)^2 + (-1)^3 + (-1)^4 + - - - + (-1)^{99} + (-1)^{100}$$

Even power = positive number

Odd power = negative number

$$= -1 + 1 - 1 \ldots \ldots . -1 + 1$$
$$= 0$$

6. Correct option: (B) $\frac{12}{7}$

$$(3^{-1} + 4^{-1})^{-1} = \left(\frac{1}{3} + \frac{1}{4}\right)^{-1}$$
$$= \left(\frac{4+3}{12}\right)^{-1}$$
$$= \left(\frac{7}{12}\right)^{-1}$$
$$= \frac{12}{7}$$

7. Correct option: (B) 1

$$(2^0 + 3^0)^0 = (1+1)^0$$
$$= 2^0$$
$$= 1$$

8. Correct option: (A) 1

Given $3^{x+1} = 9$

$$3^{x+1} = 3^2$$

Equating powers, since base is equal $x + 1 = 2$

$$x = 1$$

9. Correct option: (C) $\frac{1}{60}$

$$\left(\begin{matrix}5^{-1} \\ \times 2^{-1}\end{matrix}\right) \times 6^{-1}$$
$$= \left(\frac{1}{5} \times \frac{1}{2}\right) \times \frac{1}{6}$$

$$= \frac{1}{10} \times \frac{1}{6}$$
$$= \frac{1}{60}$$

10. Correct option: (D) $\frac{5}{2}$

Given $4^x - 4^{x-1} = 24$

$$4^x - 4^x 4^{-1} = 24$$
$$4^x \left(1 - \frac{1}{4}\right) = 24$$
$$4^x \times \frac{3}{4} = 24$$
$$4^x = \frac{24 \times 4}{3}$$
$$4^x = 8 \times 4$$
$$4^x = 32$$
$$2^{2x} = 2^5$$
$$2x = 5$$
$$x = \frac{5}{2}$$

11. Correct option: (C) and (D)

$$(5^{-2})^3 = (5^3)^{-2} = 5^{-6}$$

12. Correct option: (A) $\frac{125}{9}$

$$\left(\frac{3}{8}\right)^{-2} \times \left(\frac{4}{5}\right)^{-3} = \left(\frac{8}{3}\right)^{2} \times \left(\frac{5}{4}\right)^{3}$$
$$= \frac{8 \times 8 \times 5 \times 5 \times 5}{3 \times 3 \times 4 \times 4 \times 4}$$
$$= \frac{2 \times 2 \times 125}{9 \times 4}$$
$$= \frac{125}{9}$$

13. Correct option: (D) 625

$$\left[\left\{\left(\frac{-1}{5}\right)^{-2}\right\}^2\right]^1 = [\{(-5)^2\}^2]^1$$
$$= [\{25\}^2]^1$$
$$= [625]^1$$
$$= 625$$

14. Correct option: (D) 0

Given

$$[(x^a)^2]^{-1} = x^b$$
$$[x^{2a}]^{-1} = x^b$$
$$\frac{1}{x^{2a}} = x^b$$
$$x^{2a} x^b = 1$$
$$x^{2a+b} = x^0$$
$$2a + b = 0$$

15. Correct option: (A) 1/2

Given $2^{-m} \times \frac{1}{2^m} = \frac{1}{4}$

$$\frac{1}{2^m} \times \frac{1}{2^m} = \frac{1}{4}$$

$$\frac{1}{4^m} = \frac{1}{4}$$
$$4^m = 4^1$$
$$m = 1$$

Now, put value of m in expression below

$$\frac{1}{14}\left\{(4^m)^{\frac{1}{2}} + \left(\frac{1}{5^m}\right)^{-1}\right\}$$
$$= \frac{1}{14}\left\{(4^1)^{\frac{1}{2}} + \left(\frac{1}{5^1}\right)^{-1}\right\}$$
$$= \frac{1}{14}\left\{(4)^{\frac{1}{2}} + \left(\frac{1}{5}\right)^{-1}\right\}$$
$$= \frac{1}{14}\left\{(2^2)^{\frac{1}{2}} + 5\right\}$$
$$= \frac{1}{14}\{2 + 5\}$$
$$= \frac{1}{14} \times 7$$
$$= \frac{1}{2}$$

16. Correct option: (C) $\frac{9}{4}$

$$\left(\frac{64}{125}\right)^{-\frac{2}{3}} \div \frac{1}{\left(\frac{256}{625}\right)^{\frac{1}{4}}} + \left[\frac{63}{80}\right]^0$$
$$= \left(\frac{64}{125}\right)^{-\frac{2}{3}} \div \frac{1}{\left(\frac{256}{625}\right)^{\frac{1}{4}}} + 1$$
$$= \left[\frac{(4)^3}{(5)^3}\right]^{-\frac{2}{3}} \div \frac{1}{\left[\frac{(4)^4}{(5)^4}\right]^{\frac{1}{4}}} + 1$$
$$= \left(\frac{4}{5}\right)^{-2} \div \frac{1}{\left(\frac{4}{5}\right)} + 1$$
$$= \left(\frac{5}{4}\right)^2 \div \frac{5}{4} + 1$$
$$= \frac{25}{16} \times \frac{4}{5} + 1$$
$$= \frac{5}{4} + 1$$
$$= \frac{9}{4}$$

17. Correct Matching is: $a \to Q; b \to R; c \to S; d \to P$

a) Given $2^{a+1} \times 5^a = 200$

$$2^a \times 2^1 \times 5^a = 200$$
$$2^a \times 5^a = \frac{200}{2}$$
$$(2 \times 5)^a = 100$$
$$10^a = 10^2$$
$$a = 2$$

b) Given

$$\frac{2^{x-1} \times 4^{2x+1}}{8^{x-1}} = 64$$
$$\frac{2^{x-1} \times (2^2)^{2x+1}}{(2^3)^{x-1}} = 2^6$$
$$\frac{2^{x-1} \times 2^{4x+2}}{2^{3x-3}} = 2^6$$
$$\frac{2^{x-1+(4x+2)}}{2^{3x-3}} = 2^6$$
$$2^{x-1+4x+2} = 2^6 \times 2^{3x-3}$$
$$2^{5x+1} = 2^{6+3x-3}$$
$$2^{5x+1} = 2^{3x+3}$$
$$5x + 1 = 3x + 3$$
$$5x - 3x = 3 - 1$$
$$2x = 2$$
$$x = 1$$

c) $\frac{2^1+2^0+2^{-1}}{2^{-2}+2^{-3}+2^{-4}}$

$$= \frac{2 + 1 + \frac{1}{2}}{\frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4}}$$
$$= \frac{3 + \frac{1}{2}}{\frac{1}{4} + \frac{1}{8} + \frac{1}{16}}$$
$$= \frac{\frac{7}{2}}{\frac{4 + 2 + 1}{16}}$$
$$= \frac{\left(\frac{7}{2}\right)}{\left(\frac{7}{16}\right)}$$
$$= \frac{16 \times 7}{2 \times 7}$$
$$= 8$$

d) $5^{-5} \times 5^{2n+1} = 5^{13} \div 5^5$

$$5^{-5+2n+1} = \frac{5^{13}}{5^5}$$
$$5^{2n-4} = 5^8$$
$$5^{2n} \times 5^{-4} = 5^8$$
$$\frac{5^{2n}}{5^4} = 5^8$$
$$5^{2n} = 5^8 \times 5^4$$
$$5^{2n} = 5^{12}$$
$$2n = 12$$
$$n = 6$$

18. Correct Answer is: 2

Given

$$2^{2x-1} = \frac{1}{8^{x-3}}$$
$$2^{2x-1} \times (2^3)^{x-3} = 1$$

$$2^{2x-1} \times (2)^{3x-9} = 1$$
$$2^{2x-1+3x-9} = 1$$
$$2^{5x-10} = 2^0$$
$$5x - 10 = 0$$
$$5x = 10$$
$$x = 2$$

19. Correct Answer is: 1

Given

$$\frac{p}{q} = \left(\frac{4}{3}\right)^2 \div \left(\frac{3}{4}\right)^{-2}$$
$$\frac{p}{q} = \frac{\left(\frac{4}{3}\right)^2}{\left(\frac{4}{3}\right)^2}$$
$$\frac{p}{q} = 1$$

Value of $\left(\frac{p}{q}\right)^{-10}$ is

$$= (1)^{-10}$$
$$= 1$$

20. Correct Answer is: 1

$$\frac{2^{m+3} \times 3^{2m-n} \times 5^{m+n+3} \times 6^{n+1}}{6^{m+1} \times 10^{n+3} \times 15^m}$$
$$= \frac{2^{m+3} \times 3^{2m-n} \times 5^{m+n+3} \times 2^{n+1} \times 3^{n+1}}{2^{m+1} \times 3^{m+1} \times 2^{n+3} \times 5^{n+3} \times 3^m \times 5^m}$$
$$= \frac{2^{m+n+4} \times 3^{2m+1} \times 5^{m+n+3}}{2^{m+n+4} \times 3^{2m+1} \times 5^{m+n+3}}$$
$$= 1$$

STUDENT PRACTICE QUESTIONS

1. Correct option: (C) 72

$3^2 \times 2^3 = 3 \times 3 \times 2 \times 2 \times 2 = 9 \times 8 = 72$

2. Correct option: (B) -288

$(-2)^5 \times 3^2 = (-2) \times (-2) \times (-2) \times (-2) \times (-2) \times 3 \times 3$
$$= (-32) \times 9$$
$$= -288$$

3. Correct option: (D) 0

$$(-1)^{2001} + (-1)^{2002} + (-1)^{2003} + (-1)^{2004} + \cdots (-1)^{2009} + (-1)^{2010}$$

Even power = positive number

Odd power = negative number

$$= -1 + 1 - 1 + 1 - \cdots - 1 + 1$$
$$= 0$$

4. Correct option: (C) 125

$\left[(3^2 - 2^2) \div \left(\frac{1}{5}\right)^2\right] = \left[(9 - 4) \div \frac{1}{25}\right]$
$$= 5 \times 25$$
$$= 125$$

5. Correct option: (B) 5^5

Exponential form of 3125 is $= 5 \times 5 \times 5 \times 5 \times 5$

$$= 5^5$$

6. Correct option: (B) 2

Given $25^{n-1} + 100 = 5^{2n-1}$

$$(5^2)^{n-1} + 100 = 5^{2n-1}$$
$$5^{2n-2} - 5^{2n-1} + 100 = 0$$
$$5^{2n-1} - 5^{2n-2} = 100$$
$$5^{2n}5^{-1} - 5^{2n}5^{-2} = 100$$
$$5^{2n}(5^{-1} - 5^{-2}) = 100$$
$$25^n\left(\frac{1}{5} - \frac{1}{25}\right) = 100$$
$$25^n\left(\frac{5-1}{25}\right) = 100$$
$$25^n = \frac{100 \times 25}{4}$$
$$25^n = 25 \times 25$$
$$25^n = 25^2$$

Equating powers, since base is equal

n = 2

7. Correct option: (C) 4

$$\frac{9^n \times 3^2 \times 3^n - 27^n}{(3^3)^5 \times 2^3} = \frac{1}{27}$$
$$\frac{9 \times 27^n - 27^n}{27^5 \times 8} = \frac{1}{27}$$
$$27^n(9-1) = \frac{27^5 \times 8}{27}$$
$$27^n \times 8 = 27^4 \times 8$$
$$27^n = 27^4$$
$$n = 4$$

8. Correct option: (A) 8.6×10^7

Standard form of 86000000 is 8.6×10^7

9. Correct option: (A) 3.87×10^{-2}

Standard form of 0.0000387×10^3 is

$$0.0000387 \times 1000$$
$$= 0.0387$$
$$= 3.87 \times 10^{-2}$$

10. Correct option: (B) 8.32×10^{-3}

Standard form of 0.00832 is 8.32×10^{-3}

11. Correct option: (B) and (D)

$$\left(\frac{25}{9}\right)^5 = \left(\frac{5 \times 5}{3 \times 3}\right)^5$$
$$= \left[\left(\frac{5}{3}\right)^2\right]^5$$
$$= \left(\frac{5}{3}\right)^{10}$$
$$= \left(\frac{3}{5}\right)^{-10}$$

12. Correct option: (A) $-25a^8$ and (C) -5^2a^8

$$(-1)^5 \times (a^2)^3 \times (5a)^2 = -1 \times a^6 \times 5^2 \times a^2$$
$$= -5^2a^8$$

$$= -25a^8$$

13. Correct option: (B) $\frac{125}{4}$

$$\frac{8^{-2} \times 5^3}{2^{-4}} = \frac{\frac{1}{8^2} \times 125}{\frac{1}{2^4}}$$

$$= \frac{125 \times 2^4}{8^2}$$

$$= \frac{125 \times 16}{64}$$

$$= \frac{125}{4}$$

14. Correct option: (C) 2

Given $5^m \div 5^{-3} = 5^5$

$$\frac{5^m}{5^{-3}} = 5^5$$

$$\frac{5^5}{\left(\frac{1}{5^3}\right)} = 5^5$$

$$5^m \times 5^3 = 5^5$$

$$5^{m+3} = 5^5$$

$$m + 3 = 5$$

$$m = 2$$

15. Correct Matching is: a → Q; b → S; c → P; d → R

a) $\frac{25 \times t^{-4}}{5^{-3} \times 10 \times t^{-8}}$

$$= \frac{25 \times 5^3 \times t^8}{10 \times t^4}$$

$$= \frac{5 \times 125 \times t^4}{2}$$

$$= \frac{625t^4}{2}$$

b) $\left[\left(\frac{1}{3}\right)^{-3} - \left(\frac{1}{2}\right)^{-3}\right] \div \left[\frac{1}{4}\right]^{-3}$

$$= [(3)^3 - (2)^3] \div (4)^3$$

$$= (27 - 8) \div 64$$

$$= \frac{19}{64}$$

c) $a^m . a^n . b^{-(m+n)}$

$$= a^{m+n} . \frac{1}{b^{m+n}}$$

$$= \left(\frac{a}{b}\right)^{m+n}$$

d) $a^m . a^n . b^{m+n}$

$$= a^{m+n} b^{m+n}$$

$$= (ab)^{m+n}$$

16. Correct Answer is: 4

Given

$$y = \left[\frac{x^{ab}}{x^{bc}}\right]^{-1} \cdot \left[\frac{x^{ca}}{x^{ab}}\right]^{-1} \cdot \left[\frac{x^{bc}}{x^{ca}}\right]^{-1}$$

$$y = \left[\frac{x^{bc}}{x^{ab}}\right] \cdot \left[\frac{x^{ab}}{x^{ca}}\right] \cdot \left[\frac{x^{ca}}{x^{bc}}\right]$$

$$y = 1$$

Now $y^2 + 2y + 1$

$$= (1)^2 + 2(1) + 1$$

$$= 1 + 2 + 1$$

$$= 4$$

17. Correct Answer is: 1

Given $$x^a = \left[\left[\left[\frac{1}{x^{-b}}\right]^{-1}\right]^{-b}\right]^{\frac{-1}{b}}$$

$$x^a = \left[\left[\left[\left[\frac{1}{\left(\frac{1}{x^b}\right)}\right]^{-1}\right]^{-b}\right]^{\frac{-1}{b}}\right]$$

$$x^a = \left[\left[\left[x^b\right]^{-1}\right]^{-b}\right]^{\frac{-1}{b}}$$

$$x^a = \left[\left[x^{-b}\right]^{-b}\right]^{\frac{-1}{b}}$$

$$x^a = \left[\left[\frac{1}{x^b}\right]^{-b}\right]^{\frac{-1}{b}}$$

$$x^a = \left[\frac{1}{x^b}\right]^1$$

$$x^a x^b = 1$$

$$x^{a+b} = 1$$

$$x^{a+b} = x^0$$

$$a + b = 0$$

$$a = -b$$

Now value of $\frac{x^{-b}.x^b}{x^a.x^b} + \frac{x^{a+b}}{x^{a^3+b^3}} - x^{a^2-b^2}$

$$= \frac{1}{x^{-b}x^b} + \frac{x^a x^b}{x^{(-b)^3+b^3}} - x^{(-b)^2-b^2}$$

$$= 1 + \frac{x^{-b}x^b}{x^{-b^3+b^3}} - x^{b^2-b^2}$$

$$= 1 + \frac{1}{x^0} - x^0$$

$$= 1 + \frac{1}{1} - 1$$

$$= 1$$

18. Correct Answer is: 9

$$\frac{3^{n+5} - 6.3^{n+2}}{3^{n+1} \times 7} = \frac{3^n \times 3^5 - 6 \times 3^n \times 3^2}{3^n \times 3^1 \times 7}$$

$$= \frac{3^n \times 3(3^4 - 6 \times 3)}{3^n \times 3 \times 7}$$

$$= \frac{81 - 18}{7}$$

$$= \frac{63}{7}$$

$$= 9$$

19. Correct Answer is: The value of $\frac{x}{y}$ is 2

Given

$$3^{x-1} = 27$$

$$3^{x-1} = 3^3$$

$$x - 1 = 3$$

$$x = 4$$

And

$$4^{y+2} = 256$$

$$4^{y+2} = 4^4$$

$$y + 2 = 4$$

$$y = 2$$

Now value of $\frac{x}{y}$ is

$$= \frac{4}{2}$$

$$= 2$$

20. Correct Answer is: 1

$$3^{3x-5} = \frac{1}{9^x}$$

$$3^{3x} \times 3^{-5} \times 9^x = 1$$

$$\frac{3^{3x} \times (3^2)^x}{3^5} = 1$$

$$3^{3x} \times 3^{2x} = 3^5$$

$$3^{3x+2x} = 3^5$$

$$5x = 5$$

$$x = 1$$

8. Perimeter and Area

Solutions

LEVEL – I

1. Correct option: (B) 16 sq cm

Given

Side of square = 4cm

Area of square = side × side

$$= 4 \times 4$$
$$= 16cm^2$$

2. Correct option: (D) 49 sq cm

Given

Diagonal of square = $7\sqrt{2}cm$

Area of square = $\frac{1}{2}d^2$

$$= \frac{1}{2}\left(7\sqrt{2}\right)^2$$
$$= \frac{1}{2} \times 49 \times 2$$
$$= 49cm^2$$

3. Correct option: (A) 128 sq cm

Given

Diagonal of square = $16cm$

Area of square = $\frac{1}{2}d^2$

$$= \frac{1}{2} \times (16)^2$$
$$= \frac{1}{2} \times 256$$
$$= 128cm^2$$

4. Correct option: (C) 19 m

Given

Area of square = 361 sq. m

Side be x

Area of square = $(side)^2$

$$361 = x^2$$
$$x = \sqrt{361}$$
$$x = 19m$$

5. Correct option: (A) $36\sqrt{2}$

Given

Area of square = 1296 sq m

Diagonal be d

Area of square = $\frac{1}{2}d^2$

$$1296 = \frac{1}{2}d^2$$
$$2592 = d^2$$
$$d = \sqrt{2592}$$
$$d = 36\sqrt{2}$$

6. Correct option: (C) $2\sqrt{2}$

Given

Area of square is double of its diagonal

$$A = 2d$$

Area of square = $\frac{1}{2}d^2$

$$2d = \frac{1}{2}d^2$$

$$d = 4$$

Area, A =2d = 8

Area of square = $(side)^2$

$$8 = side^2$$

$$side = \sqrt{8}$$

$$= 2\sqrt{2}$$

7. Correct option: (B) 96 sq cm

Given

Length of rectangle = 12 cm

Breadth of rectangle = 8 cm

Area of rectangle = $length \times breadth$

$$= 12 \times 8$$

$$= 96cm^2$$

8. Correct option: (A) 60 sq. cm

Given, Base of triangle = 12 cm

Height of triangle = 10 cm

Area of triangle = $\frac{1}{2} \times base \times height$

$$= \frac{1}{2} \times 12 \times 10$$

$$= 6 \times 10$$

$$= 6\ cm^2$$

9. Correct option: (D) 16 cm

Given, Area of triangle = 40 sq. cm

Base of triangle = 5 cm

Height = h

Area of triangle = $\frac{1}{2} \times base \times height$

$$40 = \frac{1}{2} \times 5 \times h$$

$$h = \frac{80}{5}$$

$$h = 16cm$$

10. Correct option: (B) $9\sqrt{3}\ cm^2$

Given, Side of equilateral triangle = 6 cm

Area of equilateral triangle = $\frac{\sqrt{3}}{4}a^2$

$$= \frac{\sqrt{3}}{4} \times 6 \times 6$$

$$= \frac{\sqrt{3}}{2} \times 3 \times 6$$

$$= \sqrt{3} \times 3 \times 3$$

$$= 9\sqrt{3}\ cm$$

11. Correct option: (D) $2 \times \sqrt{A\sqrt{3}}cm$

Given, Area of equilateral triangle = A

Side = a

Area of equilateral triangle = $\frac{\sqrt{3}}{4}a^2$

$$A = \frac{\sqrt{3}}{4}a^2$$
$$a^2 = 4A\sqrt{3}$$
$$a = \sqrt{4A\sqrt{3}}$$
$$a = 2\sqrt{A\sqrt{3}}cm$$

12. Correct option: (A) b^2

Given, Base of triangle is half of its height $b = \frac{h}{2}$

$$h = 2b$$

Area of triangle = $\frac{1}{2} \times base \times height$

$$= \frac{1}{2} \times b \times 2b$$
$$= b^2$$

13. Correct option: (D) 243 sq. cm

Given, Height of triangle = 18 cm

Base is $\frac{3}{2}$ of height $b = \frac{3}{2} \times 18$

$$b = 3 \times 9$$
$$b = 27\ cm$$

Area of triangle = $\frac{1}{2} \times base \times height$

$$= \frac{1}{2} \times 18 \times 27$$
$$= 9 \times 27$$
$$= 243cm^2$$

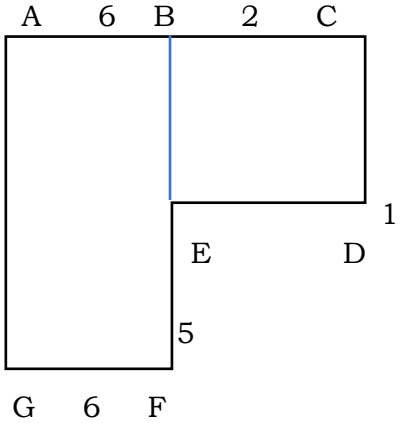
14. Correct option: (B) 12

Given, Area of triangle is 6 times the height $A = 6h$

Area of triangle = $\frac{1}{2} \times base \times height$

$$6h = \frac{1}{2} \times b \times h$$
$$12 = b$$

15. Correct option: (B) 38

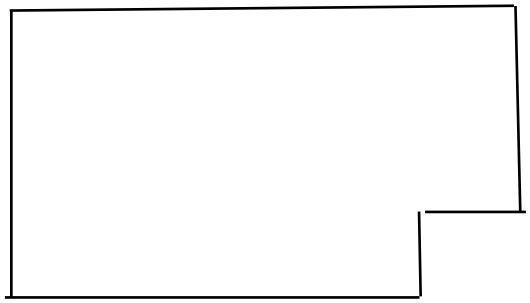


Area of figure = area of rectangle BCDE + area of square ABGF

$$= 2 \times 1 + 6 \times 6$$
$$= 2 + 36$$
$$= 38$$

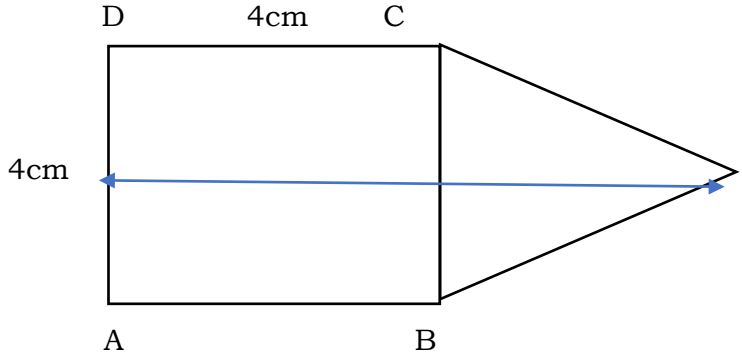
16. Correct option: (A) 63

8



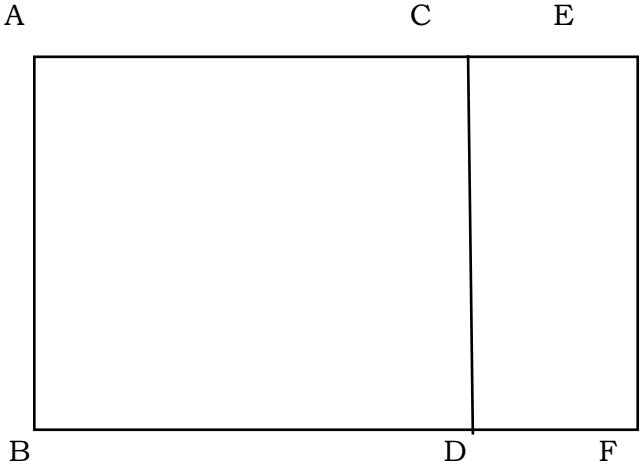
Given
Side of square = 8
Area of square = $8 \times 8 = 64$
Side of square cut = 1
Area of square = $1 \times 1 = 1$
Area of portion left = $64 - 1 = 63$

17. Correct option: (A)



Area of square ABCD = $4 \times 4 = 16cm^2$
Area of triangle CEB = $\frac{1}{2} \times 4 \times 1 = 2cm^2$
Area of figure = area of square + area of triangle
$$= 16 + 2$$
$$= 18cm^2$$

18. Correct option: (C) 12 cm



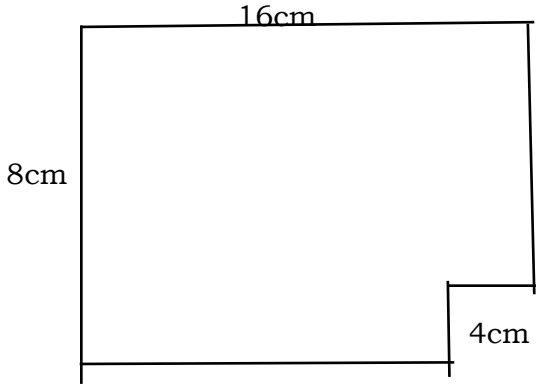
Given, Area of rectangle ABEF = 96 sq. cm
Area of square ABCD = 64 sq. cm
Side of square = a
Area of square = side ×side $64 = a^2$

$$a = \sqrt{64}$$
$$a = 8cm$$

Breadth of rectangle ABEF = 8 cm
Area of rectangle = $length \times breadth$ $96 = l \times 8$

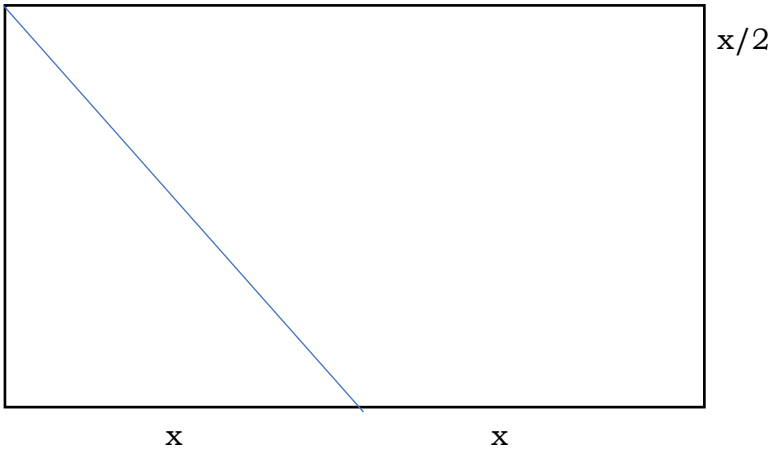
$$l = 12\ cm$$

19. Correct option: (A) 112 sq. cm



Area of cut of square = $4 \times 4 = 16cm^2$
Area of rectangle = $16 \times 8 = 128cm^2$
Area of figure = $128 - 16 = 112cm^2$

20. Correct option: (D) 64 sq. cm



Area of right-angled triangle = 16 sq. cm
Area of triangle = $\frac{1}{2} \times base \times height$

$$16 = \frac{1}{2} \times (x) \times \left(\frac{x}{2}\right)$$
$$64 = x^2$$
$$x = \sqrt{64}$$
$$x = 8$$

Length of rectangle = 2x = 2×8 = 16 cm
Breadth of rectangle = $\frac{x}{2} = \frac{8}{2} = 4\ cm$
Area of rectangle = $lenght \times breadth$ $= 16 \times 4$

$$= 64\ cm^2$$

LEVEL – II

1. Correct option: (A) 16

Given Area of square is 16 times side of square $A = 16 \times side$

Area of square = $(side)^2$ $16 \times side = (side)^2$

$side = 16$

2. Correct option: (C) 50 sq. cm

Given Area of square is 5 times its diagonal $A = 5d$

$$d = \frac{A}{5}$$

Area of square = $\frac{1}{2}d^2$

$$A = \frac{1}{2}\left(\frac{A}{5}\right)^2$$

$$A = \frac{1}{2} \times \frac{A^2}{25}$$

$$A = 50$$

3. Correct option: (D) 20 cm

Given , Area of triangle = area of rectangle

Length of rectangle = 20 cm

Breadth of rectangle = 15 cm

Area of rectangle = length × breadth

$$= 20 \times 15$$

$$= 300\ cm^2$$

Base of triangle = 30 cm

Area of triangle = $\frac{1}{2} \times base \times height$

$$300 = \frac{1}{2} \times 30 \times h$$

$$300 = 15h$$

$$h = 20\ cm$$

4. Correct option: (A) 40 cm

Given, Area of triangle = 220 sq. cm

Height of triangle = 11 cm

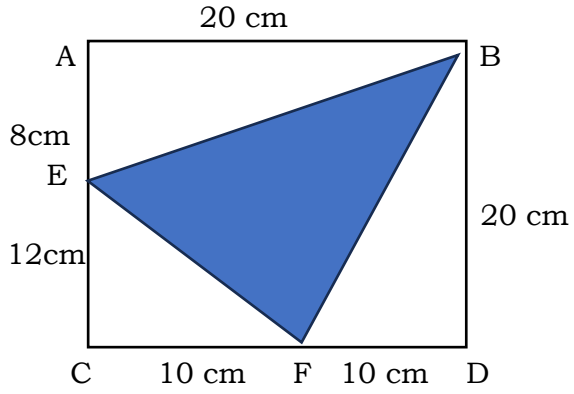
Area of triangle = $\frac{1}{2} \times base \times height$

$$220 = \frac{1}{2} \times b \times 11$$

$$b = \frac{440}{11}$$

$$b = 40\ cm$$

5. Correct option: (B) 160 sq cm



Area of square ABCD = 20×20

$$= 400cm^2$$

Area of triangle ABE = $\frac{1}{2} \times AE \times AB$

$$= \frac{1}{2} \times 8 \times 20$$
$$= 8 \times 10$$
$$= 80cm^2$$

Area of triangle CEF = $\frac{1}{2} \times CE \times CF$

$$= \frac{1}{2} \times 12 \times 10$$
$$= 6 \times 10$$
$$= 60cm^2$$

Area of triangle BDF = $\frac{1}{2} \times BD \times DF$

$$= \frac{1}{2} \times 20 \times 10$$
$$= 10 \times 10$$
$$= 100cm^2$$

Area of shaded region = Area of square – (Area of triangle ABE +Area of triangle CEF + Area of triangle BDF)

$$= 400 - (60 + 80 + 100)$$
$$= 400 - 240$$
$$= 160cm^2$$

6. Correct option: (A) p + q + r

Given, Side AB = p, BC = r, AC = q of a triangle ABC

Perimeter of triangle = sum of it sides length $= p + q + r$

7. Correct option: (B) 45 cm

Given, Area of rectangle = 1125 sq. cm

Breadth, b = 25 cm

Length l

Area of rectangle = $lenght \times breadth$

$$1125 = l \times 25$$
$$l = 45\ cm$$

8. Correct option: (A) 1000

Given Length of rectangle = 50 cm

Breadth of rectangle = 20 cm

Area of rectangle = $lenght \times breadth$

$$A = 50 \times 20$$
$$= 1000cm^2$$

9. Correct option: (D) increase 4 times

Area of square will increase 4 times if its sides are doubled.

10. Correct option: (A) 440

Given Length of rectangle = 130 m

Breadth of rectangle = 90 m

Perimeter of rectangle = $2(l + b)$

$$= 2(130 + 90)$$
$$= 2(220)$$
$$= 440\ m$$

11. Correct option: (A) and (B)

Given

Area of square = A

Side = $\sqrt{A}$

Diagonal = $\sqrt{2A}$

12. Correct option: (C) Area increases and (D) Perimeter increases

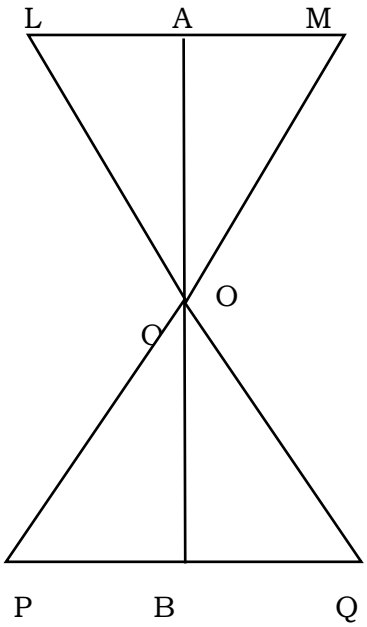
If length of rectangle increases its perimeter and area also increases.

13. Correct option: (A) 6 sq. m and (C) 6×10^4 sq cm

Given,

AB = 6 m

LM = PQ = 4 m



LMO forms a triangle

O is mid-point of AB, AO = $\frac{AB}{2} = \frac{6}{2} = 3\ m$

Area of triangle LMO = $\frac{1}{2} \times base\ LM \times height\ AO$

$$= \frac{1}{2} \times 3 \times 4$$

$$= 2 \times 3$$

$$= 6\ m^2$$

$$= 6 \times 10^4\ cm^2$$

14. Correct option: (B) 40 cm

Let length be l

Area of rectangle = $length \times breadth$

$$1200 = l \times 30$$

$$l = 40\ cm$$

15. Correct option: (C) 4800 sq cm

Length and breadth become twice, so

l = $2 \times 40 = 80\ cm$

b = $2 \times 30 = 60\ cm$

Area of rectangle = $length \times breadth$

$$= 80 \times 60$$

$$= 4800 cm^2$$

16. Correct option: (A) 20 cm

Given Area of rectangle = 1200 sq. cm

Breadth is twice, b = $2 \times 30 = 60\ cm$

Area of rectangle = $length \times breadth$

$$1200 = l \times 60$$
$$l = 20\ cm$$

17. Correct Matching is: $a \to R$; $b \to S$; $c \to Q$; $d \to P$

a) Given Side of square = a

Area = a^2

Area = $\frac{1}{2}d^2$

$$a^2 = \frac{1}{2}d^2$$
$$\sqrt{2a^2} = d$$
$$d = \sqrt{2}a$$

b) Side of equilateral triangle = a

Height = $\sqrt{a^2 - \left(\frac{a}{2}\right)^2}$

$$= \sqrt{a^2 - \frac{a^2}{4}}$$
$$= \frac{\sqrt{3}a}{2}$$

c) Side of equilateral triangle = a

Height = $\frac{\sqrt{3}a}{2}$

Area = $\frac{1}{2} \times base \times height$

$$= \frac{1}{2} \times a \times \frac{\sqrt{3}a}{2}$$
$$= \frac{\sqrt{3}a^2}{4}$$

d) Given Area = a

Length = l

Area of rectangle = length × breadth

$$a = l \times b$$
$$\frac{a}{l} = b$$

18. Correct Answer is: 8 cm

Given Side of square = 8 cm

Diagonal = D

Area of square = $(side)^2$

$$= 8 \times 8$$
$$= 64\ cm^2$$

Also, area of square = $\frac{1}{2}D^2$

$$64 = \frac{1}{2}D^2$$
$$D = \sqrt{64 \times 2}$$
$$D = 8\sqrt{2}$$

Now, $\frac{D}{\sqrt{2}} = \frac{8\sqrt{2}}{\sqrt{2}} = 8\ cm$

19. Correct Answer is: 6 cm

Given Area of rectangle = 72 sq. cm

Length of rectangle is twice its breadth, l = 2b

Area of rectangle = length × breadth

$$72 = 2b \times b$$
$$36 = b^2$$
$$b = 6cm$$

20. Correct Answer is: 2 (The area becomes twice times)

Given

Base of triangle is twice area

Then area gets doubled

STUDENT PRACTICE QUESTIONS

1. Correct option: (C) 144 sq m

Given

Side of square = 12 m

Area of square = side × side

$$= 12 \times 12$$
$$= 144m^2$$

2. Correct option: (B) 35 cm

Given

Area of square = 1225 sq. cm

Side be x

Area of square = $(side)^2$

$$1225 = (x)^2$$
$$x = \sqrt{1225}$$
$$x = 35cm$$

3. Correct option: (B) $29\sqrt{2}$

Given

Area of square = 841 sq m

Diagonal be d

Area of square = $\frac{1}{2}d^2$

$$841 = \frac{1}{2}d^2$$
$$d^2 = 1682$$
$$d = \sqrt{1682}$$
$$d = 29\sqrt{2}$$

4. Correct option: (A) 1634 sq cm

Given

Length of rectangle = 43 cm

Breadth of rectangle = 38 cm

Area of rectangle = $length \times breadth$

$$= 43 \times 38$$
$$= 1634cm^2$$

5. Correct option: (C) 13 cm

Given

Length of rectangle = 12 cm

Breadth of rectangle = 5 cm

Now diagonal is

$$(diagonal)^2 = (length)^2 + (breadth)^2$$
$$diagonal = \sqrt{(12)^2 + (5)^2}$$
$$diagonal = \sqrt{144 + 25}$$
$$diagonal = \sqrt{169}$$
$$diagonal = 13$$

6. Correct option: (B) 12 cm

Given

Length of rectangle = 20 cm

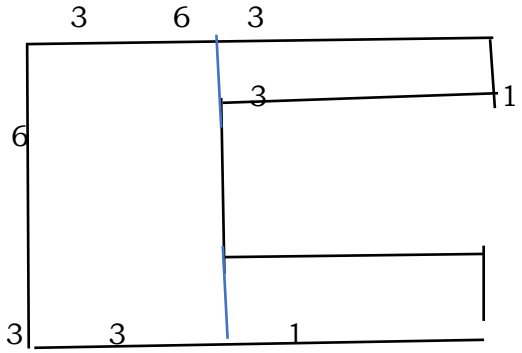
Area of rectangle = 240 sq cm

Breadth of rectangle = b

Area of rectangle = $length \times breadth$

$$240 = 20 \times b$$
$$b = \frac{240}{20}$$
$$b = 12\ cm$$

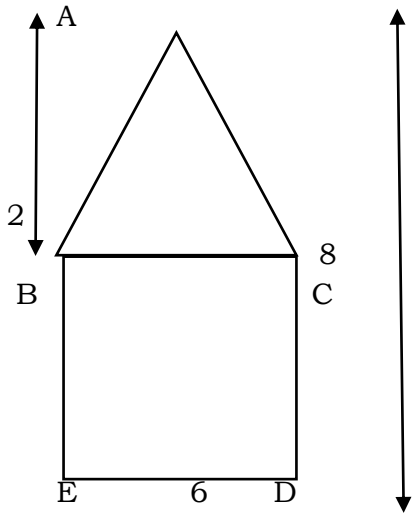
7. Correct option: (C) 24



Area of figure = area of rectangles

$$= lenght \times breadth$$
$$= 6 \times 3 + 3 \times 1 + 3 \times 1$$
$$= 18 + 3 + 3$$
$$= 24$$

8. Correct option: (B) 42

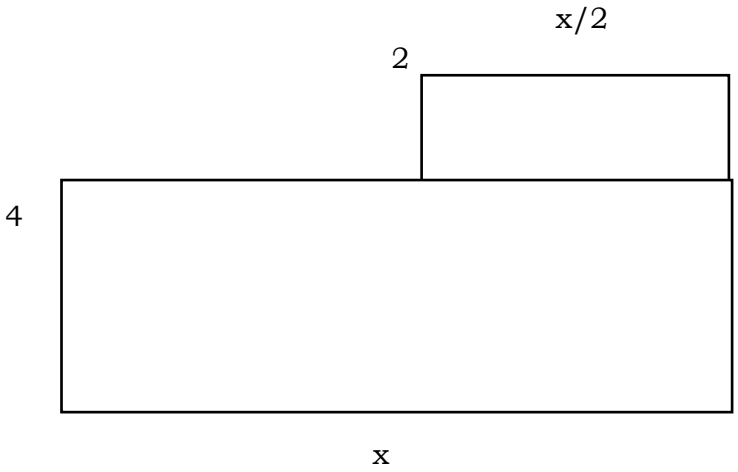


BCDE is a square
BC = CD = ED = BE = 6
Area of square = $6 \times 6 = 36$
In triangle ABC
Base BC = 6
Height = 2
Area of triangle = $\frac{1}{2} \times 6 \times 2 = 6$
Area of figure = area of square + area of triangle

$$= 36 + 6$$
$$= 42$$

9. Correct option: (D) 12 cm

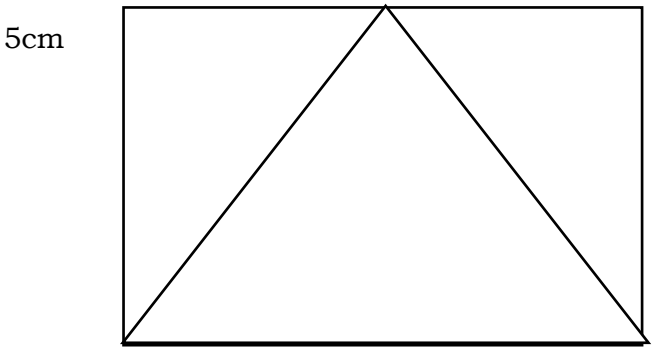
Given Area of figure = 60 sq. cm



Area of figure = area of rectangles

$$60 = 4 \times (x) + 2 \times \frac{x}{2}$$
$$60 = 4x + x$$
$$5x = 60$$
$$x = 12cm$$

10. Correct option: (A) 20 sq. cm



Area of triangle = $\frac{1}{2} \times base \times height$

$$= \frac{1}{2} \times 8 \times 5$$
$$= 4 \times 5$$
$$= 20\ cm^2$$

11. Correct option: (A) Base and (B) Height

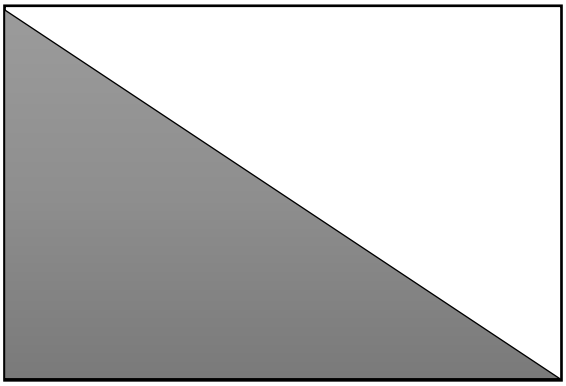
Area of triangle depends on base and height

12. Correct option: (B) 23 sq. m and (D) 23×10^4 sq cm

Given

Area of shaded region = 23 sq. m

Which is a triangle



Area of unshaded region is also a triangle

So, area of unshaded triangle = 23 sq m = 23×10^4 sq cm

13. Correct option: (D) 10 cm

Given Length of rectangle = 3 cm

Breadth of rectangle = 2 cm

Perimeter = $2(l + b)$

$$= 2(3 + 2)$$
$$= 2(5)$$
$$= 10\ cm$$

14. Correct option: (A) 16 m

Given Length of rectangle = 4.5 m

Breadth of rectangle = 3.5 m

Perimeter = $2(l + b)$

$$= 2(4.5 + 3.5)$$
$$= 2(8)$$
$$= 16\ m$$

15. Correct Matching is: $a \rightarrow Q; b \rightarrow P; c \rightarrow R;\ d \rightarrow S;$

a) Each angle in a square is 90°

b) Each angle in an equilateral triangle is 60°

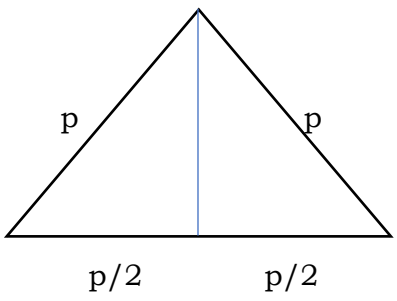
c) Perimeter of given rectangle of length p and breadth q is 2(p + q)

d) P = 9 units

16. Correct Answer is: $p = \frac{2q}{\sqrt{3}}$

Given Side and height ratio is p:q

Sides of equilateral triangle = p



Height = $\sqrt{p^2 - \left(\frac{p}{2}\right)^2}$

$$q = \sqrt{p^2 - \frac{p^2}{4}}$$

$$q = \frac{p\sqrt{3}}{2}$$

$$p = \frac{2q}{\sqrt{3}}$$

17. Correct Answer is: 4 cm

Given

Area of rectangle = 30 sq. cm

Square is inside rectangle such that side of square is equal to breadth of rectangle

Area of square = 25 sq. cm

Side = a

Area of square = a^2

$$25 = a^2$$

$$a = 5$$

Breadth = 5 cm

Area of rectangle = length × breadth

$$20 = l \times 5$$

$$l = 4\ cm$$

18. Correct Answer is: Area of square is 4 sq. cm

Given

Side of square = 2 cm

Area of square = $(side)^2$

$$= 2 \times 2$$

$$= 4cm^2$$

19. Correct Answer is: perimeter of equilateral triangle is 3 cm

Given

Side of equilateral triangle = 1 cm

Perimeter = $1 + 1 + 1 = 3cm$

20. Correct Answer is: Perimeter of rectangle is 4

Given

Length of rectangle = $3/2$

Breadth of rectangle = $1/2$

Perimeter = $2(l + b)$

$$= 2\left(\frac{3}{2} + \frac{1}{2}\right)$$

$$= 2\left(\frac{3+1}{2}\right)$$

$$= 2\left(\frac{4}{2}\right)$$

$$= 4$$